# CACTUS
# REV 0.991
## FAB G RETAIL

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RULES: (APPLIED WHEN POSSIBLE)
1. MSB TO LSB IS TOP TO BOTTOM
2. WHEN POSSIBLE: INPUTS ON LEFT, OUTPUTS ON RIGHT
3. ORDER OF PAGES=CHIP INTERFACES, TERMINATION, POWER, DECOUPLING
4. AVOID USING OFF PAGE CONNECTORS FOR ON PAGE CONNECTIONS
5. LANED SIGNALS ARE GROUPED ON SYMBOLS
6. TRANSIMITTER NAME USED AS PREFIX WITH RX AND TX CONNECTIONS
7. SUFFIX _V  IS USED FOR VOLTAGE RAIL SIGNAL NAMES
8. SUFFIX _DP AND _DN ARE USED FOR DIFFERIENTAL PAIRS
9. UNNAMED NETS ARE NAMED WITH /2 TEXT SIZE
10. SUFFIX _N FOR ACTIVE LOW OR N JUNCTION
12. SUFFIX _P FOR P JUNCTION
13. SUFFIX _EN FOR ENABLE
14. 'CLK' FOR CLOCKS, 'RST' FOR RESETS
15. PWRGD FOR POWER GOOD
16. REV AND FAB ARE SET USING CUSTOM VARIABLES
    TOOLS>OPTIONS>VARIABLES

DRAWING
Thu Apr 20 16:18:47 2017

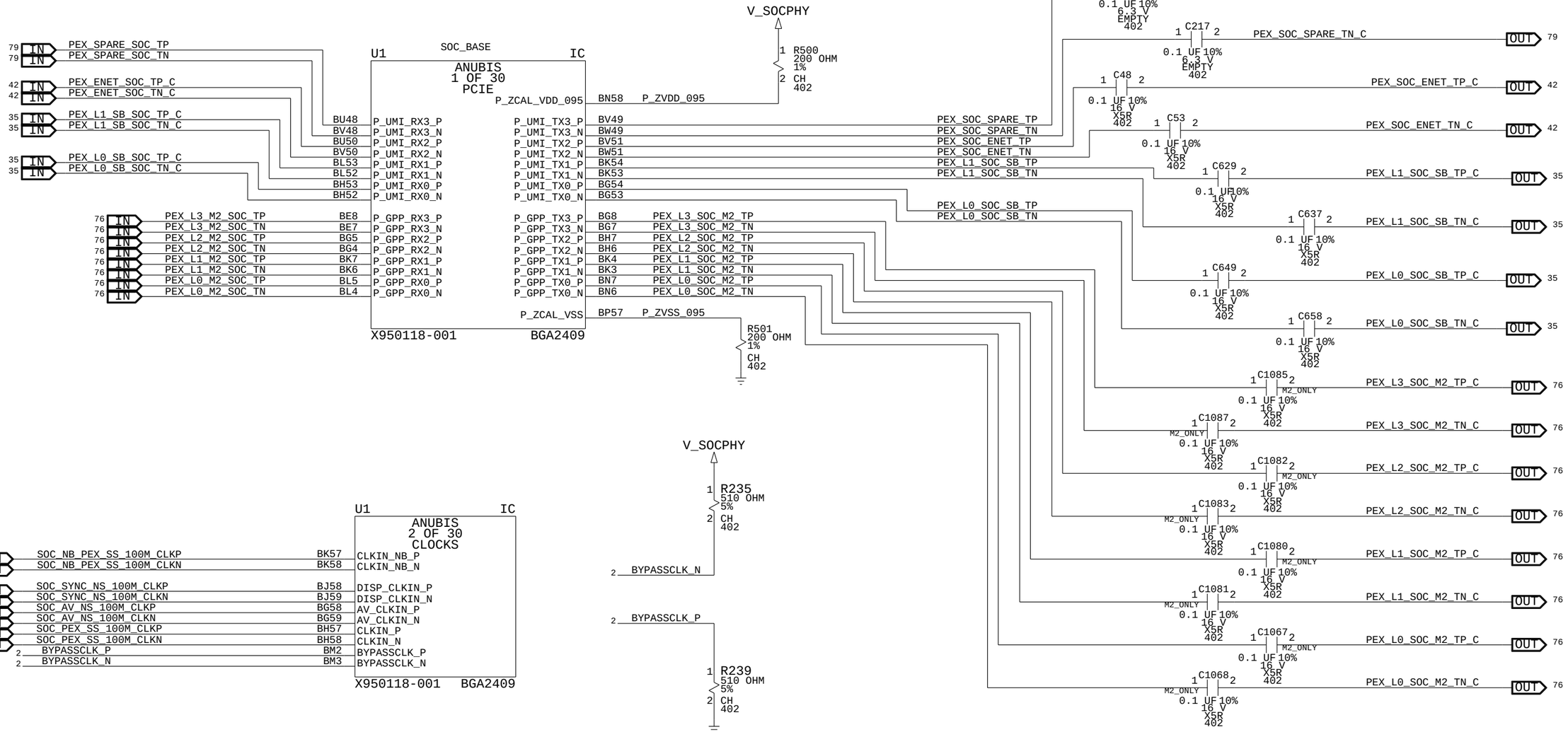
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 1/82 | 1/82 | G-R | 0.991 |

# SOC:PCIEX,CLOCKS

NOTES:

1.TO SUPPORT PCIE SPARE LANE INTERFACE (J28), POPULATE C215 AND C217

2.SEE PAGES 77 AND 79 FOR ADDITIONAL PCIE SPARE LANE INFORMATION



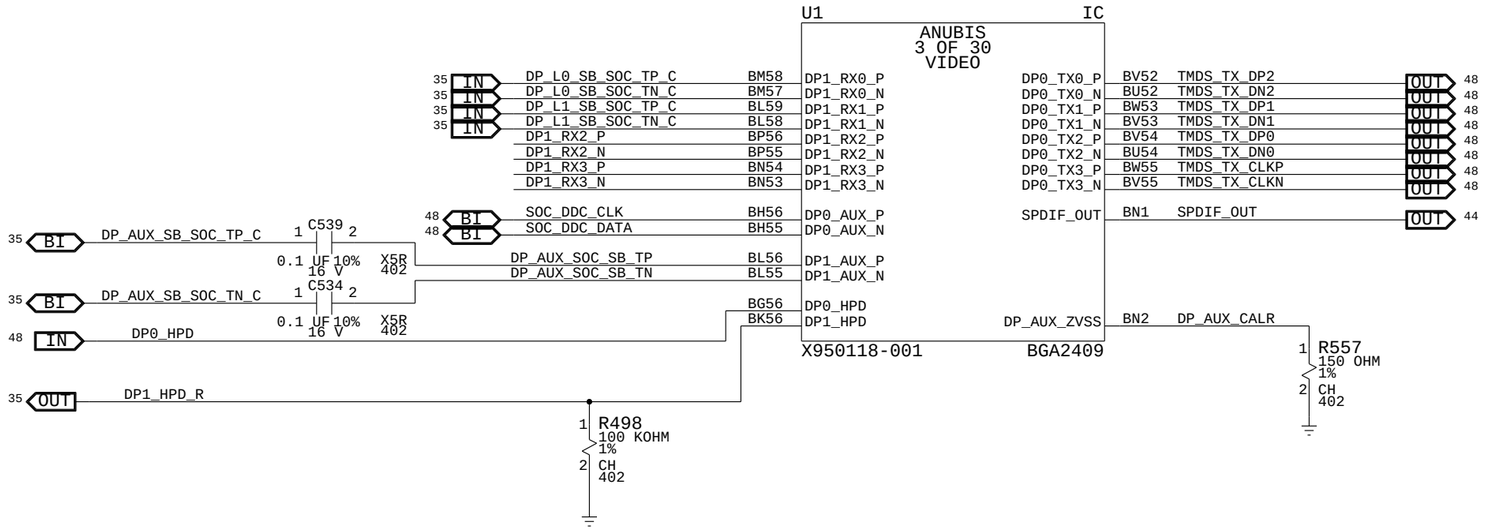




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| X950118-001 | IC | U1 | PROCSR-CPU-GPU,SM,1.0GHZ,BGA2049,ANUBIS | SOC_INCLUDE |
| X950118-001 | EMPTY | U1 | PROCSR-CPU-GPU,SM,1.0GHZ,BGA2049,ANUBIS | SOC_EMPTY |

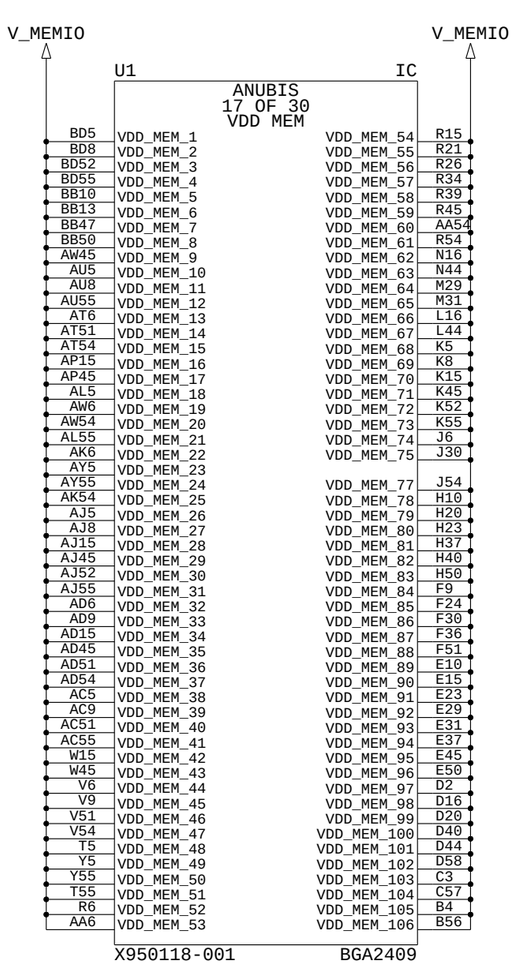
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 2/82 | 2/82 | G-R | 0.991 |

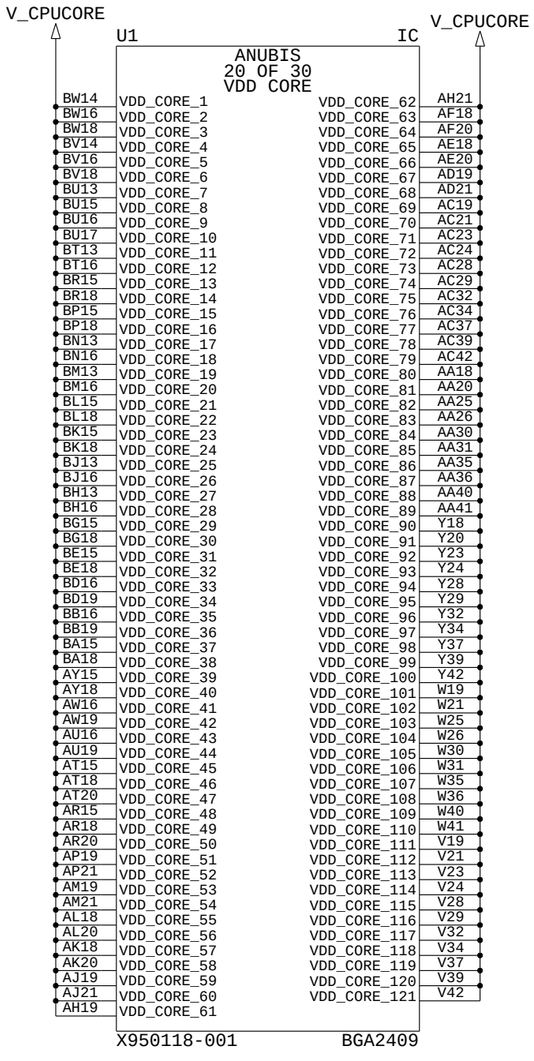
# SOC: VIDEO

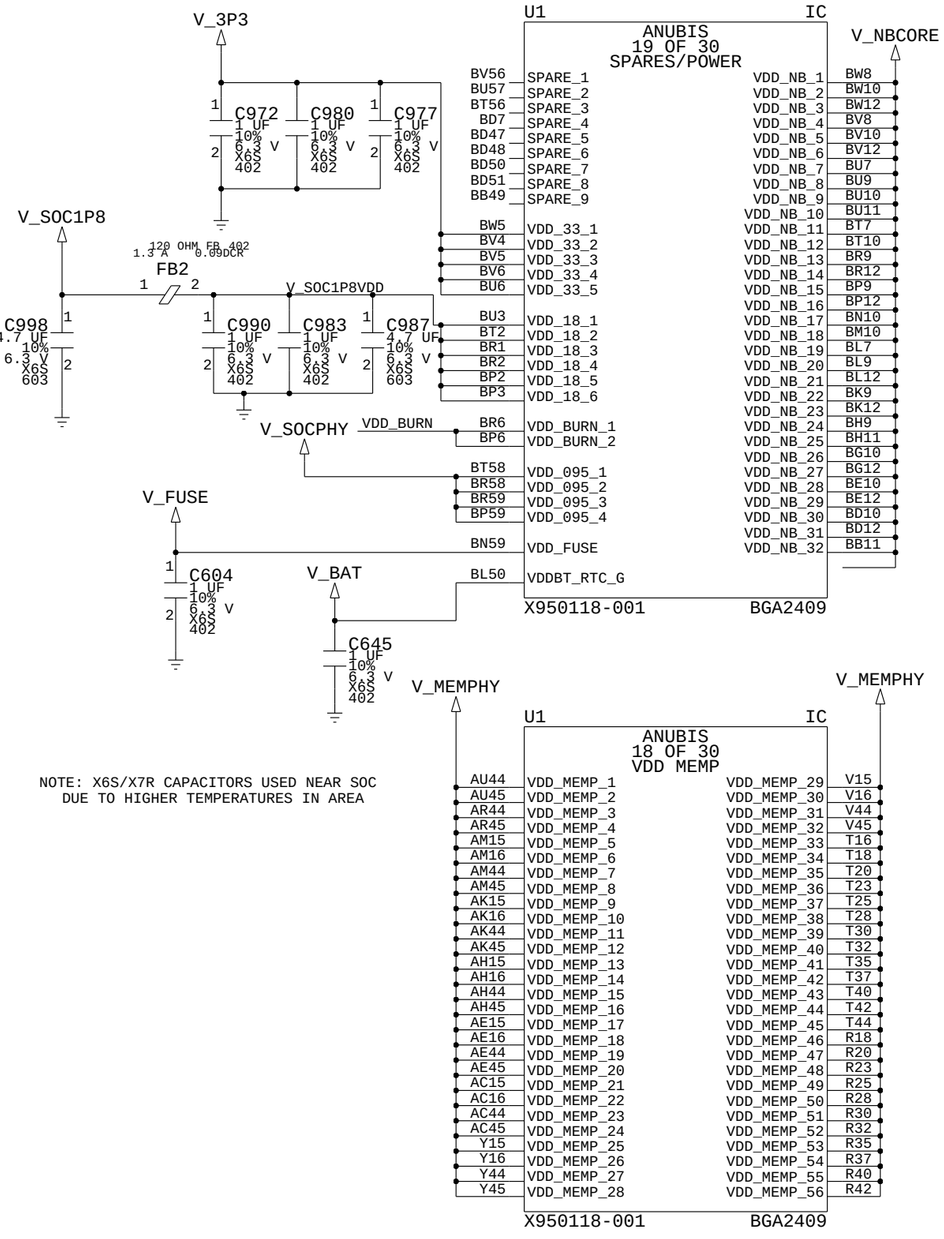



| DVI PCB ROUTING ORDERING | DP PCB ROUTING ORDERING | PIN NAME |
|---|---|---|
| TMDS CLOCK - | DP LANE 3 - | DP0_TX3_N |
| TMDS CLOCK + | DP LANE 3 + | DP0_TX3_P |
| TMDS DATA0 - | DP LANE 2 - | DP0_TX2_N |
| TMDS DATA0 + | DP LANE 2 + | DP0_TX2_P |
| TMDS DATA1 - | DP LANE 1 - | DP0_TX1_N |
| TMDS DATA1 + | DP LANE 1 + | DP0_TX1_P |
| TMDS DATA2 - | DP LANE 0 - | DP0_TX0_N |
| TMDS DATA2 + | DP LANE 0 + | DP0_TX0_P |

# SOC: POWER: MEMIO,MEMPHY,CPUCORE,NBCORE,MISC



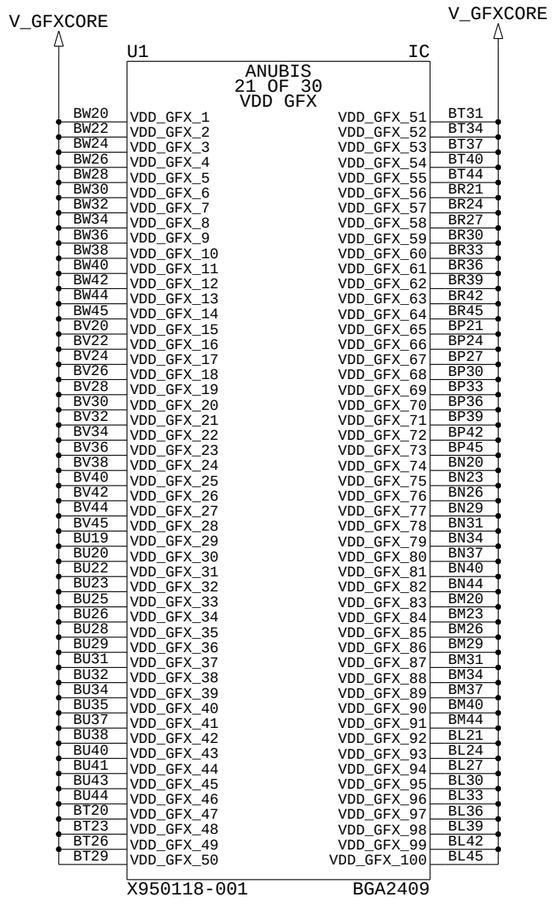



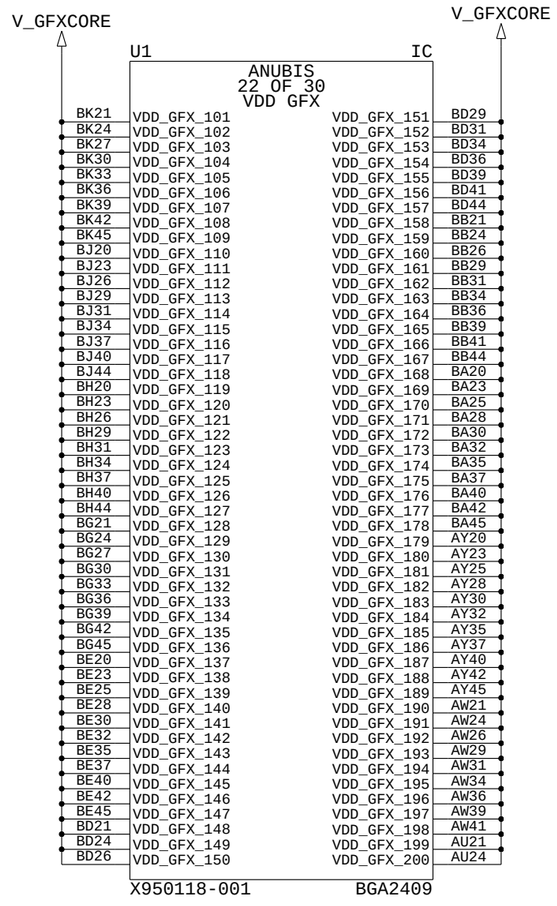



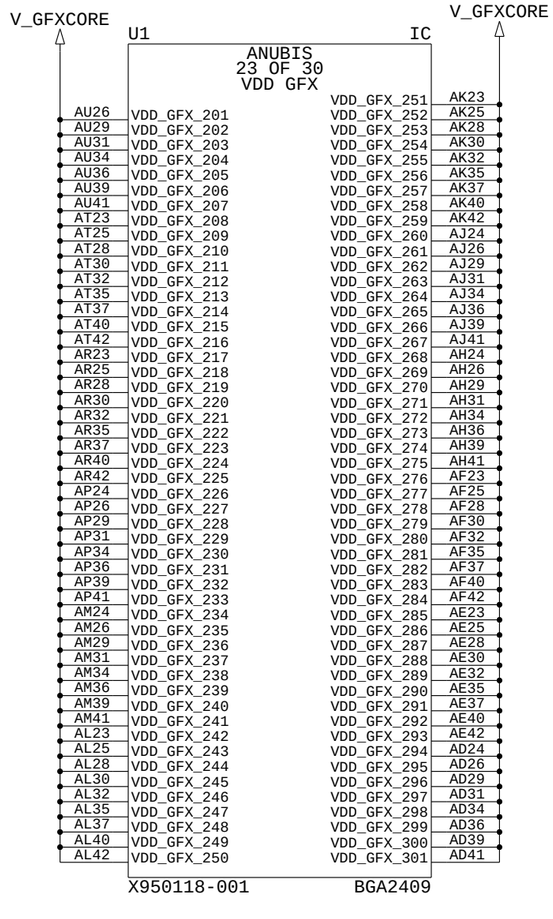

NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
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# SOC: POWER: GFXCORE

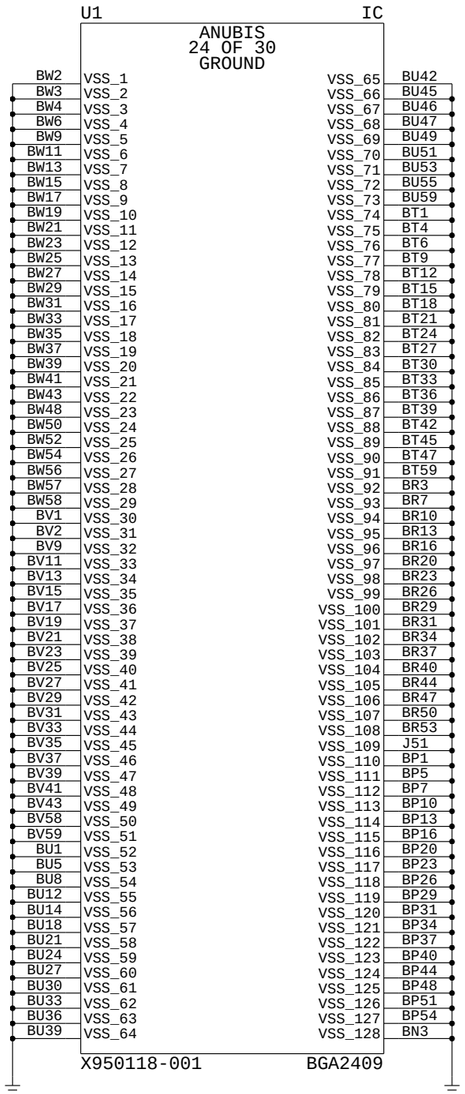


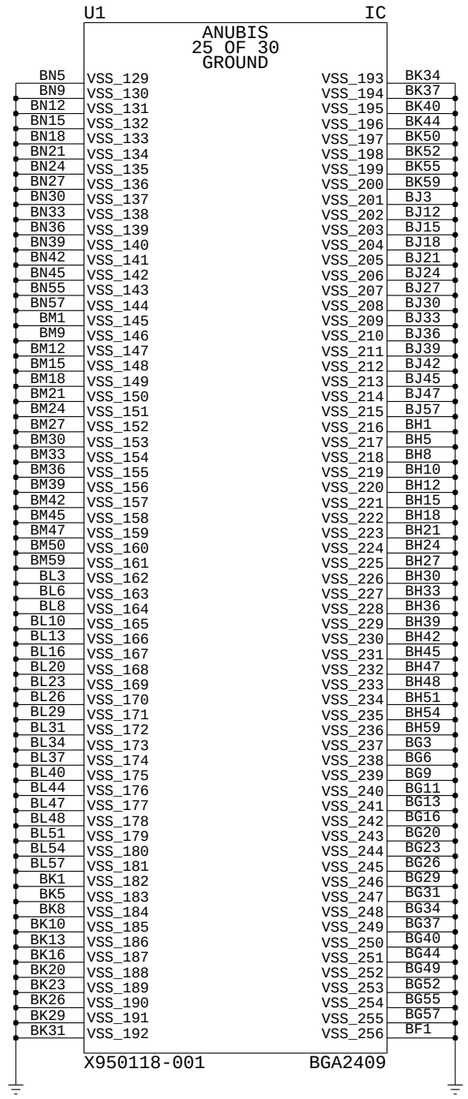



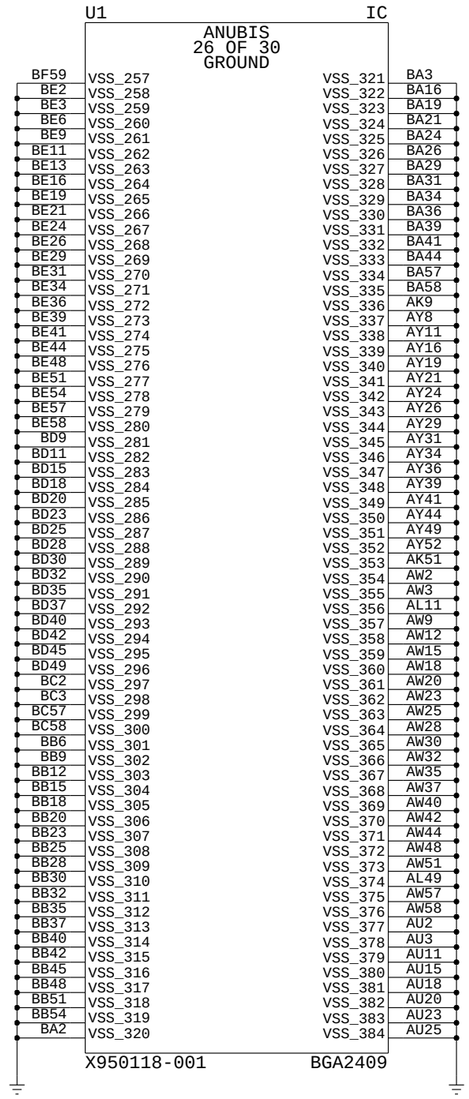



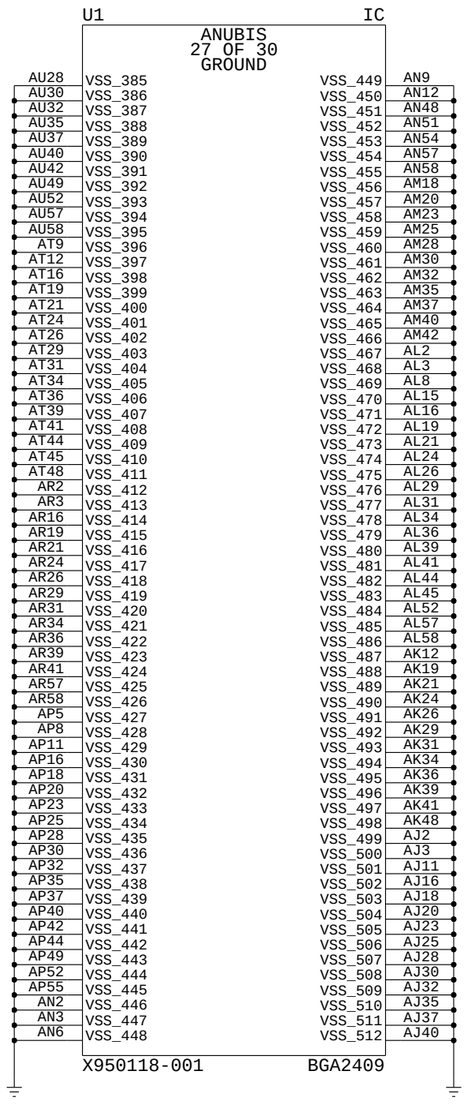

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 5/82 | 5/82 | G-R | 0.991 |

# SOC: POWER: VSS



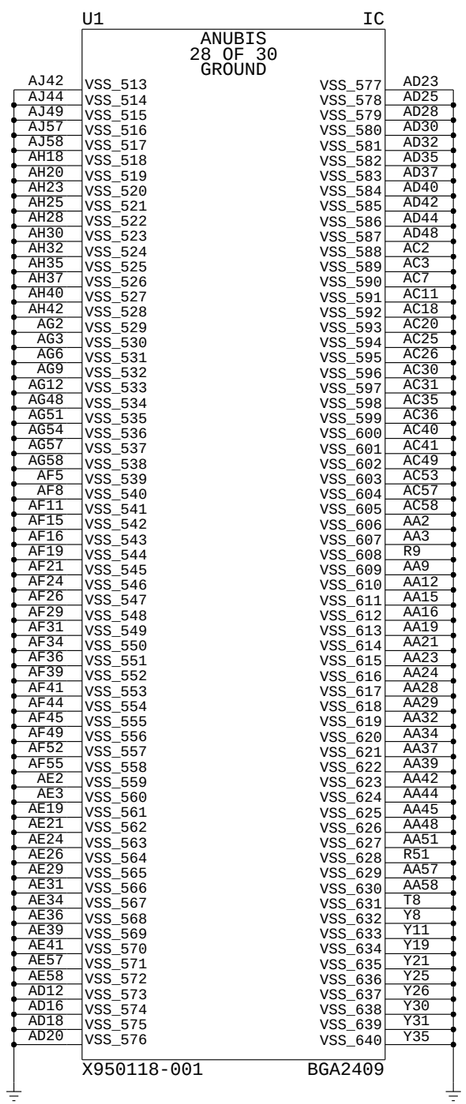



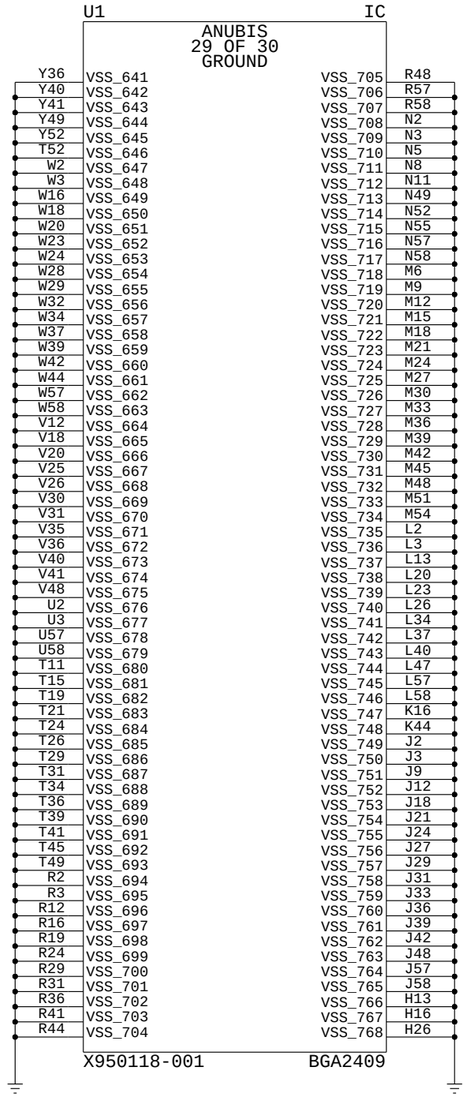



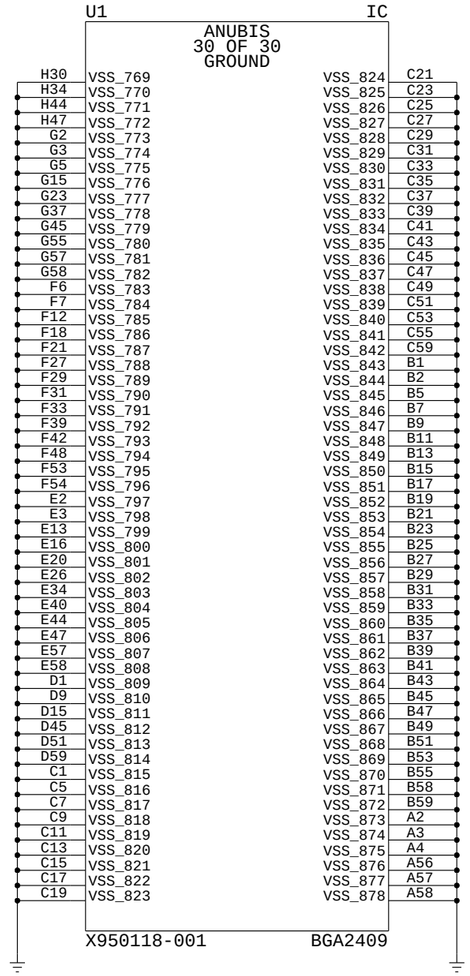




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 6/82 | 6/82 | G-R | 0.991 |

# SOC: POWER: VSS

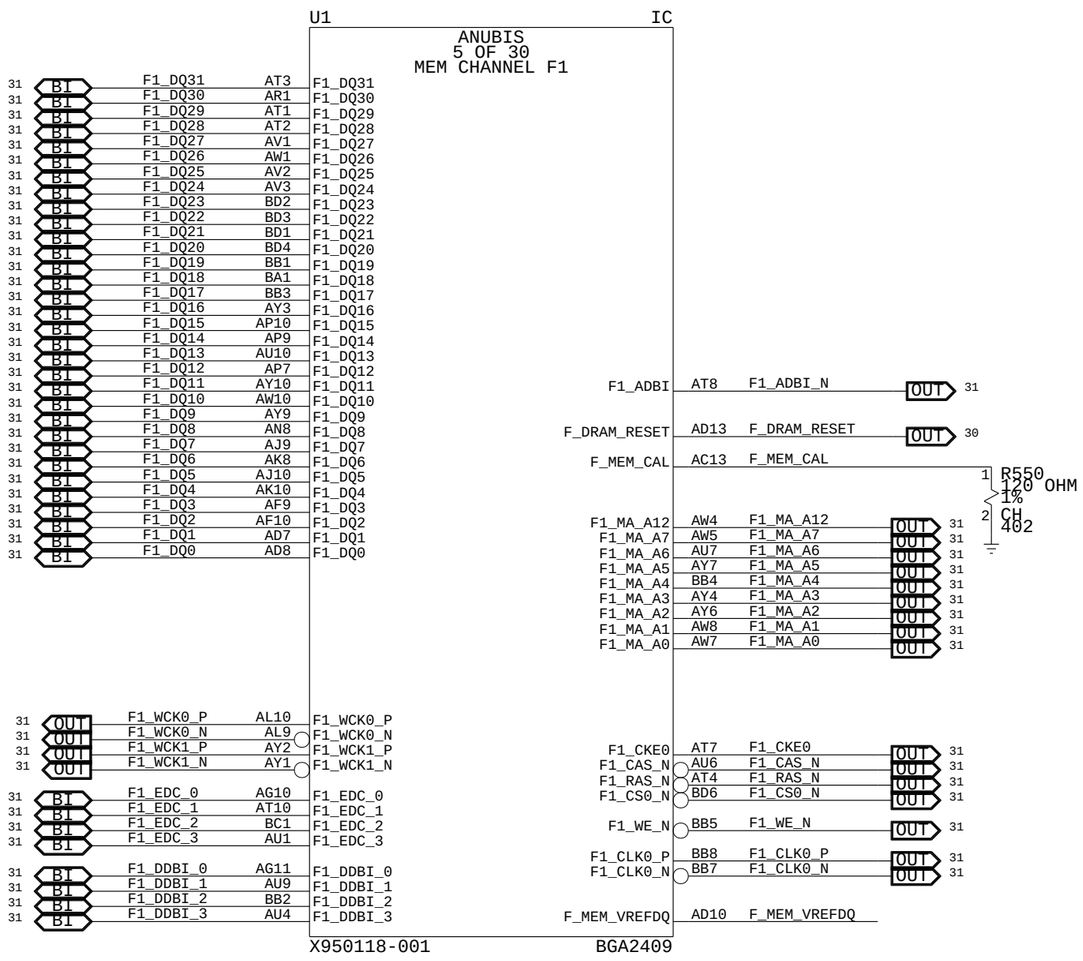


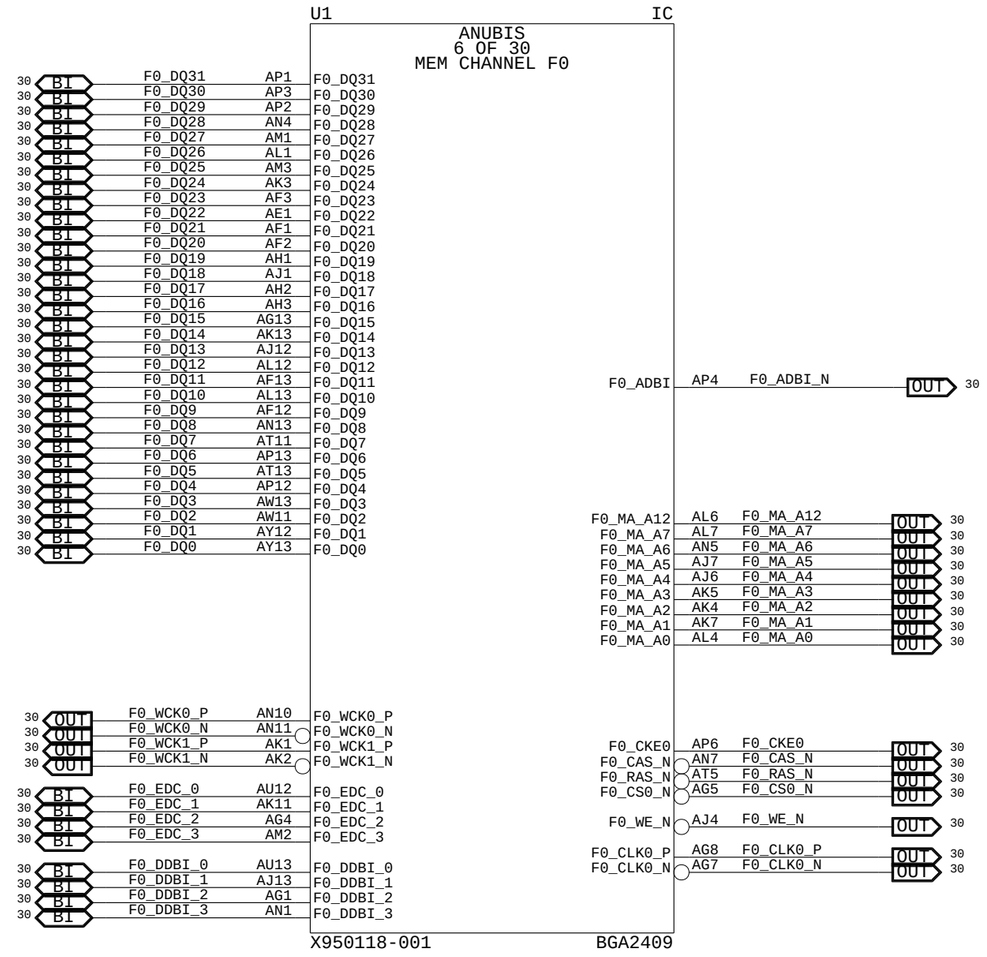




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 7/82 | 7/82 | G-R | 0.991 |

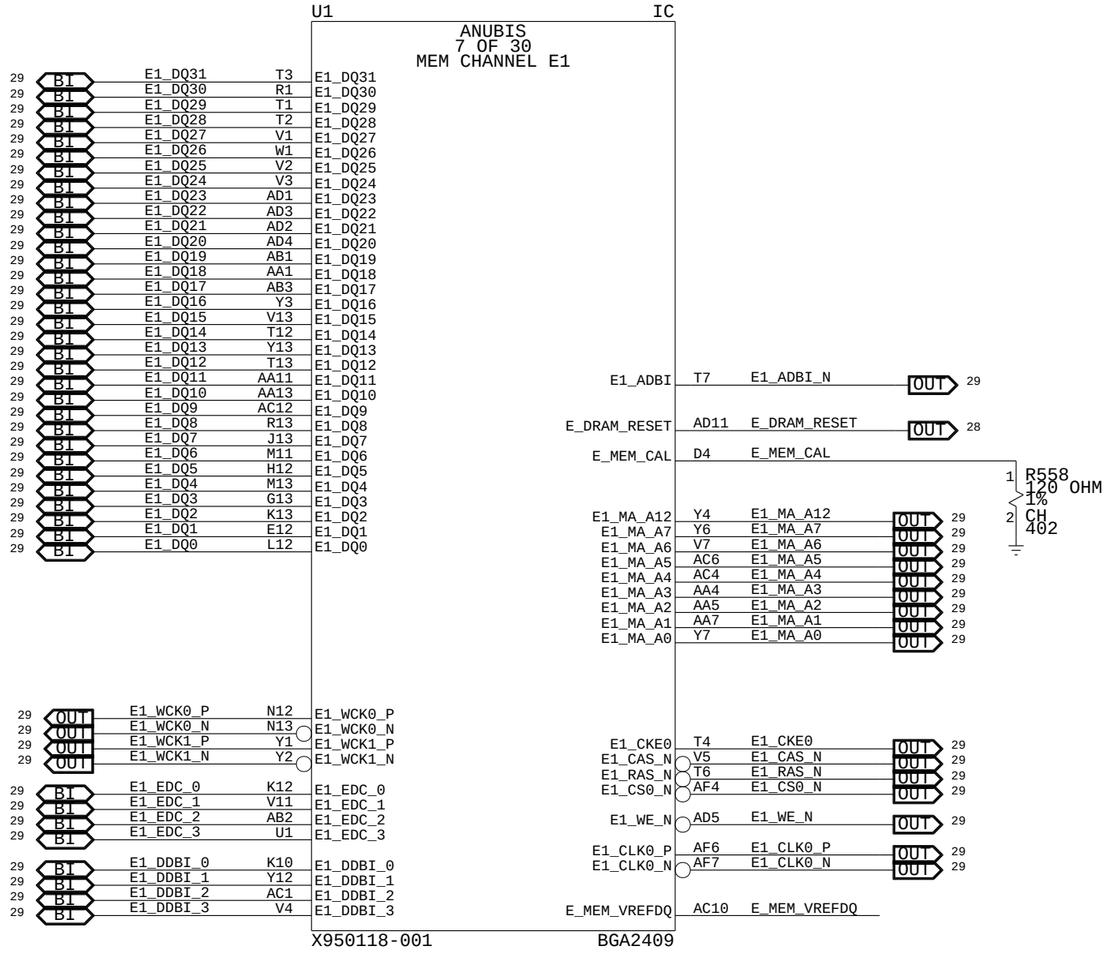
# MEMORY: PARTITION F1/F0

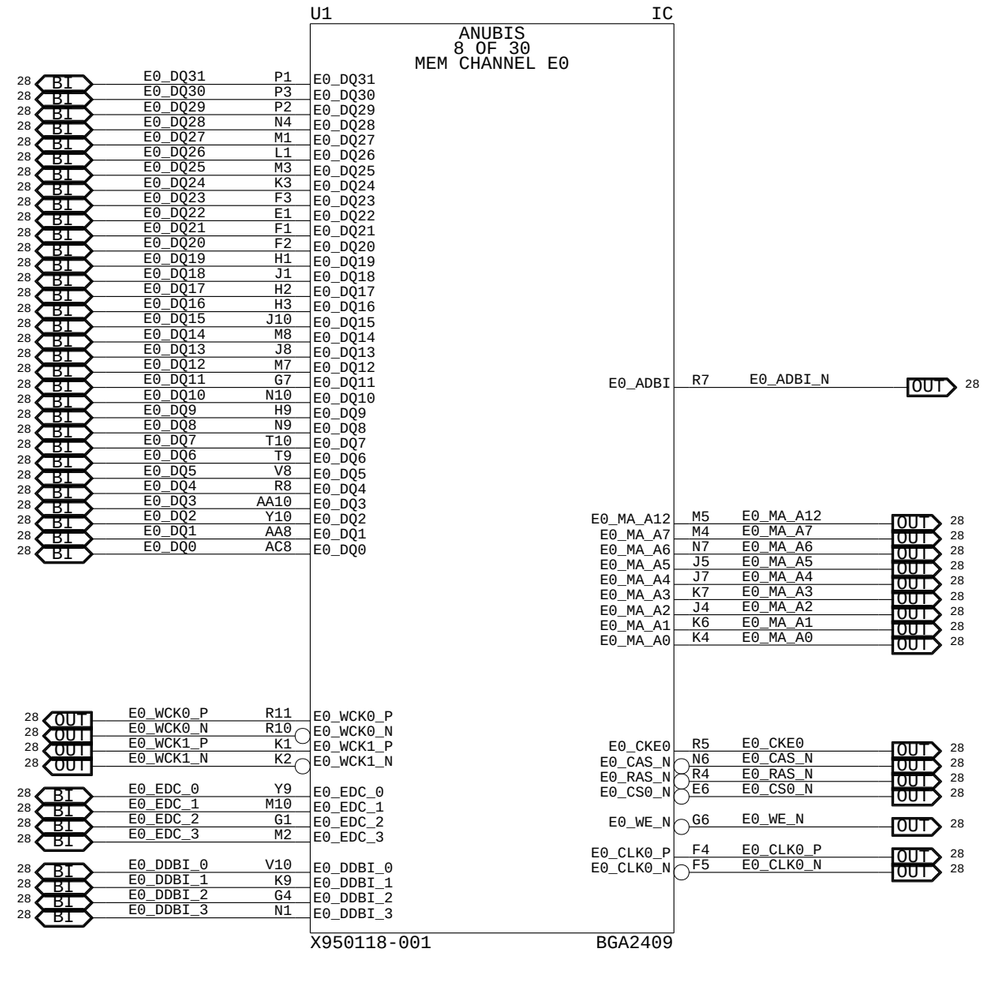



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
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| | Cactus | 8/82 | 8/82 | G-R | 0.991 |

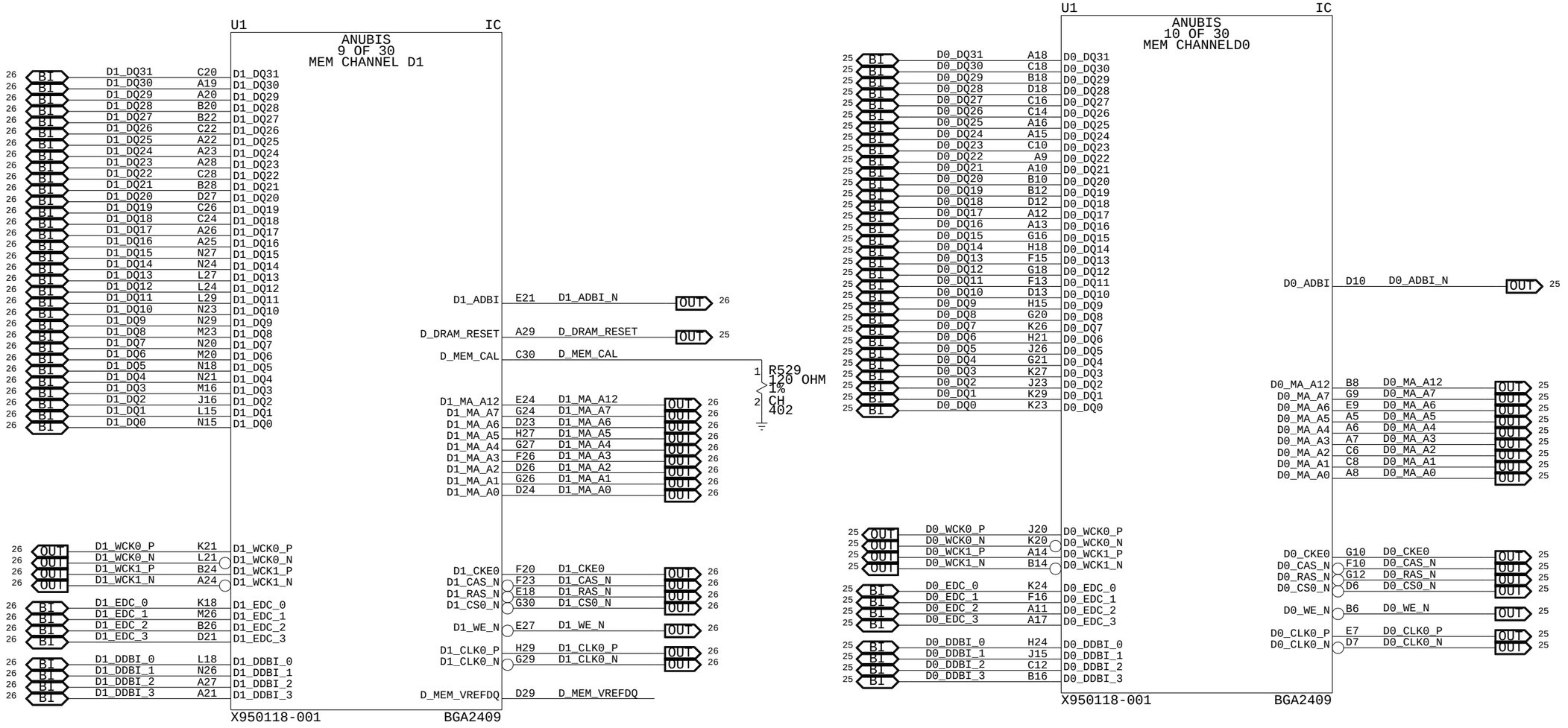
# MEMORY: PARTITION E1/E0



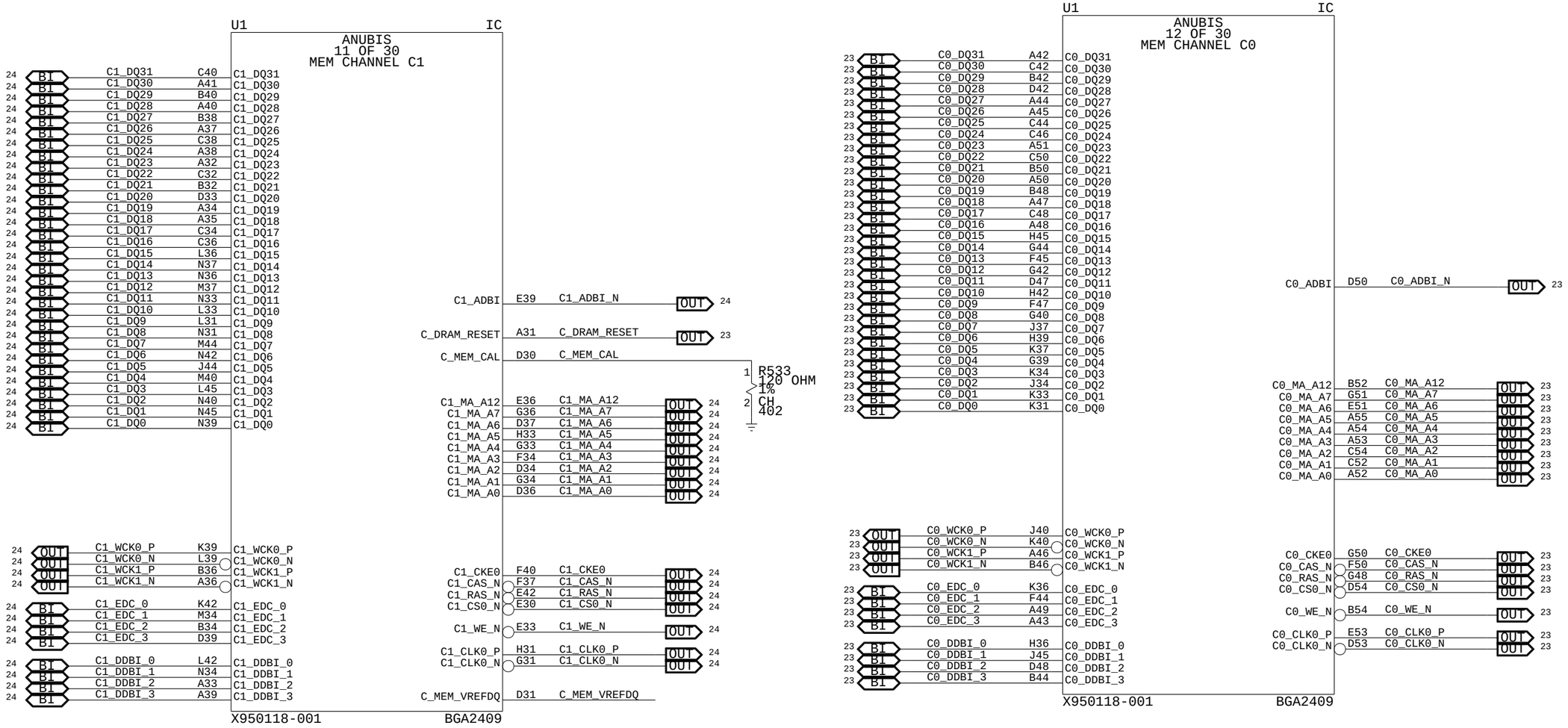

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 9/82 | 9/82 | G-R | 0.991 |

# MEMORY: PARTITION D1/D0



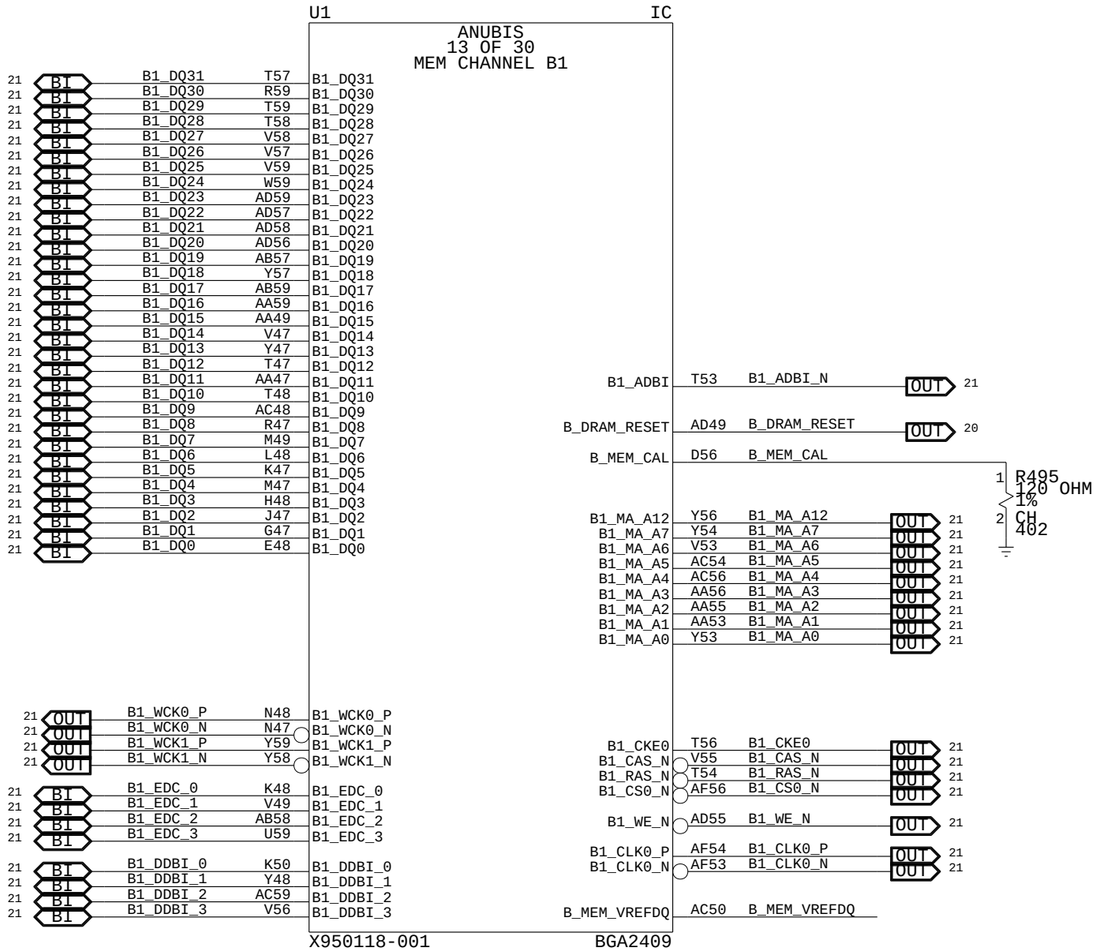

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
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| | Cactus | 10/82 | 10/82 | G-R | 0.991 |

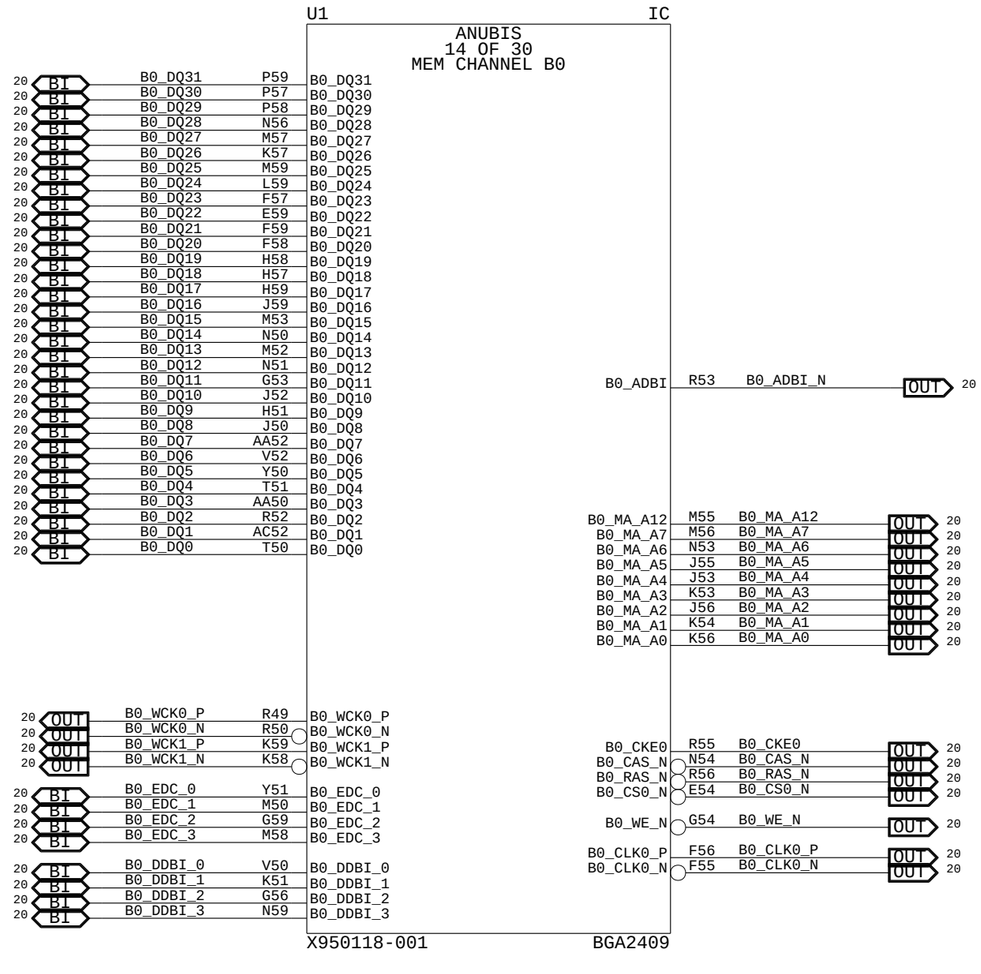
# MEMORY: PARTITION C1/C0




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 11/82 | 11/82 | G-R | 0.991 |

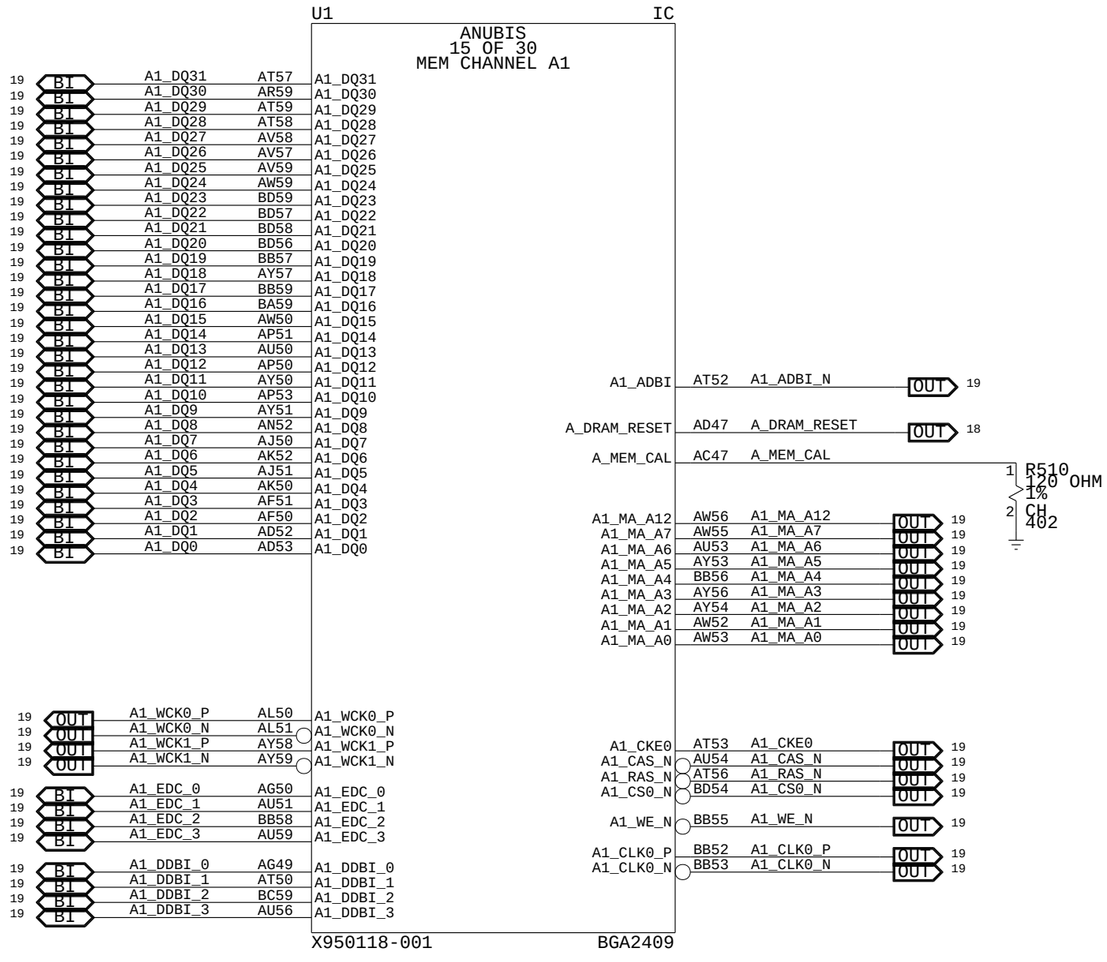
# MEMORY: PARTITION B1/B0

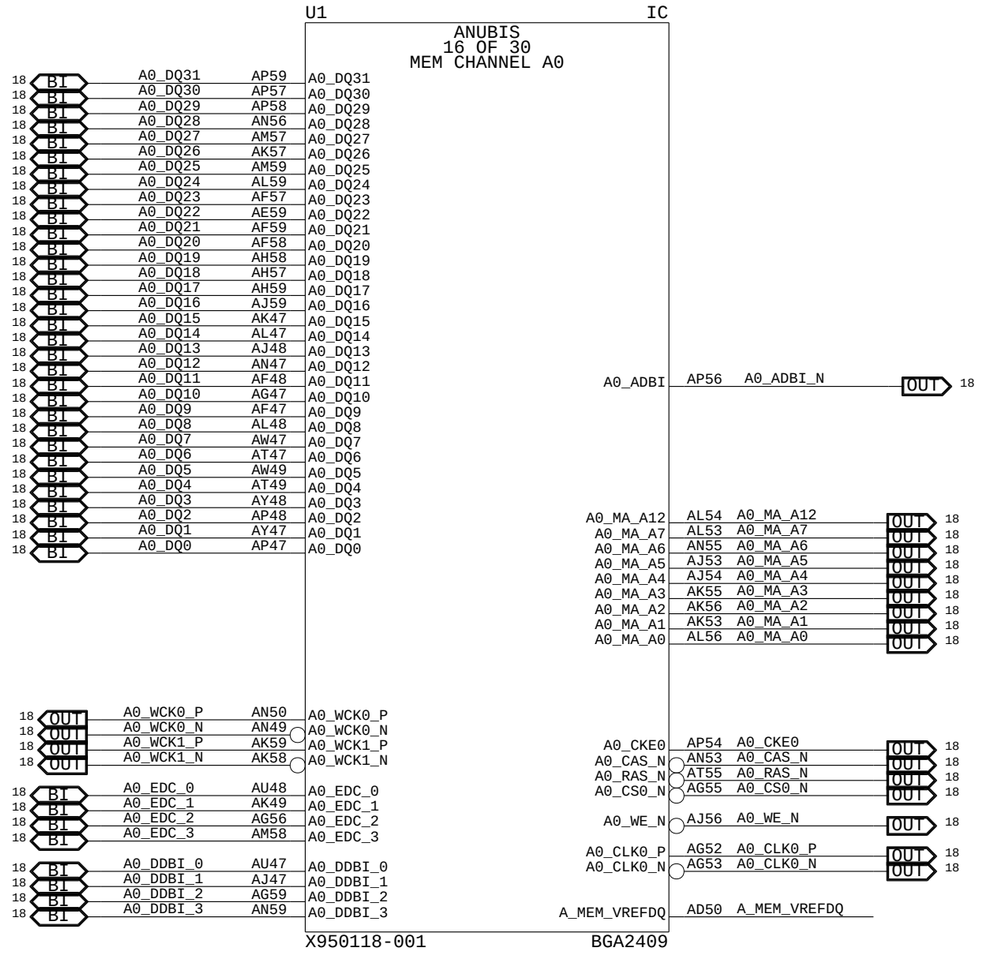



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
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# MEMORY: PARTITION A1/A0




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# SOC:DEBUG,SB SIGNALS,V_BAT

U1 IC
ANUBIS
4 OF 30
JTAG/DEBUG

ATE_TSTCLK_EN BE47 ATE_TSTCLK_EN
BF58 DLY_PSP_RESET
BD13 PSEN
BF57 A0_BYPASS
36 OUT SOC_THERMTRIP_N BL1 THERMTRIP_N
39 IN X32K_X2_R R84 1 2 0 OHM 5% 402 CH
X32K_X2 BN50 X32K_X2
X32K_X1 BP50 X32K_X1
R97 1 2 20 MOHM 5% 402 EMPTY
GPIO_3P3_3 BD53 3P3V_GPIO3
GPIO_3P3_2 BE55 3P3V_GPIO2
GPIO_3P3_1 BE56 3P3V_GPIO1
GPIO_3P3_0 BE59 3P3V_GPIO0
58 OUT GPIO_1P8_3 BN56 1P8V_GPIO9
GPIO_1P8_2 BK51 1P8V_GPIO8
Y1 2 1 SM EMPTY 32.768 KHZ
GPIO_1P8_1 BT48 1P8V_GPIO7
C144 EMPTY 2.2 PF 402 50 V +/-0.1PF
C155 EMPTY 402 2.2 PF 50 V +/-0.1PF
FT415 FTP 1GPIO_1P8_0 BT50 1P8V_GPIO6
GPIO_POSTOUT_5 BR51 1P8V_GPIO5
GPIO_POSTOUT_4 BT51 1P8V_GPIO4
GPIO_POSTOUT_3 BR54 1P8V_GPIO3
GPIO_POSTOUT_2 BP53 1P8V_GPIO2
GPIO_POSTOUT_1 BT54 1P8V_GPIO1
GPIO_POSTOUT_0 BT53 1P8V_GPIO0
C144, C155 VALUE SELECTED TO CENTER XTAL FREQ
V_SOC1P8
75 IN DBREQ_L BF2 DBREQ_N
75 OUT DBRDY BE1 DBRDY
75 IN TRST_L BJ2 TRST_N
75 36 IN TMS BH2 TMS
75 36 IN TDO BF3 TDO
75 36 OUT TDI BG2 TDI
75 36 IN TCK BG1 TCK
R392 10 KOHM 5% 2 EMPTY 402
DB16
R391 10 KOHM 1% 2 CH 402
75 36 IN SOC_RST_N BJ1 RESET_N
54 36 75 IN SOC_PWR_OK BL2 PWROK
FT414 FTP 1

TMON_CAL1 BH4 TMON_CAL1 1 FTP FT75
TMON_CAL0 BH3 TMON_CAL0 1 FTP FT78
PLLTEST0 BP4 PLLTEST0
PLLTEST1 BN4 PLLTEST1
OUT 75
OUT 75
THERMDA BE4 THERMDA
THERMDC BE5 THERMDC
R524 1 KOHM 1% CH 402 DEBUG
R525 1 KOHM 1% CH 402 DEBUG
ANALOGOUT BE50 ANALOGOUT
DIECRACKMON BE49 DIECRACKMON
BP_0 BG47 BP_0
BP_1 BG48 BP_1
BP_2 BH49 BP_2
BP_3 BH50 BP_3
SVT IN 54
SVD BI 54
SVC OUT 54
ANATSTIN_P BG50 ANATSTIN_P
ANATSTIN_N BG51 ANATSTIN_N
R626 0 OHM 5% 2 CH 402
R627 0 OHM 5% 2 CH 402
R628 0 OHM 5% 2 CH 402
ANATSTOUT_P BE52 ANATSTOUT_P
ANATSTOUT_N BE53 ANATSTOUT_N
ALERT_N BK2 SP_SMC_INT_N OUT 36
SVT BM48 SVT_R_SOC
SVD BR48 SVD_R_SOC
SVC BN48 SVC_R_SOC
SID BM51 SID
SIC BN51 SIC
RTCCLK BJ48 RTCCLK

V_1P8STBY
R430 2 1 1% 243 OHM CH 402
SB_SOC_DATA
DEBUG R405 475 OHM 1% 2 EMPTY 402
BI 36
C553 47 PF 5% 50 V 2 EMPTY 402
V_1P8STBY
DEBUG R406 475 OHM 1% 2 EMPTY 402
R431 2 1 1% 243 OHM CH 402
SB_SOC_CLK
IN 36
C554 47 PF 5% 50 V 2 EMPTY 402

VDD_MEM_SENSE K30 SOC_MEMIO_S BI 14 58
VDD_MEMP_SENSE N30 SOC_MEMPHY_S BI 14 58
VSS_MEM_SENSE L30 SOC_MEMIO_S_RTN BI 14 58
VSS_GFX_SENSE BV46 SOC_GFXCORE_S_RTN OUT 14 54
VDD_GFX_SENSE BW46 SOC_GFXCORE_S 1 DB127 OUT 14 54
VDD_NB_SENSE BV7 SOC_NBCORE_S BI 61
VSS_NB_SENSE BW7 SOC_NBCORE_S_RTN 61
VDD_CORE_SENSE BW47 SOC_CPUCORE_S OUT 14 54
VSS_CORE_SENSE BV47 SOC_CPUCORE_S_RTN BI 14 54
VDD_095_SENSE BP58 SOC_SOCPHY_S OUT 64
VSS_095_SENSE BR57

VDD_CORE_PROBE BP47 VDD_CORE_PROBE
VSS_CORE_PROBE BN47 VSS_CORE_PROBE
VDD_GFX_PROBE BK47 VDD_GFX_PROBE
VSS_GFX_PROBE BK48 VSS_GFX_PROBE
VDD_MEMP_PROBE B30 VDD_MEMP_PROBE
VSS_MEMP_PROBE A30 VSS_MEMP_PROBE
54 OUT
54 OUT

X950118-001 BGA2409

NOTE: SOC_MEMIO_S_RTN IS ACTUALLY SENSE RETURN FROM SOC FOR MEMPHY. NOT CHANGING NETNAME TO STAY CONSISTENT WITH SCB

58 14 SOC_MEMIO_S C1220 1 2 SOC_MEMIO_S_RTN 14 58 1 UF 10% 6.3 V EMPTY 402
58 14 SOC_MEMPHY_S C1221 1 2 SOC_MEMIO_S_RTN 14 58 1 UF 10% 6.3 V EMPTY 402
54 14 SOC_GFXCORE_S C1222 1 2 SOC_GFXCORE_S_RTN 14 54 1 UF 10% 6.3 V EMPTY 402
54 14 SOC_CPUCORE_S C1223 1 2 SOC_CPUCORE_S_RTN 14 54 1 UF 10% 6.3 V EMPTY 402

NOTE: SINCE C1222 AND C1223 ARE NEAR SOC, CHANGE THEM TO X6S/X7R IF EVER STUFFED DUE TO HIGHER TEMPERATURES IN AREA

C1220, C1221, C1222, AND C1223 ARE THE QUALIFICATION/MEASUREMENT TEST POINTS FOR THE RELEVANT SOC VR, AND MUST BE MEASURED WITH A DIFFERENTIAL PROBE

DB58 1 V_CPUCORE
DB63 1 V_GFXCORE
DB59 1
SENSE CONNECTION FOR VRTB

V_3P3STBY
U77 SOT-523 EMPTY
DB93 1 LARGE-TP RTC_VREG_IN
DB92 1 LARGE-TP
3.0V BATTERY INPUT
RTC_VREG_IN_D
C512 0.1 UF 10% 16 V EMPTY 402
Q1 EMPTY NCP663 VIN VOUT NC GND X899201-001 SC82
RTC_VREG_OUT
C513 0.1 UF 10% 16 V EMPTY 402
R77 1 2 0 OHM 5% 402 EMPTY
V_1P8STBY
R387 0 OHM 5% 2 CH 402
V_BAT
R388 221 KOHM 1% EMPTY 402
STUFF R388 IF SOC IS NOT STUFFED

V_BAT
NOM.VOLTAGE: 1.8V

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 14/82 | 14/82 | G-R | 0.991 |

# SOC: DECOUPLING

V_GFXCORE

55 0805 22UF

C847 1 2 22 UF 20% 6.3 V X6S 805
C771 1 2 22 UF 20% 6.3 V X6S 805
C843 1 2 22 UF 20% 6.3 V X6S 805
C735 1 2 22 UF 20% 6.3 V X6S 805
C305 1 2 22 UF 20% 6.3 V X6S 805
C772 1 2 22 UF 20% 6.3 V X6S 805
C767 1 2 22 UF 20% 6.3 V X6S 805
C692 1 2 22 UF 20% 6.3 V X6S 805
C736 1 2 22 UF 20% 6.3 V X6S 805
C279 1 2 22 UF 20% 6.3 V X6S 805
C805 1 2 22 UF 20% 6.3 V X6S 805
C804 1 2 22 UF 20% 6.3 V X6S 805
C801 1 2 22 UF 20% 6.3 V X6S 805
C846 1 2 22 UF 20% 6.3 V X6S 805
C278 1 2 22 UF 20% 6.3 V X6S 805
C693 1 2 22 UF 20% 6.3 V X6S 805
C737 1 2 22 UF 20% 6.3 V X6S 805
C845 1 2 22 UF 20% 6.3 V X6S 805
C800 1 2 22 UF 20% 6.3 V X6S 805
C277 1 2 22 UF 20% 6.3 V X6S 805
C694 1 2 22 UF 20% 6.3 V X6S 805
C802 1 2 22 UF 20% 6.3 V X6S 805
C770 1 2 22 UF 20% 6.3 V X6S 805
C768 1 2 22 UF 20% 6.3 V X6S 805
C257 1 2 22 UF 20% 6.3 V X6S 805
C293 1 2 22 UF 20% 6.3 V X6S 805
C733 1 2 22 UF 20% 6.3 V X6S 805
C803 1 2 22 UF 20% 6.3 V X6S 805
C769 1 2 22 UF 20% 6.3 V X6S 805
C690 1 2 22 UF 20% 6.3 V X6S 805
C294 1 2 22 UF 20% 6.3 V X6S 805
C734 1 2 22 UF 20% 6.3 V X6S 805
C842 1 2 22 UF 20% 6.3 V X6S 805
C844 1 2 22 UF 20% 6.3 V X6S 805
C295 1 2 22 UF 20% 6.3 V X6S 805
C280 1 2 22 UF 20% 6.3 V X6S 805
C695 1 2 22 UF 20% 6.3 V X6S 805
C691 1 2 22 UF 20% 6.3 V X6S 805
C296 1 2 22 UF 20% 6.3 V X6S 805
C281 1 2 22 UF 20% 6.3 V X6S 805
C309 1 2 22 UF 20% 6.3 V X6S 805
C732 1 2 22 UF 20% 6.3 V X6S 805
C297 1 2 22 UF 20% 6.3 V X6S 805
C264 1 2 22 UF 20% 6.3 V X6S 805
C308 1 2 22 UF 20% 6.3 V X6S 805
C262 1 2 22 UF 20% 6.3 V X6S 805
C261 1 2 22 UF 20% 6.3 V X6S 805
C254 1 2 22 UF 20% 6.3 V X6S 805
C263 1 2 22 UF 20% 6.3 V X6S 805
C260 1 2 22 UF 20% 6.3 V X6S 805
C255 1 2 22 UF 20% 6.3 V X6S 805
C253 1 2 22 UF 20% 6.3 V X6S 805
C256 1 2 22 UF 20% 6.3 V X6S 805
C307 1 2 22 UF 20% 6.3 V X6S 805
C306 1 2 22 UF 20% 6.3 V X6S 805

V_GFXCORE

55 0805 22UF

C723 1 2 22 UF 20% 6.3 V X6S 805
C703 1 2 22 UF 20% 6.3 V X6S 805
C700 1 2 22 UF 20% 6.3 V X6S 805
C722 1 2 22 UF 20% 6.3 V X6S 805
C817 1 2 22 UF 20% 6.3 V X6S 805
C727 1 2 22 UF 20% 6.3 V X6S 805
C806 1 2 22 UF 20% 6.3 V X6S 805
C697 1 2 22 UF 20% 6.3 V X6S 805
C827 1 2 22 UF 20% 6.3 V X6S 805
C834 1 2 22 UF 20% 6.3 V X6S 805
C728 1 2 22 UF 20% 6.3 V X6S 805
C698 1 2 22 UF 20% 6.3 V X6S 805
C835 1 2 22 UF 20% 6.3 V X6S 805
C701 1 2 22 UF 20% 6.3 V X6S 805
C752 1 2 22 UF 20% 6.3 V X6S 805
C838 1 2 22 UF 20% 6.3 V X6S 805
C704 1 2 22 UF 20% 6.3 V X6S 805
C758 1 2 22 UF 20% 6.3 V X6S 805
C750 1 2 22 UF 20% 6.3 V X6S 805
C837 1 2 22 UF 20% 6.3 V X6S 805
C726 1 2 22 UF 20% 6.3 V X6S 805
C751 1 2 22 UF 20% 6.3 V X6S 805
C724 1 2 22 UF 20% 6.3 V X6S 805
C808 1 2 22 UF 20% 6.3 V X6S 805
C702 1 2 22 UF 20% 6.3 V X6S 805
C836 1 2 22 UF 20% 6.3 V X6S 805
C725 1 2 22 UF 20% 6.3 V X6S 805
C782 1 2 22 UF 20% 6.3 V X6S 805
C811 1 2 22 UF 20% 6.3 V X6S 805
C749 1 2 22 UF 20% 6.3 V X6S 805
C696 1 2 22 UF 20% 6.3 V X6S 805
C780 1 2 22 UF 20% 6.3 V X6S 805
C807 1 2 22 UF 20% 6.3 V X6S 805
C699 1 2 22 UF 20% 6.3 V X6S 805
C781 1 2 22 UF 20% 6.3 V X6S 805

2 0402 4.7UF

V_GFXCORE

C746 1 2 4.7 UF 20% 6.3 V X6S 402
C747 1 2 4.7 UF 20% 6.3 V X6S 402

V_GFXCORE

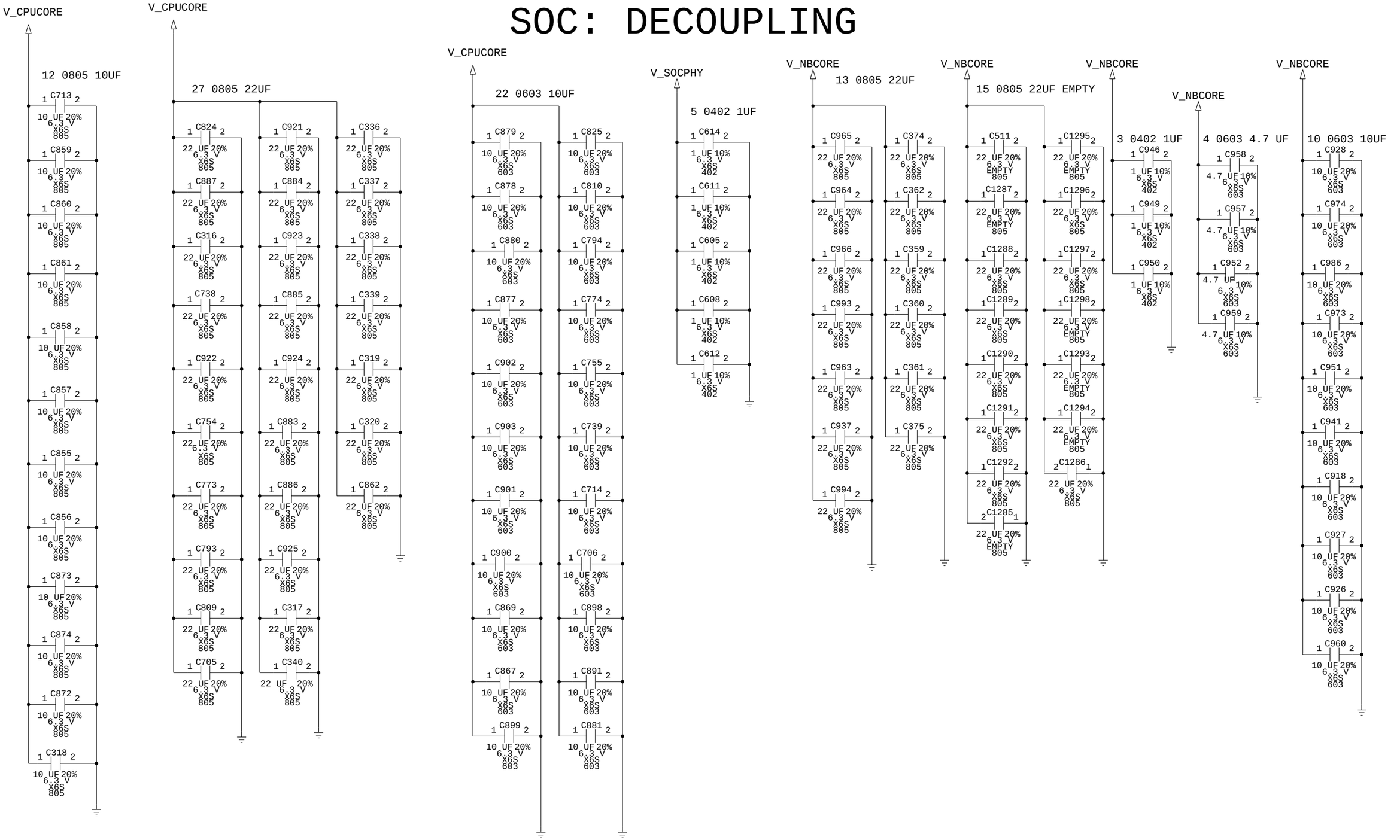
57 0603 10UF

C742 1 2 10 UF 20% 6.3 V X6S 603
C830 1 2 10 UF 20% 6.3 V X6S 603
C687 1 2 10 UF 20% 6.3 V X6S 603
C785 1 2 10 UF 20% 6.3 V X6S 603
C761 1 2 10 UF 20% 6.3 V X6S 603
C850 1 2 10 UF 20% 6.3 V X6S 603
C718 1 2 10 UF 20% 6.3 V X6S 603
C716 1 2 10 UF 20% 6.3 V X6S 603
C789 1 2 10 UF 20% 6.3 V X6S 603
C851 1 2 10 UF 20% 6.3 V X6S 603
C744 1 2 10 UF 20% 6.3 V X6S 603
C670 1 2 10 UF 20% 6.3 V X6S 603
C678 1 2 10 UF 20% 6.3 V X6S 603
C831 1 2 10 UF 20% 6.3 V X6S 603
C763 1 2 10 UF 20% 6.3 V X6S 603
C685 1 2 10 UF 20% 6.3 V X6S 603
C684 1 2 10 UF 20% 6.3 V X6S 603
C814 1 2 10 UF 20% 6.3 V X6S 603
C791 1 2 10 UF 20% 6.3 V X6S 603
C854 1 2 10 UF 20% 6.3 V X6S 603
C712 1 2 10 UF 20% 6.3 V X6S 603
C852 1 2 10 UF 20% 6.3 V X6S 603
C815 1 2 10 UF 20% 6.3 V X6S 603
C841 1 2 10 UF 20% 6.3 V X6S 603
C730 1 2 10 UF 20% 6.3 V X6S 603
C671 1 2 10 UF 20% 6.3 V X6S 603
C719 1 2 10 UF 20% 6.3 V X6S 603
C818 1 2 10 UF 20% 6.3 V X6S 603
C741 1 2 10 UF 20% 6.3 V X6S 603
C686 1 2 10 UF 20% 6.3 V X6S 603
C688 1 2 10 UF 20% 6.3 V X6S 603
C826 1 2 10 UF 20% 6.3 V X6S 603
C759 1 2 10 UF 20% 6.3 V X6S 603
C717 1 2 10 UF 20% 6.3 V X6S 603
C745 1 2 10 UF 20% 6.3 V X6S 603
C812 1 2 10 UF 20% 6.3 V X6S 603
C766 1 2 10 UF 20% 6.3 V X6S 603
C743 1 2 10 UF 20% 6.3 V X6S 603
C764 1 2 10 UF 20% 6.3 V X6S 603
C721 1 2 10 UF 20% 6.3 V X6S 603
C784 1 2 10 UF 20% 6.3 V X6S 603
C762 1 2 10 UF 20% 6.3 V X6S 603
C792 1 2 10 UF 20% 6.3 V X6S 603
C720 1 2 10 UF 20% 6.3 V X6S 603
C798 1 2 10 UF 20% 6.3 V X6S 603
C790 1 2 10 UF 20% 6.3 V X6S 603
C816 1 2 10 UF 20% 6.3 V X6S 603
C753 1 2 10 UF 20% 6.3 V X6S 603
C832 1 2 10 UF 20% 6.3 V X6S 603
C673 1 2 10 UF 20% 6.3 V X6S 603
C833 1 2 10 UF 20% 6.3 V X6S 603
C783 1 2 10 UF 20% 6.3 V X6S 603
C813 1 2 10 UF 20% 6.3 V X6S 603
C672 1 2 10 UF 20% 6.3 V X6S 603
C853 1 2 10 UF 20% 6.3 V X6S 603
C828 1 2 10 UF 20% 6.3 V X6S 603

NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

| | | | | | |
|---|---|---|---|---|---|
| MICROSOFT CONFIDENTIAL | PROJECT NAME Cactus | PAGE 15/82 | CSA PAGE 15/82 | FAB G-R | VER 0.991 |

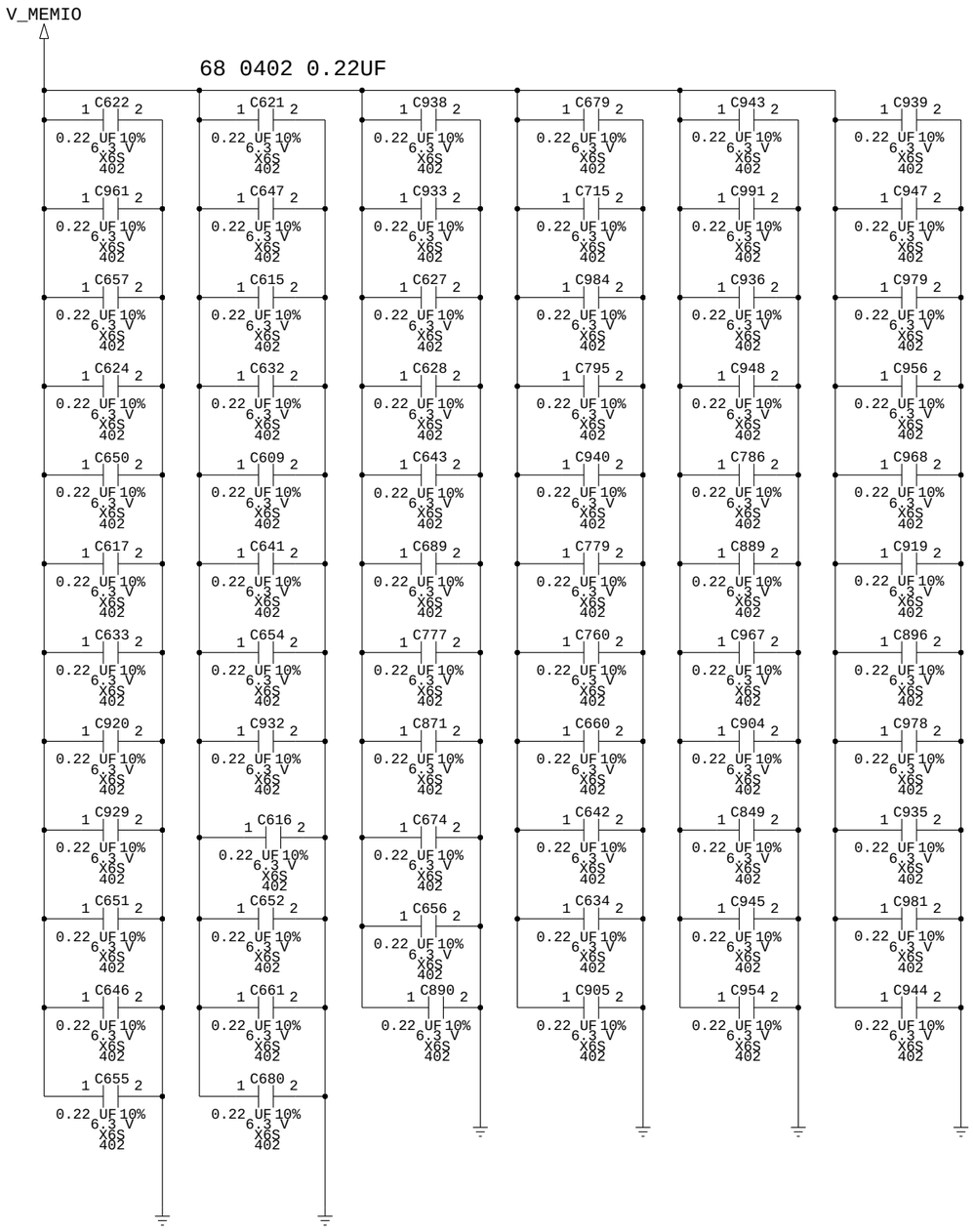
# SOC: DECOUPLING

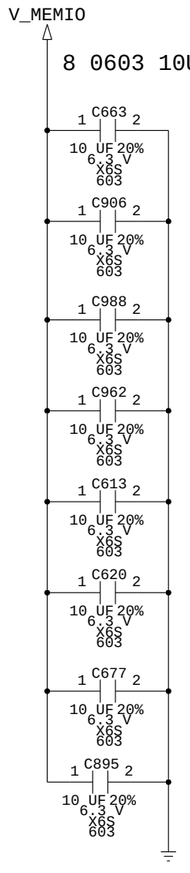


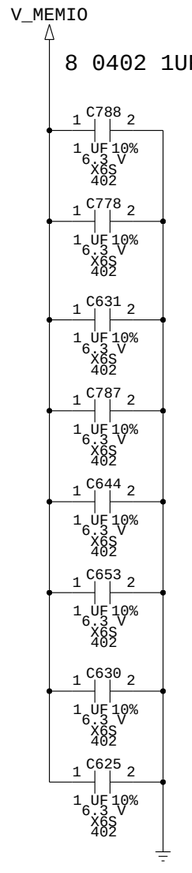

NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

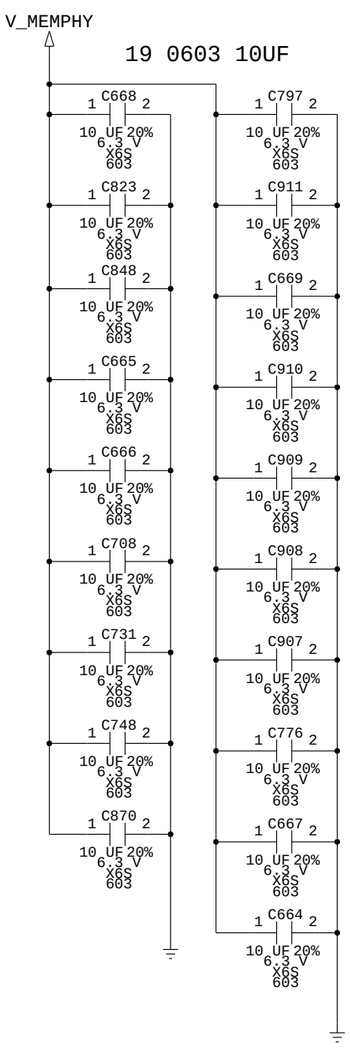
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 16/82 | 16/82 | G-R | 0.991 |

# SOC: DECOUPLING



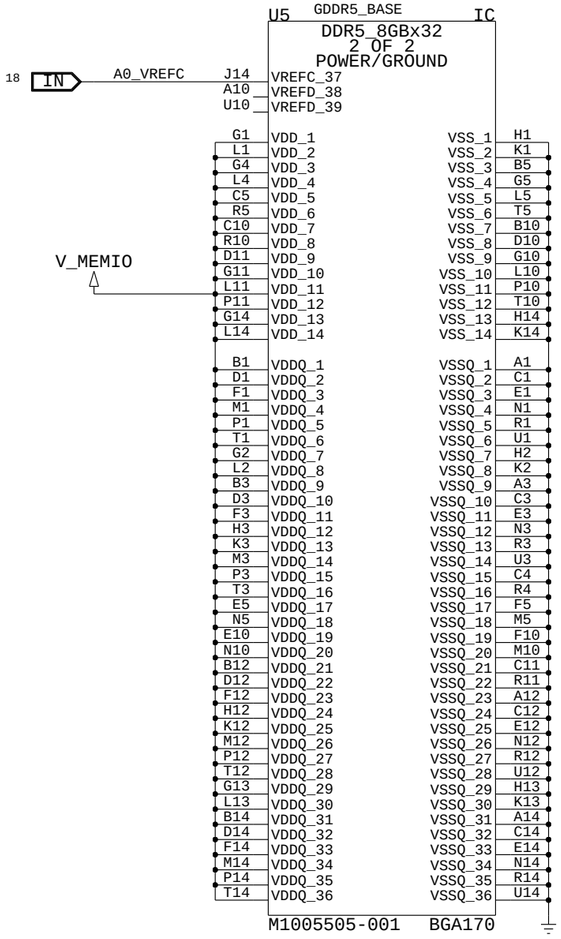



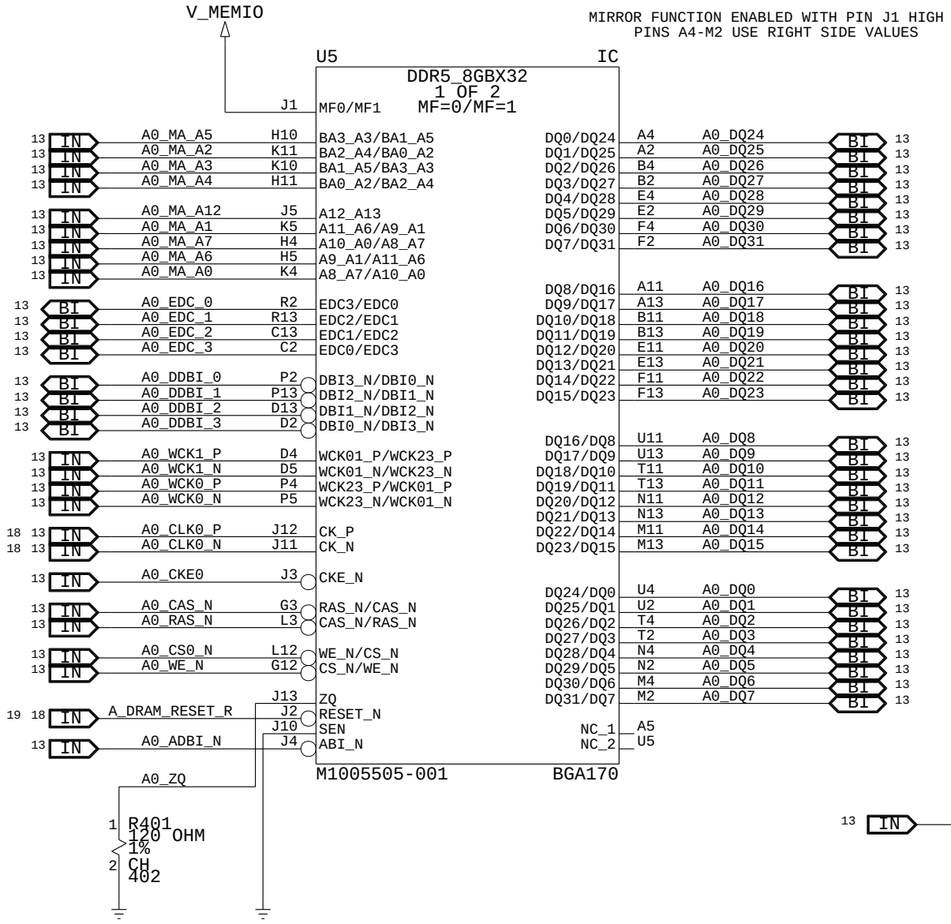



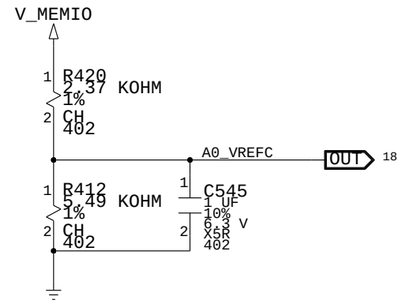

NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

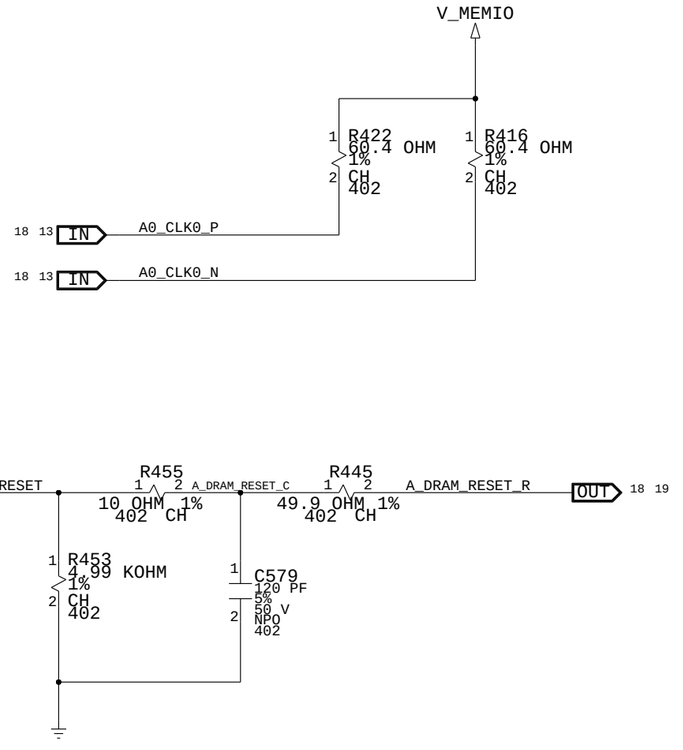
| | | | | | |
|---|---|---|---|---|---|
| MICROSOFT CONFIDENTIAL | PROJECT NAME Cactus | PAGE 17/82 | CSA PAGE 17/82 | FAB G-R | VER 0.99 |

# MEMORY: CHANNEL A0




| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| M1005505-001 | IC | U4,U5,U6,U7,U8,U9,U10,U11,U12,U13,U14,U15 | MEM,SM,8GB,16MX32,GDDR5,170FBGA | GDDR5_SAMSUNG |
| M1005717-001 | IC | U4,U5,U6,U7,U8,U9,U10,U11,U12,U13,U14,U15 | MEM,SM,8GB,16MX32,GDDR5,170FBGA | GDDR5_HYNIX |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 18/82 | 18/82 | G-R | 0.991 |

# MEMORY: CHANNEL A1

GDDR5_BASE

U4 IC

DDR5_8GBx32
2 OF 2
POWER/GROUND

| | | | |
|---|---|---|---|
| 19 IN A1_VREFC | J14 | VREFC_37 | |
| | A10 | VREFD_38 | |
| | U10 | VREFD_39 | |
| | G1 | VDD_1 | |
| | L1 | VDD_2 | |
| | G4 | VDD_3 | |
| | L4 | VDD_4 | |
| | C5 | VDD_5 | |
| | R5 | VDD_6 | |
| | C10 | VDD_7 | |
| | R10 | VDD_8 | |
| V_MEMIO | D11 | VDD_9 | |
| | G11 | VDD_10 | |
| | L11 | VDD_11 | |
| | P11 | VDD_12 | |
| | G14 | VDD_13 | |
| | L14 | VDD_14 | |
| | B1 | VDDQ_1 | |
| | D1 | VDDQ_2 | |
| | F1 | VDDQ_3 | |
| | M1 | VDDQ_4 | |
| | P1 | VDDQ_5 | |
| | T1 | VDDQ_6 | |
| | G2 | VDDQ_7 | |
| | L2 | VDDQ_8 | |
| | B3 | VDDQ_9 | |
| | D3 | VDDQ_10 | |
| | F3 | VDDQ_11 | |
| | H3 | VDDQ_12 | |
| | K3 | VDDQ_13 | |
| | M3 | VDDQ_14 | |
| | P3 | VDDQ_15 | |
| | T3 | VDDQ_16 | |
| | E5 | VDDQ_17 | |
| | N5 | VDDQ_18 | |
| | E10 | VDDQ_19 | |
| | N10 | VDDQ_20 | |
| | B12 | VDDQ_21 | |
| | D12 | VDDQ_22 | |
| | F12 | VDDQ_23 | |
| | H12 | VDDQ_24 | |
| | K12 | VDDQ_25 | |
| | M12 | VDDQ_26 | |
| | P12 | VDDQ_27 | |
| | T12 | VDDQ_28 | |
| | G13 | VDDQ_29 | |
| | L13 | VDDQ_30 | |
| | B14 | VDDQ_31 | |
| | D14 | VDDQ_32 | |
| | F14 | VDDQ_33 | |
| | M14 | VDDQ_34 | |
| | P14 | VDDQ_35 | |
| | T14 | VDDQ_36 | |
| | H1 | VSS_1 | |
| | K1 | VSS_2 | |
| | B5 | VSS_3 | |
| | G5 | VSS_4 | |
| | L5 | VSS_5 | |
| | T5 | VSS_6 | |
| | B10 | VSS_7 | |
| | D10 | VSS_8 | |
| | G10 | VSS_9 | |
| | L10 | VSS_10 | |
| | P10 | VSS_11 | |
| | T10 | VSS_12 | |
| | H14 | VSS_13 | |
| | K14 | VSS_14 | |
| | A1 | VSSQ_1 | |
| | C1 | VSSQ_2 | |
| | E1 | VSSQ_3 | |
| | N1 | VSSQ_4 | |
| | R1 | VSSQ_5 | |
| | U1 | VSSQ_6 | |
| | H2 | VSSQ_7 | |
| | K2 | VSSQ_8 | |
| | A3 | VSSQ_9 | |
| | C3 | VSSQ_10 | |
| | E3 | VSSQ_11 | |
| | N3 | VSSQ_12 | |
| | R3 | VSSQ_13 | |
| | U3 | VSSQ_14 | |
| | C4 | VSSQ_15 | |
| | R4 | VSSQ_16 | |
| | F5 | VSSQ_17 | |
| | M5 | VSSQ_18 | |
| | F10 | VSSQ_19 | |
| | M10 | VSSQ_20 | |
| | C11 | VSSQ_21 | |
| | R11 | VSSQ_22 | |
| | A12 | VSSQ_23 | |
| | C12 | VSSQ_24 | |
| | E12 | VSSQ_25 | |
| | N12 | VSSQ_26 | |
| | R12 | VSSQ_27 | |
| | U12 | VSSQ_28 | |
| | H13 | VSSQ_29 | |
| | K13 | VSSQ_30 | |
| | A14 | VSSQ_31 | |
| | C14 | VSSQ_32 | |
| | E14 | VSSQ_33 | |
| | N14 | VSSQ_34 | |
| | R14 | VSSQ_35 | |
| | U14 | VSSQ_36 | |

M1005505-001 BGA170

MIRROR FUNCTION DISABLED WITH PIN J1 LOW
PINS A4-M2 USE LEFT SIDE VALUES

U4 IC

DDR5_8GBX32
1 OF 2
MF=0/MF=1

| | | | |
|---|---|---|---|
| | J1 | MF0/MF1 | |
| 13 IN A1_MA_A3 | H10 | BA3_A3/BA1_A5 | |
| 13 IN A1_MA_A4 | K11 | BA2_A4/BA0_A2 | |
| 13 IN A1_MA_A5 | K10 | BA1_A5/BA3_A3 | |
| 13 IN A1_MA_A2 | H11 | BA0_A2/BA2_A4 | |
| 13 IN A1_MA_A12 | J5 | A12_A13 | |
| 13 IN A1_MA_A6 | K5 | A11_A6/A9_A1 | |
| 13 IN A1_MA_A0 | H4 | A10_A0/A8_A7 | |
| 13 IN A1_MA_A1 | H5 | A9_A1/A11_A6 | |
| 13 IN A1_MA_A7 | K4 | A8_A7/A10_A0 | |
| 13 BI A1_EDC_3 | R2 | EDC3/EDC0 | |
| 13 BI A1_EDC_2 | R13 | EDC2/EDC1 | |
| 13 BI A1_EDC_1 | C13 | EDC1/EDC2 | |
| 13 BI A1_EDC_0 | C2 | EDC0/EDC3 | |
| 13 BI A1_DDBI_3 | P2 | DBI3_N/DBI0_N | |
| 13 BI A1_DDBI_2 | P13 | DBI2_N/DBI1_N | |
| 13 BI A1_DDBI_1 | D13 | DBI1_N/DBI2_N | |
| 13 BI A1_DDBI_0 | D2 | DBI0_N/DBI3_N | |
| 13 IN A1_WCK0_P | D4 | WCK01_P/WCK23_P | |
| 13 IN A1_WCK0_N | D5 | WCK01_N/WCK23_N | |
| 13 IN A1_WCK1_P | P4 | WCK23_P/WCK01_P | |
| 13 IN A1_WCK1_N | P5 | WCK23_N/WCK01_N | |
| 19 13 IN A1_CLK0_P | J12 | CK_P | |
| 19 13 IN A1_CLK0_N | J11 | CK_N | |
| 13 IN A1_CKE0 | J3 | CKE_N | |
| 13 IN A1_RAS_N | G3 | RAS_N/CAS_N | |
| 13 IN A1_CAS_N | L3 | CAS_N/RAS_N | |
| 13 IN A1_WE_N | L12 | WE_N/CS_N | |
| 13 IN A1_CS0_N | G12 | CS_N/WE_N | |
| | J13 | ZQ | |
| 18 IN A_DRAM_RESET_R | J2 | RESET_N | |
| | J10 | SEN | |
| 13 IN A1_ADBI_N | J4 | ABI_N | |

| | | | |
|---|---|---|---|
| DQ0/DQ24 | A4 | A1_DQ0 | BI 13 |
| DQ1/DQ25 | A2 | A1_DQ1 | BI 13 |
| DQ2/DQ26 | B4 | A1_DQ2 | BI 13 |
| DQ3/DQ27 | B2 | A1_DQ3 | BI 13 |
| DQ4/DQ28 | E4 | A1_DQ4 | BI 13 |
| DQ5/DQ29 | E2 | A1_DQ5 | BI 13 |
| DQ6/DQ30 | F4 | A1_DQ6 | BI 13 |
| DQ7/DQ31 | F2 | A1_DQ7 | BI 13 |
| DQ8/DQ16 | A11 | A1_DQ8 | BI 13 |
| DQ9/DQ17 | A13 | A1_DQ9 | BI 13 |
| DQ10/DQ18 | B11 | A1_DQ10 | BI 13 |
| DQ11/DQ19 | B13 | A1_DQ11 | BI 13 |
| DQ12/DQ20 | E11 | A1_DQ12 | BI 13 |
| DQ13/DQ21 | E13 | A1_DQ13 | BI 13 |
| DQ14/DQ22 | F11 | A1_DQ14 | BI 13 |
| DQ15/DQ23 | F13 | A1_DQ15 | BI 13 |
| DQ16/DQ8 | U11 | A1_DQ16 | BI 13 |
| DQ17/DQ9 | U13 | A1_DQ17 | BI 13 |
| DQ18/DQ10 | T11 | A1_DQ18 | BI 13 |
| DQ19/DQ11 | T13 | A1_DQ19 | BI 13 |
| DQ20/DQ12 | N11 | A1_DQ20 | BI 13 |
| DQ21/DQ13 | N13 | A1_DQ21 | BI 13 |
| DQ22/DQ14 | M11 | A1_DQ22 | BI 13 |
| DQ23/DQ15 | M13 | A1_DQ23 | BI 13 |
| DQ24/DQ0 | U4 | A1_DQ24 | BI 13 |
| DQ25/DQ1 | U2 | A1_DQ25 | BI 13 |
| DQ26/DQ2 | T4 | A1_DQ26 | BI 13 |
| DQ27/DQ3 | T2 | A1_DQ27 | BI 13 |
| DQ28/DQ4 | N4 | A1_DQ28 | BI 13 |
| DQ29/DQ5 | N2 | A1_DQ29 | BI 13 |
| DQ30/DQ6 | M4 | A1_DQ30 | BI 13 |
| DQ31/DQ7 | M2 | A1_DQ31 | BI 13 |
| NC_1 | A5 | | |
| NC_2 | U5 | | |

M1005505-001 BGA170

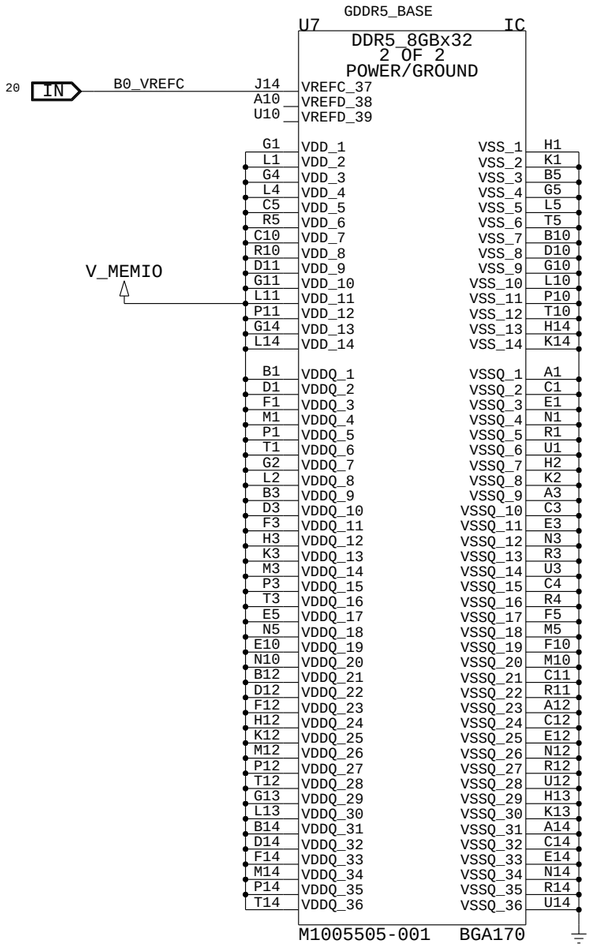
A1_ZQ
1 R82 120 OHM 1% 2 CH 402

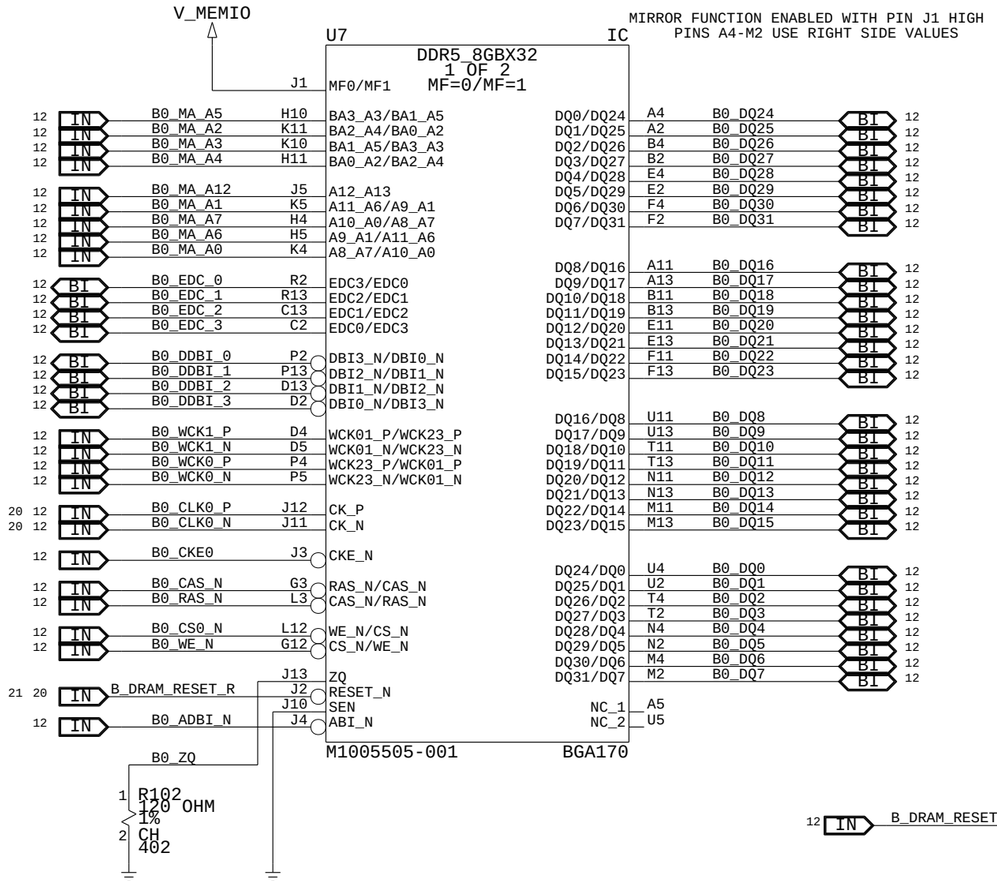
V_MEMIO
1 R93 2.37 KOHM 1% 2 CH 402
A1_VREFC OUT 19
R89 5.49 KOHM 1% CH 402
1 C129 1UF 10% 6.3 V X5R 2 402

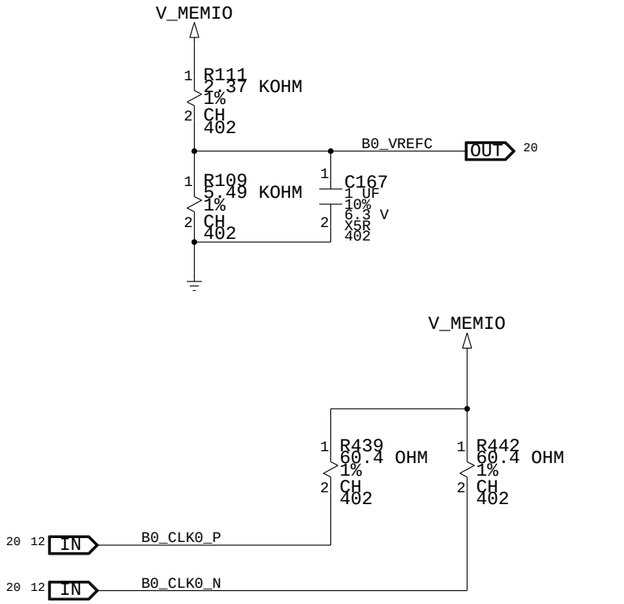
V_MEMIO
1 R425 60.4 OHM 1% 2 CH 402
1 R433 60.4 OHM 1% 2 CH 402
19 13 IN A1_CLK0_P
19 13 IN A1_CLK0_N

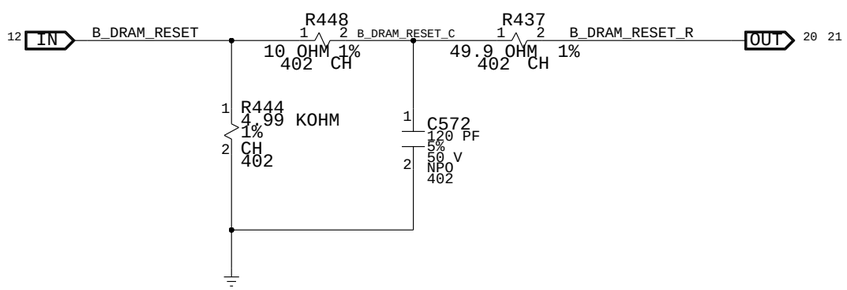
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 19/82 | 19/82 | G-R | 0.991 |

# MEMORY: CHANNEL B0



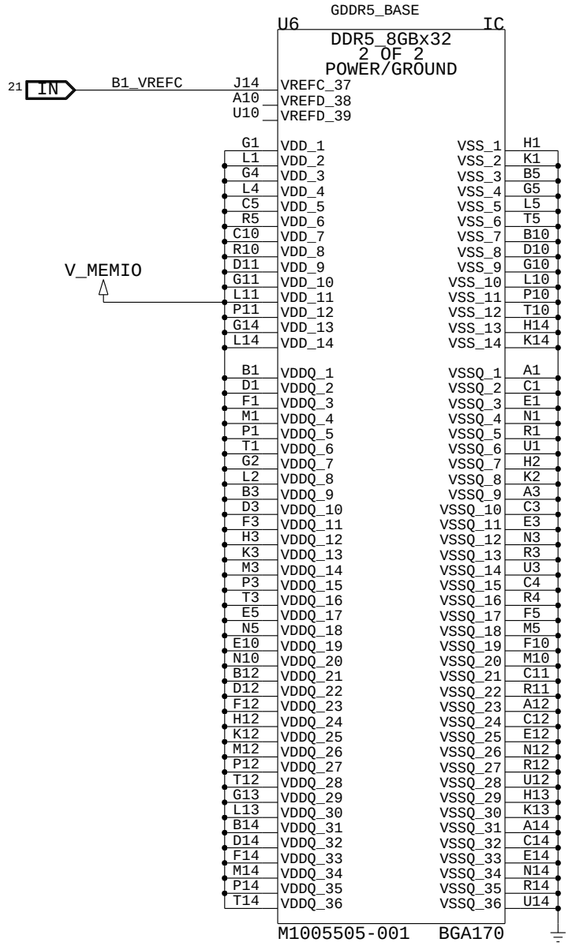



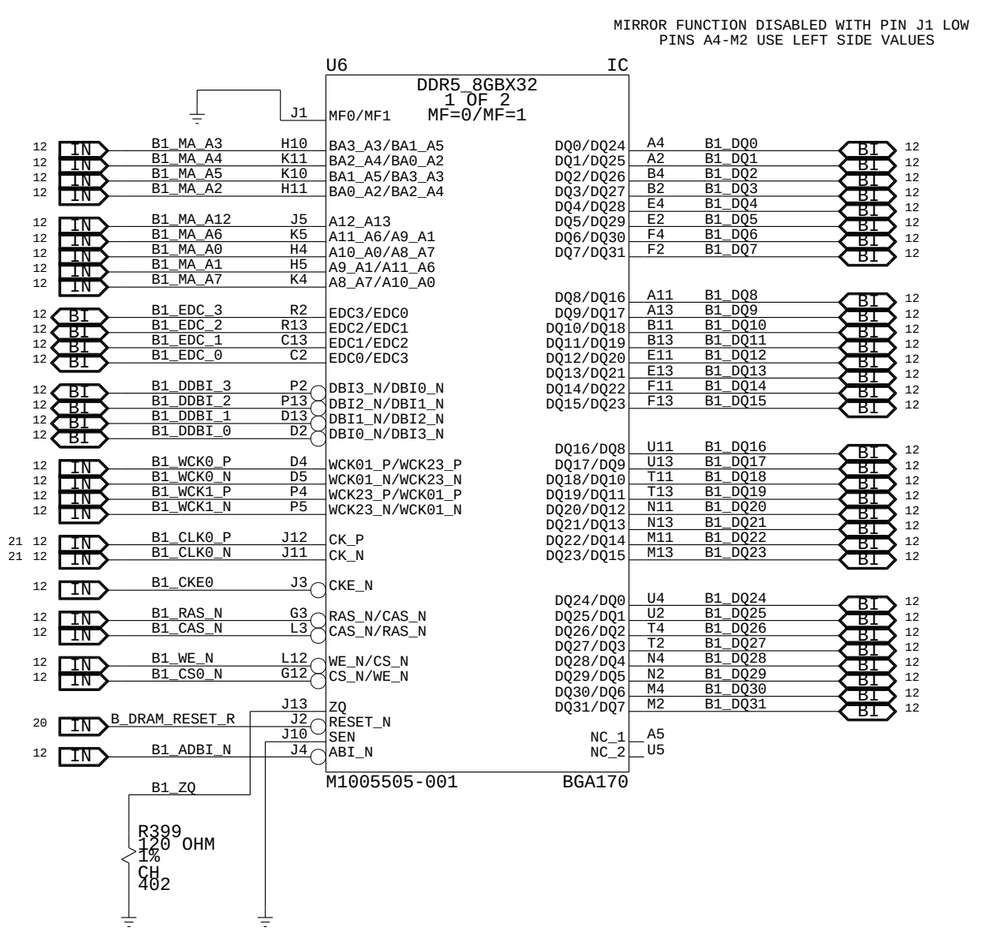



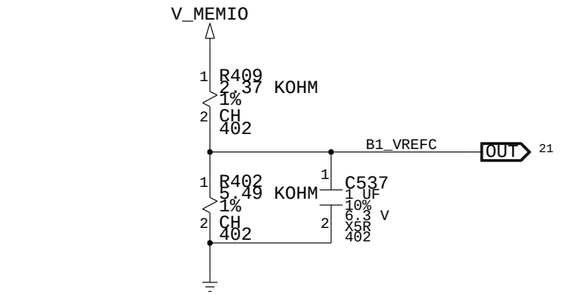



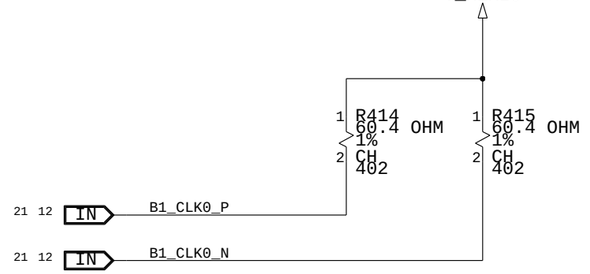

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 20/82 | 20/82 | G-R | 0.991 |

# MEMORY: CHANNEL B1



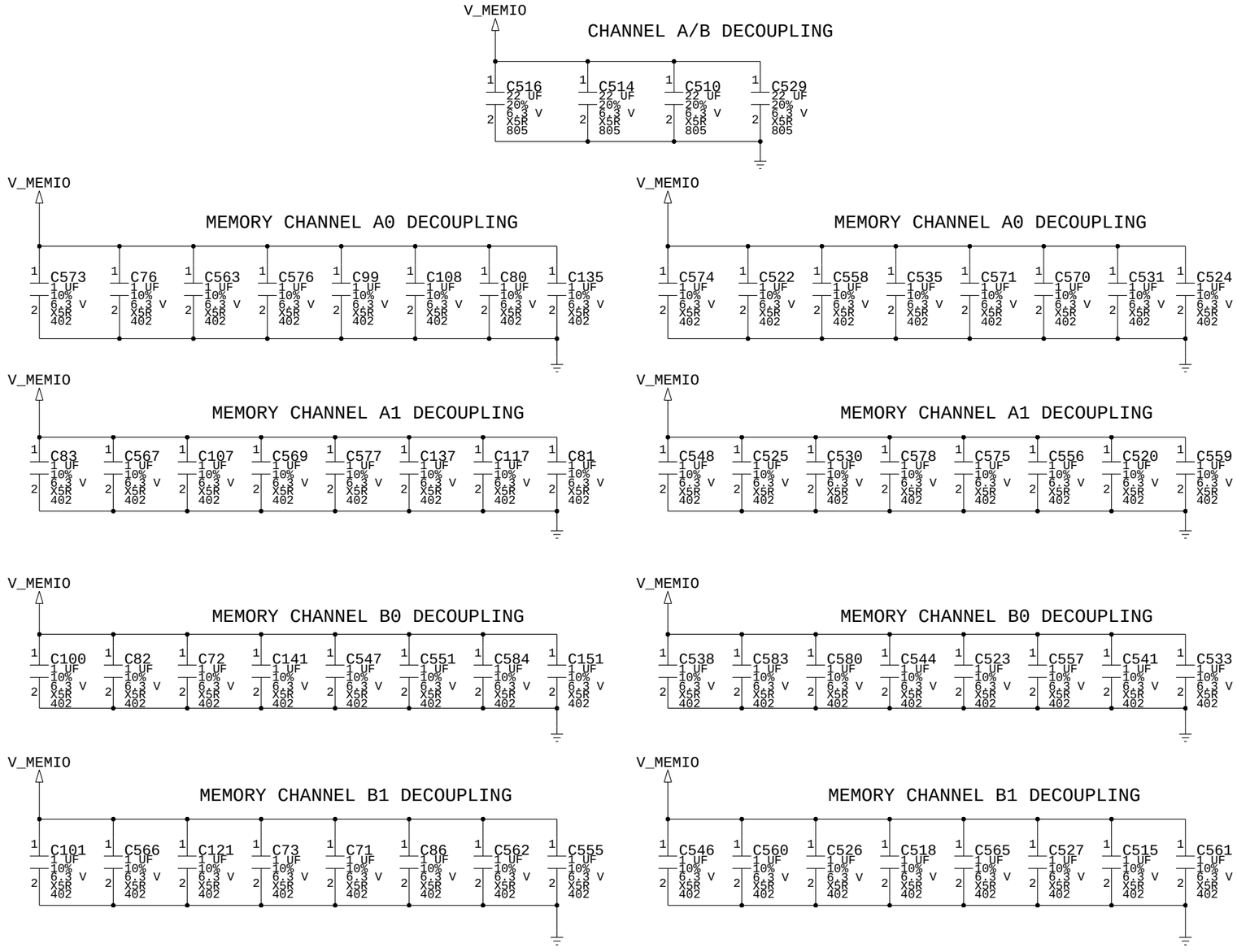




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 21/82 | 21/82 | G-R | 0.991 |

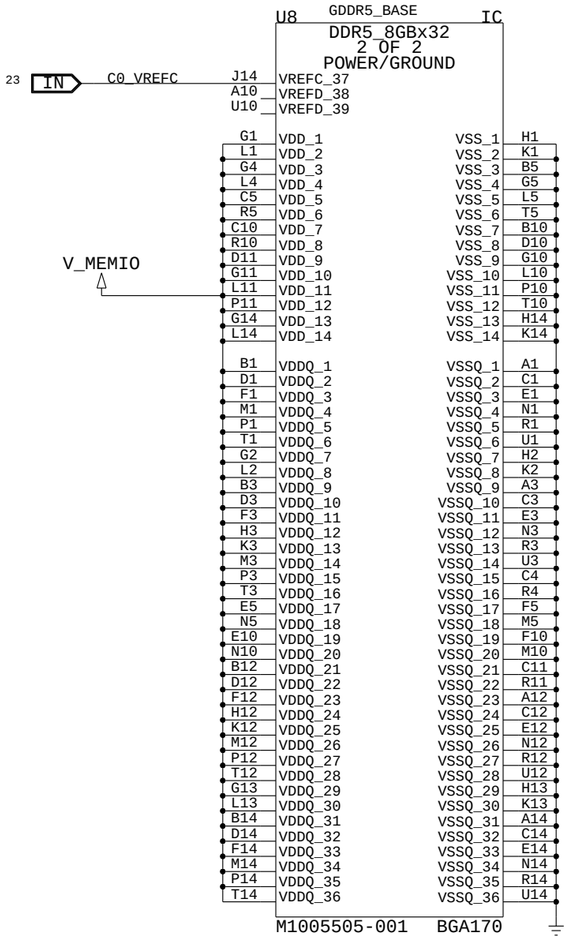
# MEMORY: CHANNEL A/B DECOUPLING



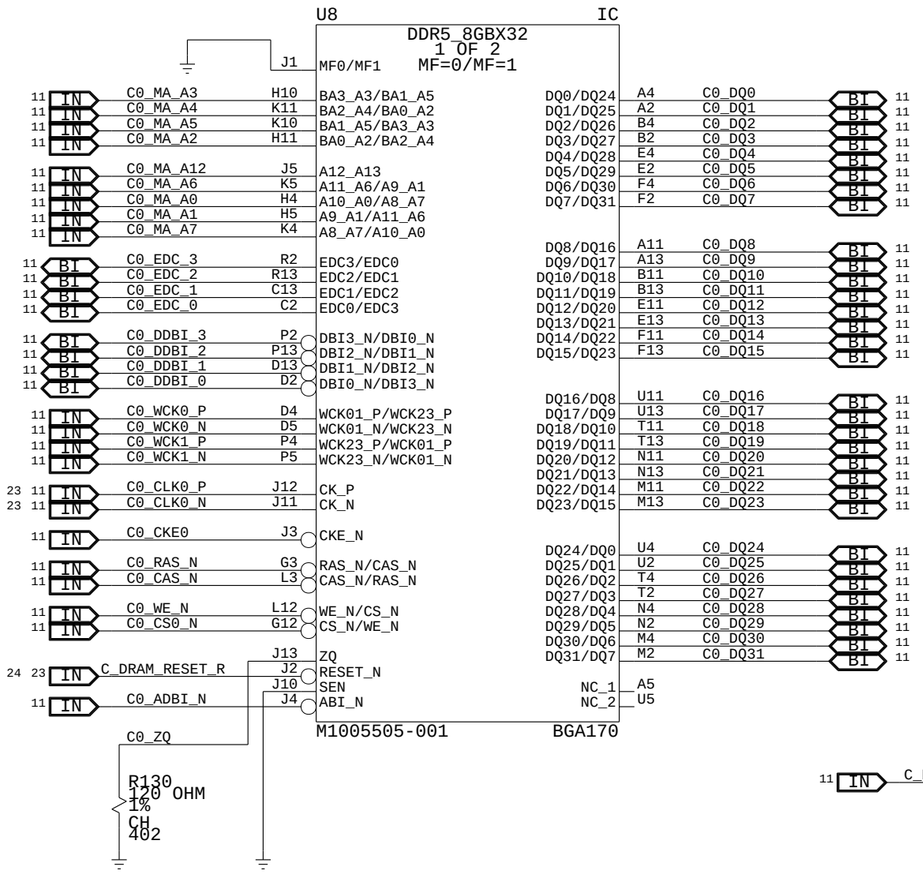

NOTE: ADDITIONAL MEMORY DECOUPLING ON PAGE 33

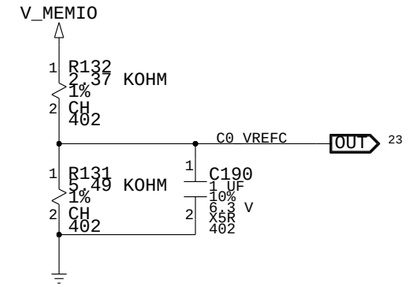
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 22/82 | 22/82 | G-R | 0.991 |

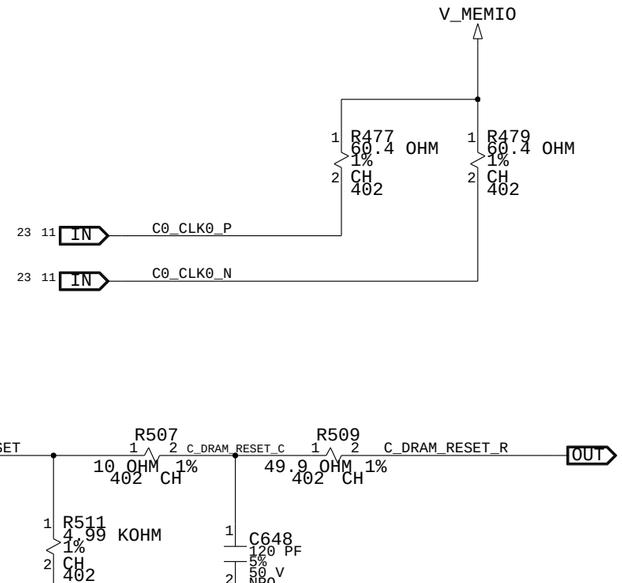
# MEMORY: CHANNEL C0

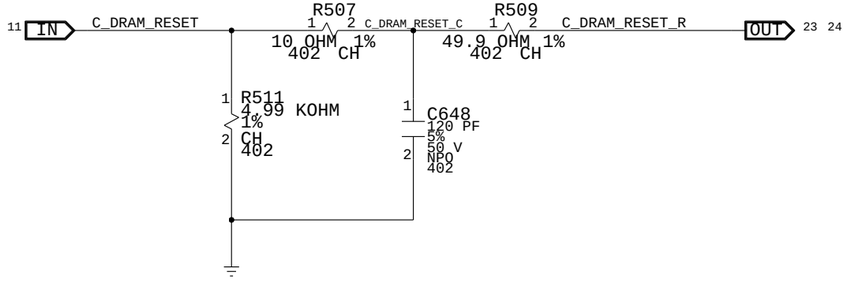



MIRROR FUNCTION DISABLED WITH PIN J1 LOW
PINS A4-M2 USE LEFT SIDE VALUES



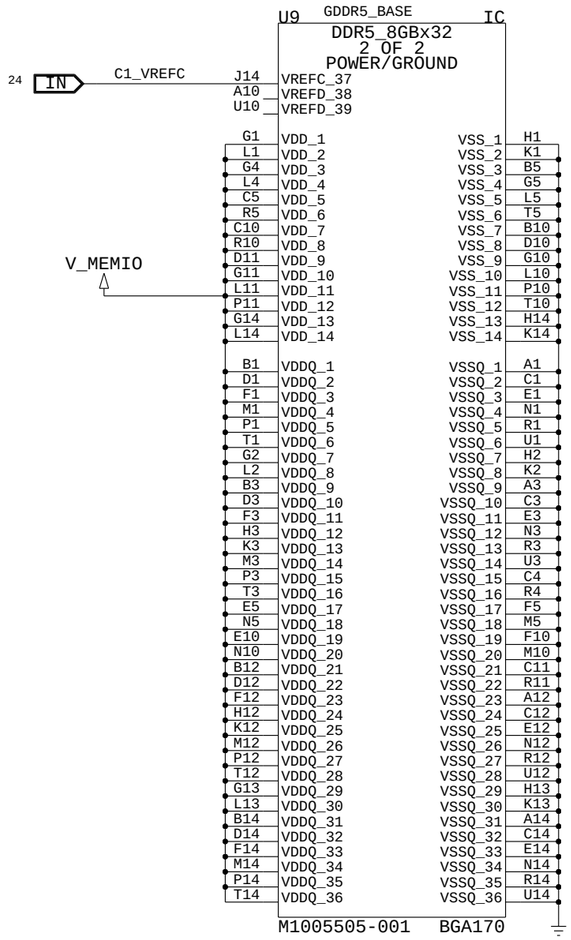



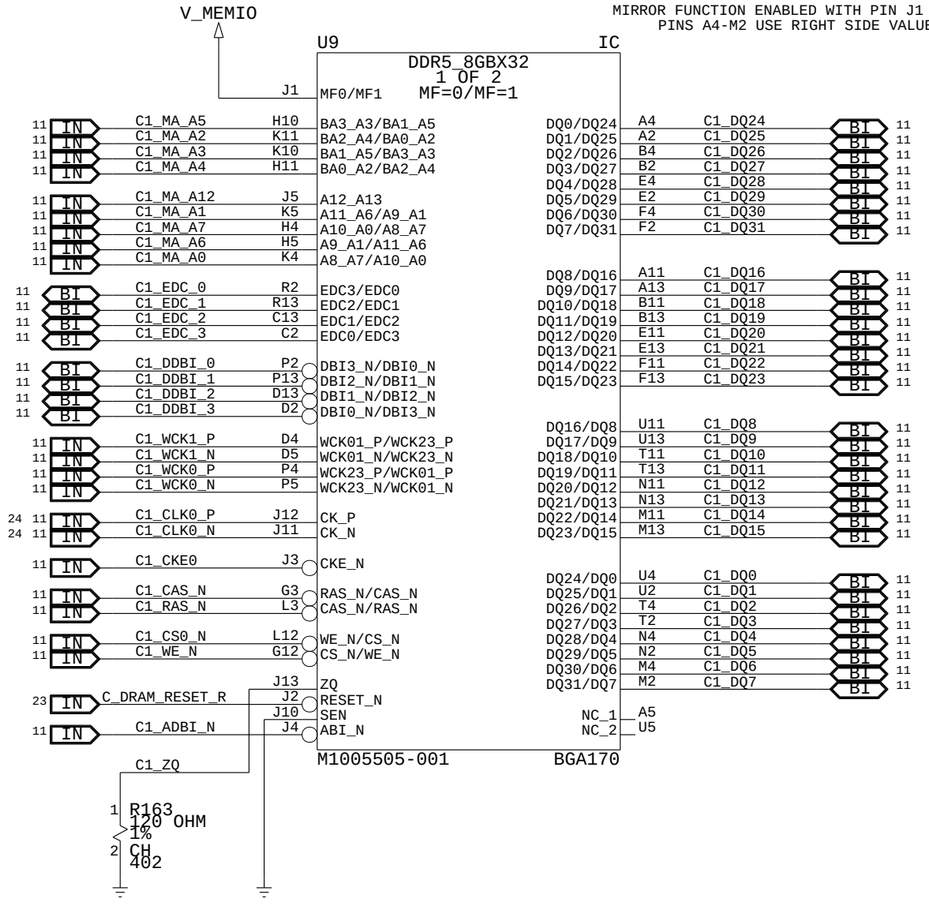



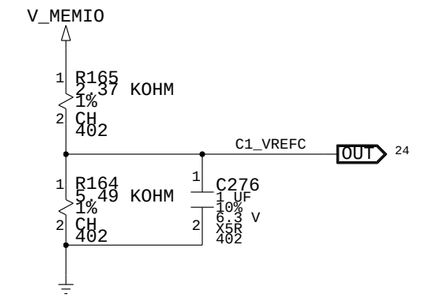



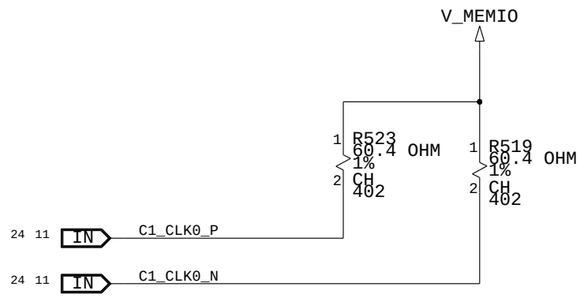

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 23/82 | 23/82 | G-R | 0.991 |

# MEMORY: CHANNEL C1




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 24/82 | 24/82 | G-R | 0.991 |

# MEMORY: CHANNEL D0

GDDR5_BASE

U11 IC

DDR5_8GBx32
2 OF 2
POWER/GROUND

25 IN D0_VREFC

| | | |
|---|---|---|
| J14 | VREFC_37 | |
| A10 | VREFD_38 | |
| U10 | VREFD_39 | |
| G1 | VDD_1 | |
| L1 | VDD_2 | |
| G4 | VDD_3 | |
| L4 | VDD_4 | |
| C5 | VDD_5 | |
| R5 | VDD_6 | |
| C10 | VDD_7 | |
| R10 | VDD_8 | |
| D11 | VDD_9 | |
| G11 | VDD_10 | |
| L11 | VDD_11 | |
| P11 | VDD_12 | |
| G14 | VDD_13 | |
| L14 | VDD_14 | |
| B1 | VDDQ_1 | |
| D1 | VDDQ_2 | |
| F1 | VDDQ_3 | |
| M1 | VDDQ_4 | |
| P1 | VDDQ_5 | |
| T1 | VDDQ_6 | |
| G2 | VDDQ_7 | |
| L2 | VDDQ_8 | |
| B3 | VDDQ_9 | |
| D3 | VDDQ_10 | |
| F3 | VDDQ_11 | |
| H3 | VDDQ_12 | |
| K3 | VDDQ_13 | |
| M3 | VDDQ_14 | |
| P3 | VDDQ_15 | |
| T3 | VDDQ_16 | |
| E5 | VDDQ_17 | |
| N5 | VDDQ_18 | |
| E10 | VDDQ_19 | |
| N10 | VDDQ_20 | |
| B12 | VDDQ_21 | |
| D12 | VDDQ_22 | |
| F12 | VDDQ_23 | |
| H12 | VDDQ_24 | |
| K12 | VDDQ_25 | |
| M12 | VDDQ_26 | |
| P12 | VDDQ_27 | |
| T12 | VDDQ_28 | |
| G13 | VDDQ_29 | |
| L13 | VDDQ_30 | |
| B14 | VDDQ_31 | |
| D14 | VDDQ_32 | |
| F14 | VDDQ_33 | |
| M14 | VDDQ_34 | |
| P14 | VDDQ_35 | |
| T14 | VDDQ_36 | |
| H1 | VSS_1 | |
| K1 | VSS_2 | |
| B5 | VSS_3 | |
| G5 | VSS_4 | |
| L5 | VSS_5 | |
| T5 | VSS_6 | |
| B10 | VSS_7 | |
| D10 | VSS_8 | |
| G10 | VSS_9 | |
| L10 | VSS_10 | |
| P10 | VSS_11 | |
| T10 | VSS_12 | |
| H14 | VSS_13 | |
| K14 | VSS_14 | |
| A1 | VSSQ_1 | |
| C1 | VSSQ_2 | |
| E1 | VSSQ_3 | |
| N1 | VSSQ_4 | |
| R1 | VSSQ_5 | |
| U1 | VSSQ_6 | |
| H2 | VSSQ_7 | |
| K2 | VSSQ_8 | |
| A3 | VSSQ_9 | |
| C3 | VSSQ_10 | |
| E3 | VSSQ_11 | |
| N3 | VSSQ_12 | |
| R3 | VSSQ_13 | |
| C4 | VSSQ_14 | |
| R4 | VSSQ_15 | |
| F5 | VSSQ_16 | |
| M5 | VSSQ_17 | |
| F10 | VSSQ_18 | |
| M10 | VSSQ_19 | |
| C11 | VSSQ_20 | |
| R11 | VSSQ_21 | |
| A12 | VSSQ_22 | |
| C12 | VSSQ_23 | |
| E12 | VSSQ_24 | |
| N12 | VSSQ_25 | |
| R12 | VSSQ_26 | |
| U12 | VSSQ_27 | |
| H13 | VSSQ_28 | |
| K13 | VSSQ_29 | |
| A14 | VSSQ_30 | |
| C14 | VSSQ_31 | |
| E14 | VSSQ_32 | |
| N14 | VSSQ_33 | |
| R14 | VSSQ_34 | |
| U14 | VSSQ_35 | |
| | VSSQ_36 | |

V_MEMIO

M1005505-001 BGA170

V_MEMIO

U11 IC

DDR5_8GBX32
1 OF 2
MF=0/MF=1

J1 MF0/MF1

| | | | | | |
|---|---|---|---|---|---|
| 10 IN | D0_MA_A5 | H10 | BA3_A3/BA1_A5 | DQ0/DQ24 | A4 | D0_DQ24 | B1 10 |
| 10 IN | D0_MA_A2 | K11 | BA2_A4/BA0_A2 | DQ1/DQ25 | A2 | D0_DQ25 | B1 10 |
| 10 IN | D0_MA_A3 | K10 | BA1_A5/BA3_A3 | DQ2/DQ26 | B4 | D0_DQ26 | B1 10 |
| 10 IN | D0_MA_A4 | H11 | BA0_A2/BA2_A4 | DQ3/DQ27 | B2 | D0_DQ27 | B1 10 |
| | | | | DQ4/DQ28 | E4 | D0_DQ28 | B1 10 |
| 10 IN | D0_MA_A12 | J5 | A12_A13 | DQ5/DQ29 | E2 | D0_DQ29 | B1 10 |
| 10 IN | D0_MA_A1 | K5 | A11_A6/A9_A1 | DQ6/DQ30 | F4 | D0_DQ30 | B1 10 |
| 10 IN | D0_MA_A7 | H4 | A10_A0/A8_A7 | DQ7/DQ31 | F2 | D0_DQ31 | B1 10 |
| 10 IN | D0_MA_A6 | H5 | A9_A1/A11_A6 | | | | |
| 10 IN | D0_MA_A0 | K4 | A8_A7/A10_A0 | | | | |
| | | | | DQ8/DQ16 | A11 | D0_DQ16 | B1 10 |
| 10 BI | D0_EDC_0 | R2 | EDC3/EDC0 | DQ9/DQ17 | A13 | D0_DQ17 | B1 10 |
| 10 BI | D0_EDC_1 | R13 | EDC2/EDC1 | DQ10/DQ18 | B11 | D0_DQ18 | B1 10 |
| 10 BI | D0_EDC_2 | C13 | EDC1/EDC2 | DQ11/DQ19 | B13 | D0_DQ19 | B1 10 |
| 10 BI | D0_EDC_3 | C2 | EDC0/EDC3 | DQ12/DQ20 | E11 | D0_DQ20 | B1 10 |
| | | | | DQ13/DQ21 | E13 | D0_DQ21 | B1 10 |
| 10 BI | D0_DDBI_0 | P2 | DBI3_N/DBI0_N | DQ14/DQ22 | F11 | D0_DQ22 | B1 10 |
| 10 BI | D0_DDBI_1 | P13 | DBI2_N/DBI1_N | DQ15/DQ23 | F13 | D0_DQ23 | B1 10 |
| 10 BI | D0_DDBI_2 | D13 | DBI1_N/DBI2_N | | | | |
| 10 BI | D0_DDBI_3 | D2 | DBI0_N/DBI3_N | | | | |
| | | | | DQ16/DQ8 | U11 | D0_DQ8 | B1 10 |
| 10 IN | D0_WCK1_P | D4 | WCK01_P/WCK23_P | DQ17/DQ9 | U13 | D0_DQ9 | B1 10 |
| 10 IN | D0_WCK1_N | D5 | WCK01_N/WCK23_N | DQ18/DQ10 | T11 | D0_DQ10 | B1 10 |
| 10 IN | D0_WCK0_P | P4 | WCK23_P/WCK01_P | DQ19/DQ11 | T13 | D0_DQ11 | B1 10 |
| 10 IN | D0_WCK0_N | P5 | WCK23_N/WCK01_N | DQ20/DQ12 | N11 | D0_DQ12 | B1 10 |
| | | | | DQ21/DQ13 | N13 | D0_DQ13 | B1 10 |
| 25 10 IN | D0_CLK0_P | J12 | CK_P | DQ22/DQ14 | M11 | D0_DQ14 | B1 10 |
| 25 10 IN | D0_CLK0_N | J11 | CK_N | DQ23/DQ15 | M13 | D0_DQ15 | B1 10 |
| 10 IN | D0_CKE0 | J3 | CKE_N | | | | |
| | | | | DQ24/DQ0 | U4 | D0_DQ0 | B1 10 |
| 10 IN | D0_CAS_N | G3 | RAS_N/CAS_N | DQ25/DQ1 | U2 | D0_DQ1 | B1 10 |
| 10 IN | D0_RAS_N | L3 | CAS_N/RAS_N | DQ26/DQ2 | T4 | D0_DQ2 | B1 10 |
| | | | | DQ27/DQ3 | T2 | D0_DQ3 | B1 10 |
| 10 IN | D0_CS0_N | L12 | WE_N/CS_N | DQ28/DQ4 | N4 | D0_DQ4 | B1 10 |
| 10 IN | D0_WE_N | G12 | CS_N/WE_N | DQ29/DQ5 | N2 | D0_DQ5 | B1 10 |
| | | | | DQ30/DQ6 | M4 | D0_DQ6 | B1 10 |
| | | J13 | ZQ | DQ31/DQ7 | M2 | D0_DQ7 | B1 10 |
| 26 25 IN | D_DRAM_RESET_R | J2 | RESET_N | | | | |
| | | J10 | SEN | NC_1 | A5 | | |
| 10 IN | D0_ADBI_N | J4 | ABI_N | NC_2 | U5 | | |

D0_ZQ

1 R222 120 OHM 1% 2 CH 402

M1005505-001 BGA170

MIRROR FUNCTION ENABLED WITH PIN J1 HIGH
PINS A4-M2 USE RIGHT SIDE VALUES

V_MEMIO

1 R224 2.37 KOHM 1% 2 CH 402

D0_VREFC OUT 25

1 R223 5.49 KOHM 1% 2 CH 402

1 C387 1 UF 10% 6.3 V X5R 402

V_MEMIO

1 R562 60.4 OHM 1% 2 CH 402

1 R561 60.4 OHM 1% 2 CH 402

25 10 IN D0_CLK0_P

25 10 IN D0_CLK0_N

10 IN D_DRAM_RESET

R551 10 OHM 1% 402 CH

D_DRAM_RESET_C

R553 49.9 OHM 1% 402 CH

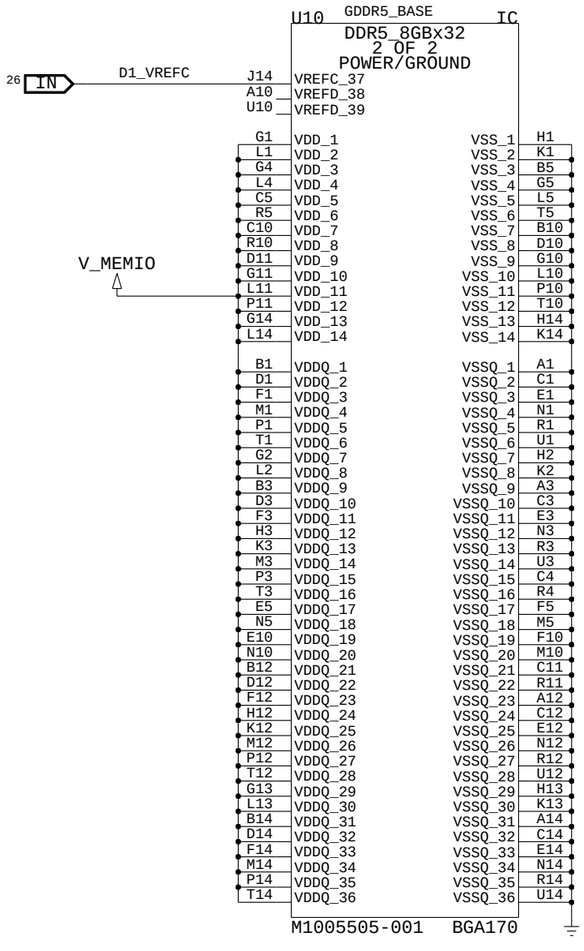
D_DRAM_RESET_R OUT 25 26

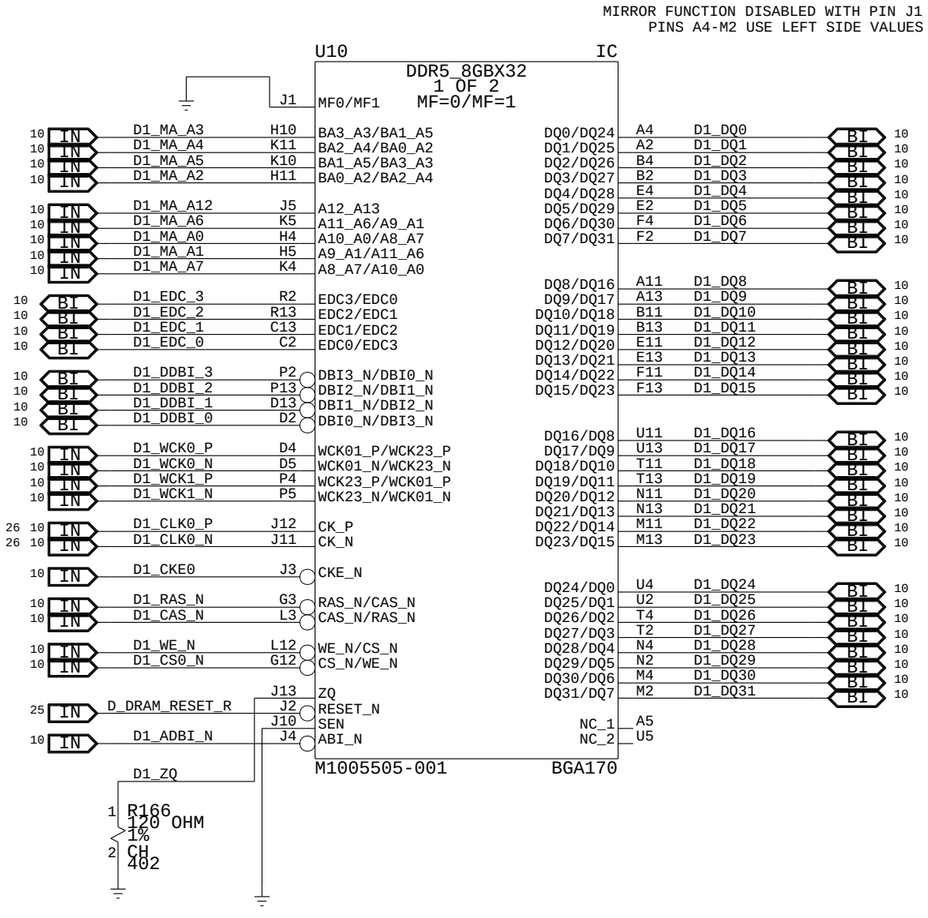
1 R554 4.99 KOHM 1% 2 CH 402

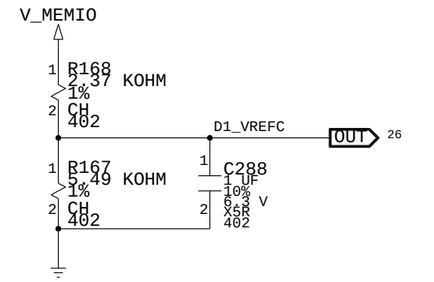
1 C894 120 PF 5% 50 V NP0 402

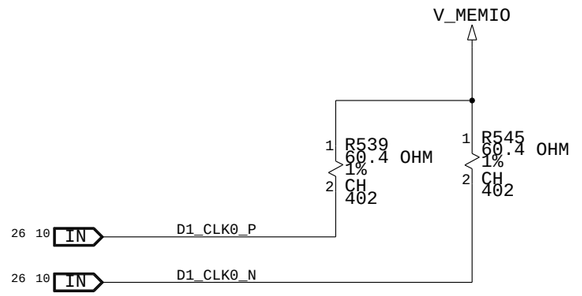
| | | | | | |
|---|---|---|---|---|---|
| MICROSOFT CONFIDENTIAL | PROJECT NAME Cactus | PAGE 25/82 | CSA PAGE 25/82 | FAB G-R | VER 0.991 |

# MEMORY: CHANNEL D1



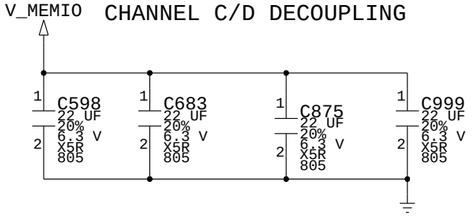




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 26/82 | 26/82 | G-R | 0.991 |

# MEMORY: CHANNEL C/D DECOUPLING

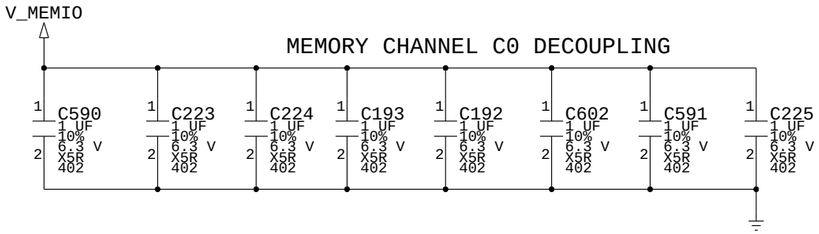
V_MEMIO

CHANNEL C/D DECOUPLING



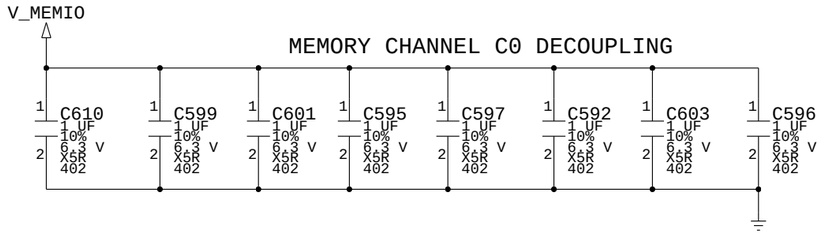

V_MEMIO

MEMORY CHANNEL C0 DECOUPLING



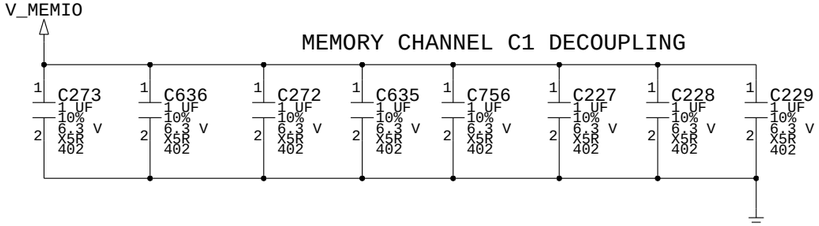

V_MEMIO

MEMORY CHANNEL C0 DECOUPLING



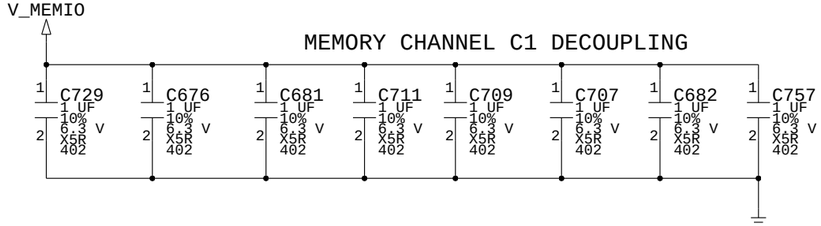

V_MEMIO

MEMORY CHANNEL C1 DECOUPLING



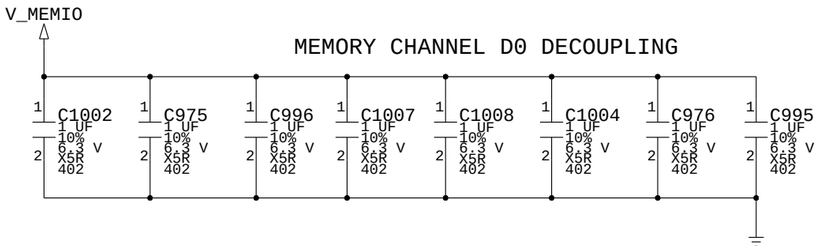

V_MEMIO

MEMORY CHANNEL C1 DECOUPLING



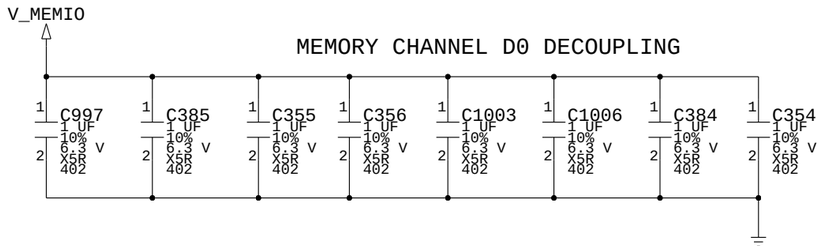

V_MEMIO

MEMORY CHANNEL D0 DECOUPLING



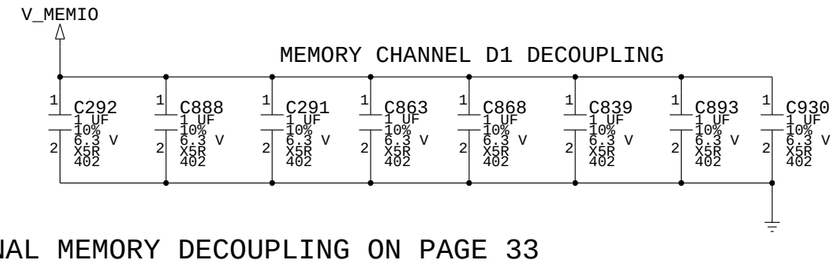

V_MEMIO

MEMORY CHANNEL D0 DECOUPLING



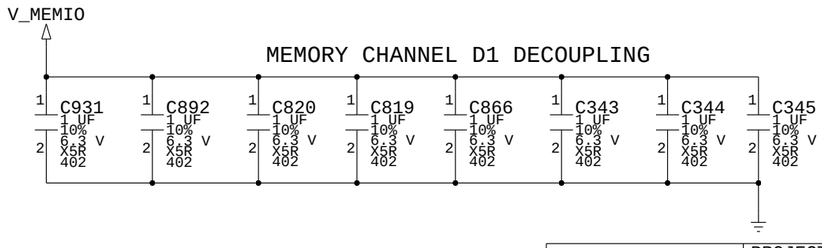

V_MEMIO

MEMORY CHANNEL D1 DECOUPLING



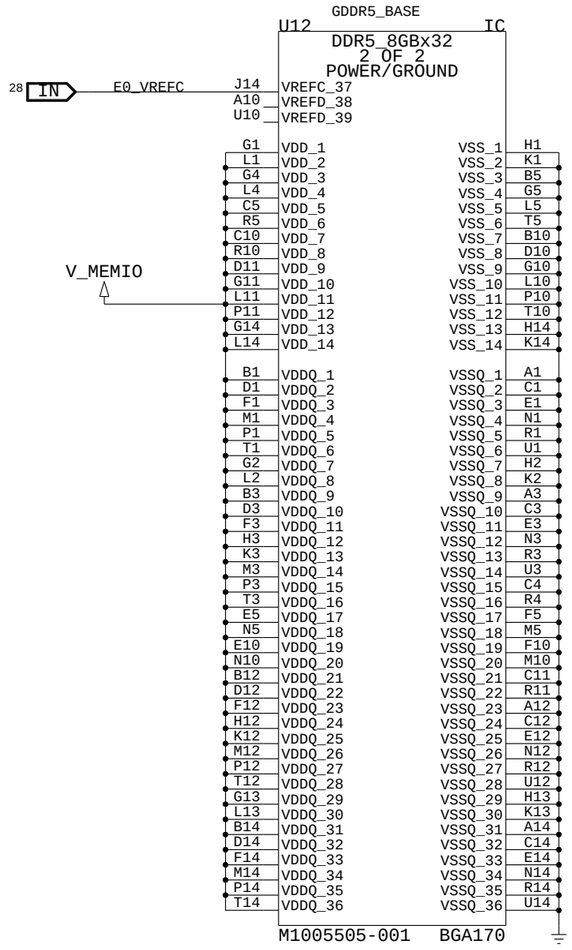

V_MEMIO

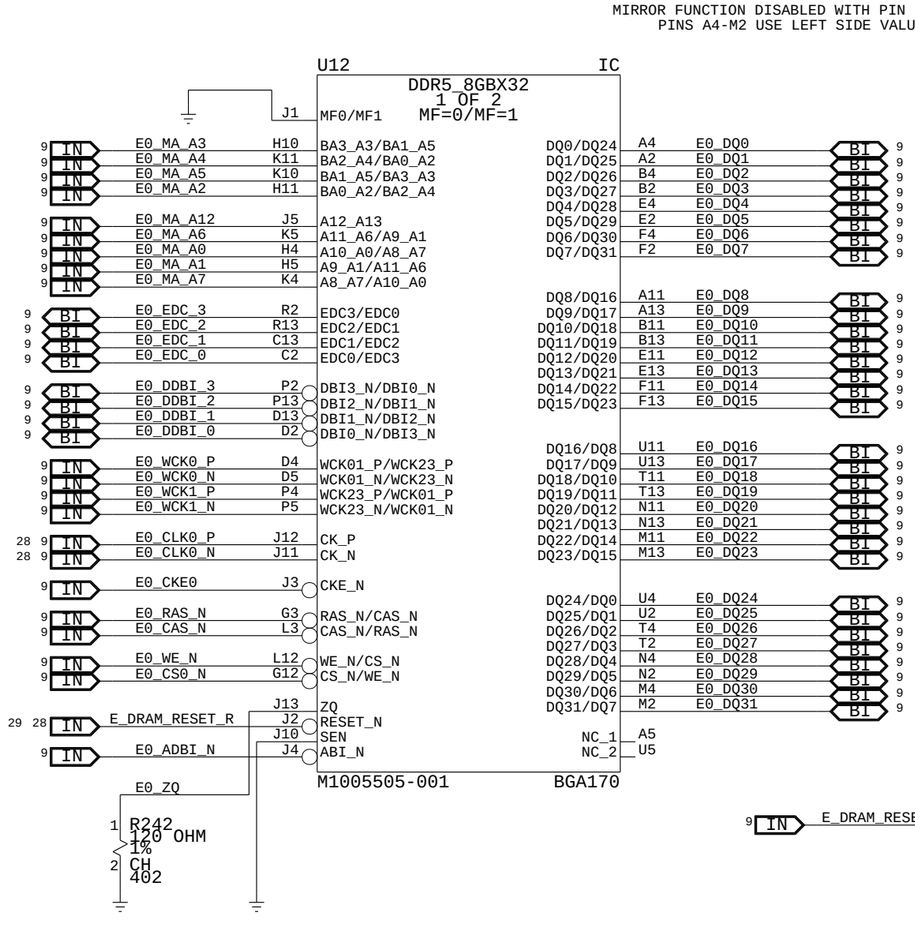
MEMORY CHANNEL D1 DECOUPLING

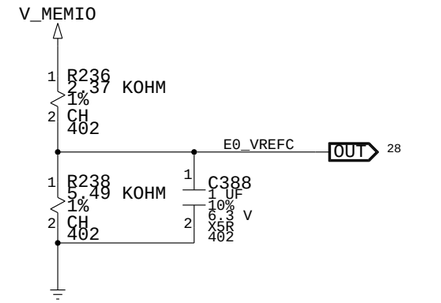


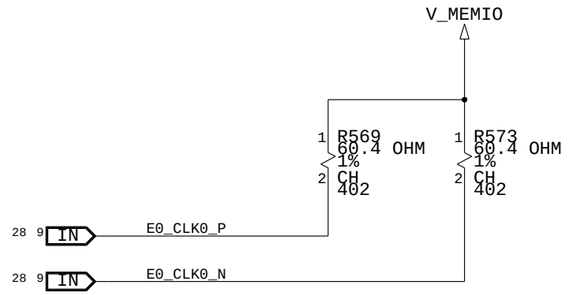

NOTE: ADDITIONAL MEMORY DECOUPLING ON PAGE 33

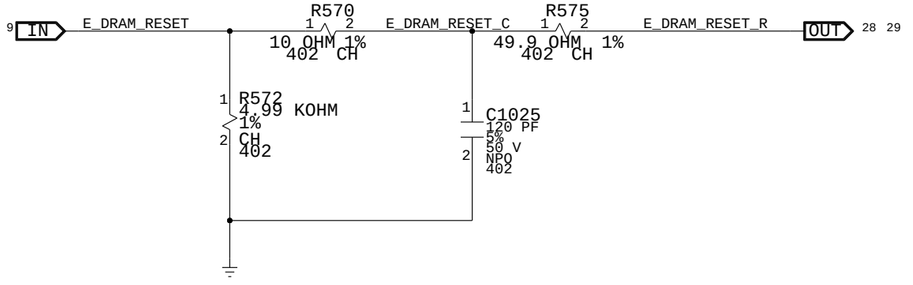
# MEMORY: CHANNEL E0



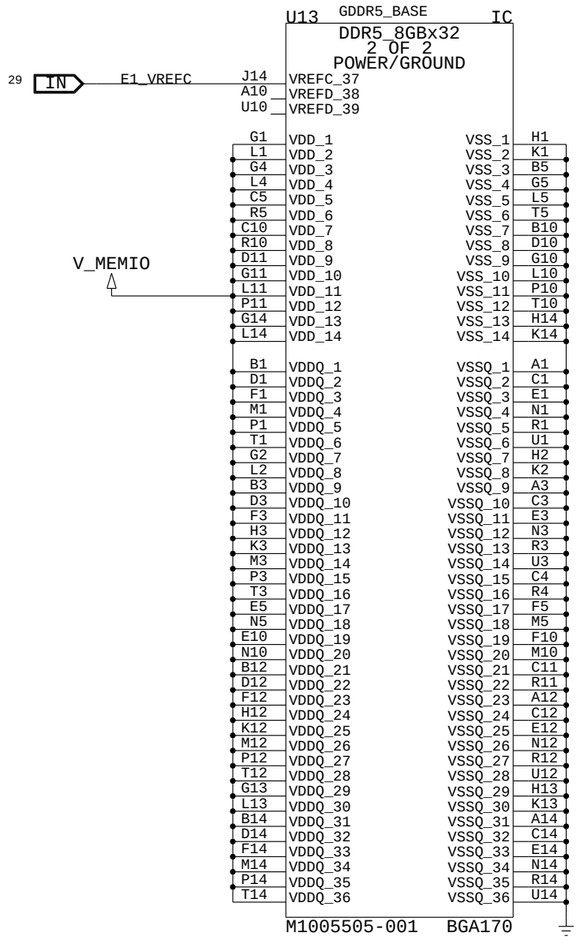



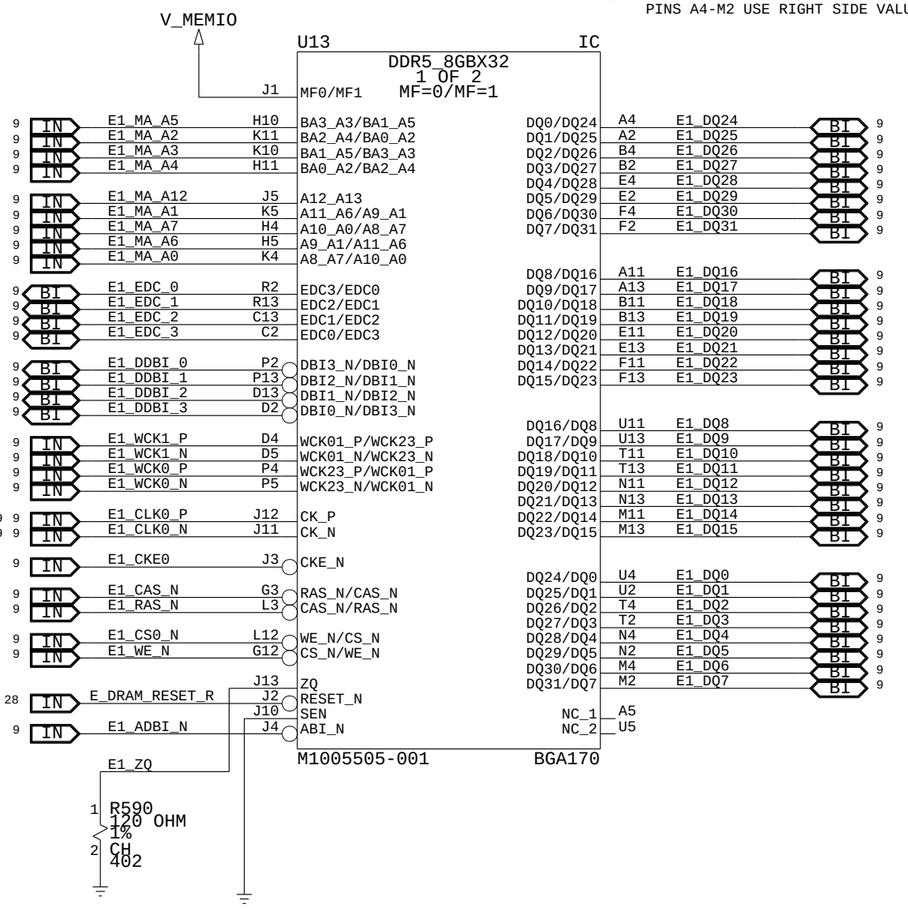



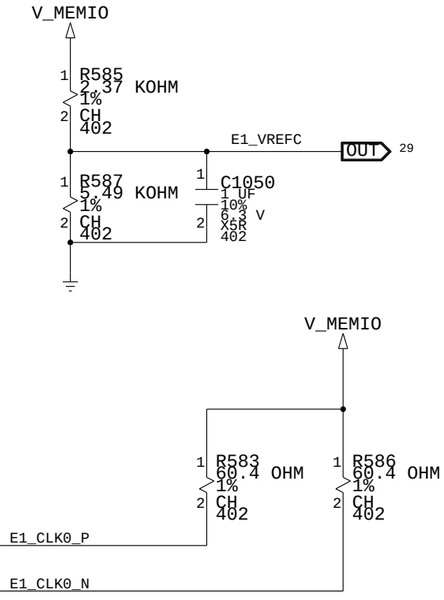




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 28/82 | 28/82 | G-R | 0.991 |

# MEMORY: CHANNEL E1

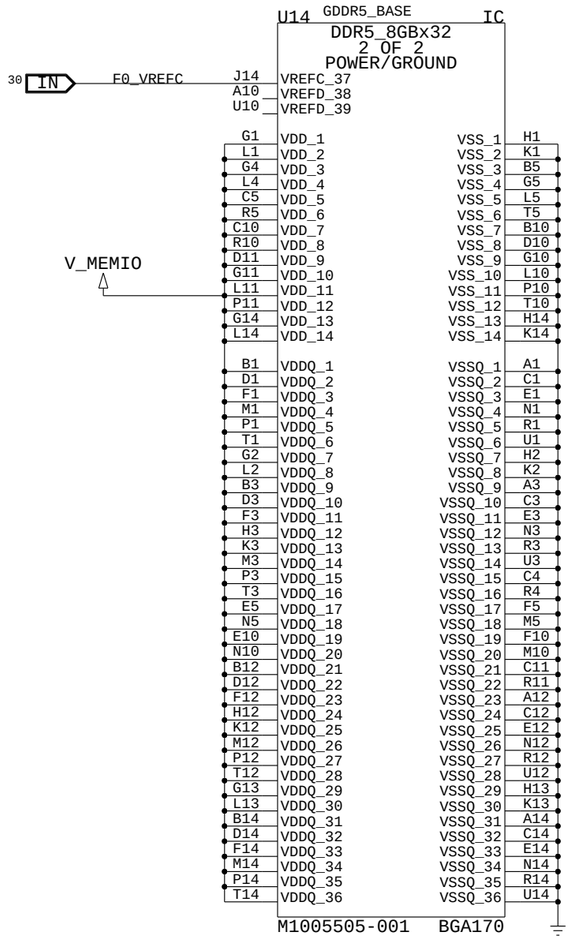


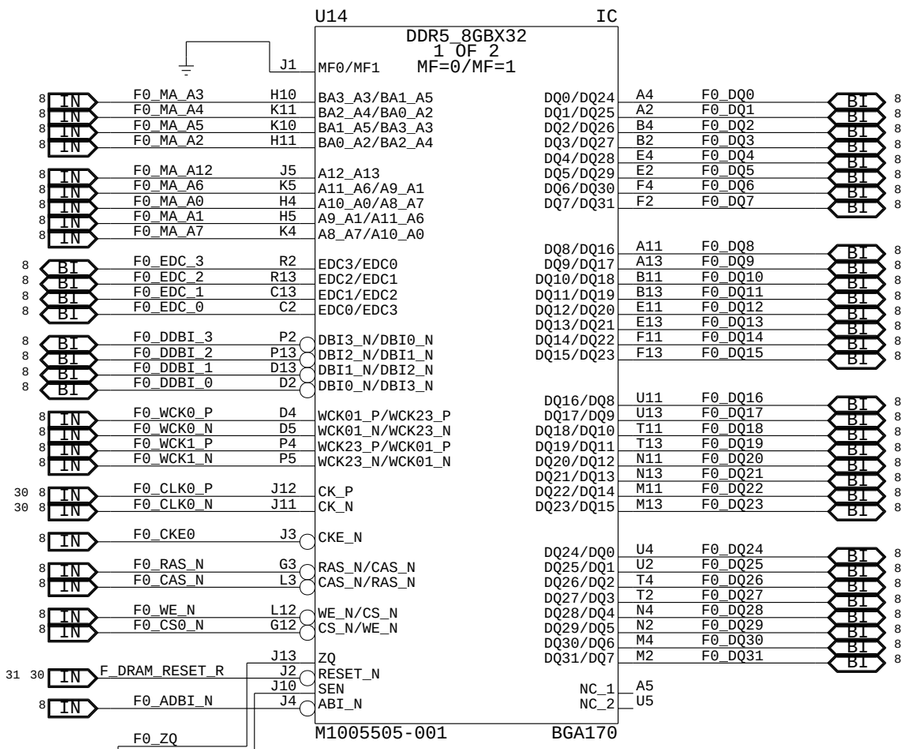



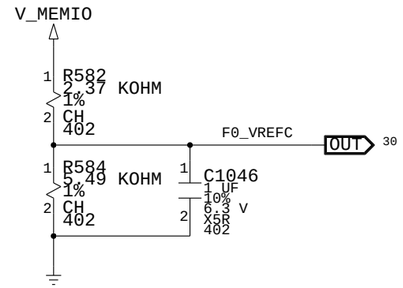

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 29/82 | 29/82 | G-R | 0.991 |

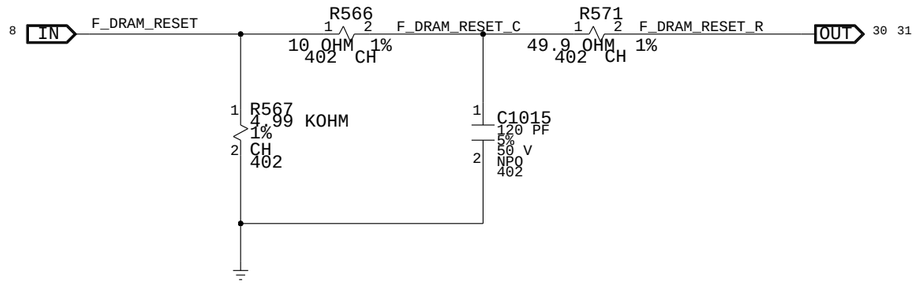
# MEMORY: CHANNEL F0




MIRROR FUNCTION DISABLED WITH PIN J1 LOW
PINS A4-M2 USE LEFT SIDE VALUES

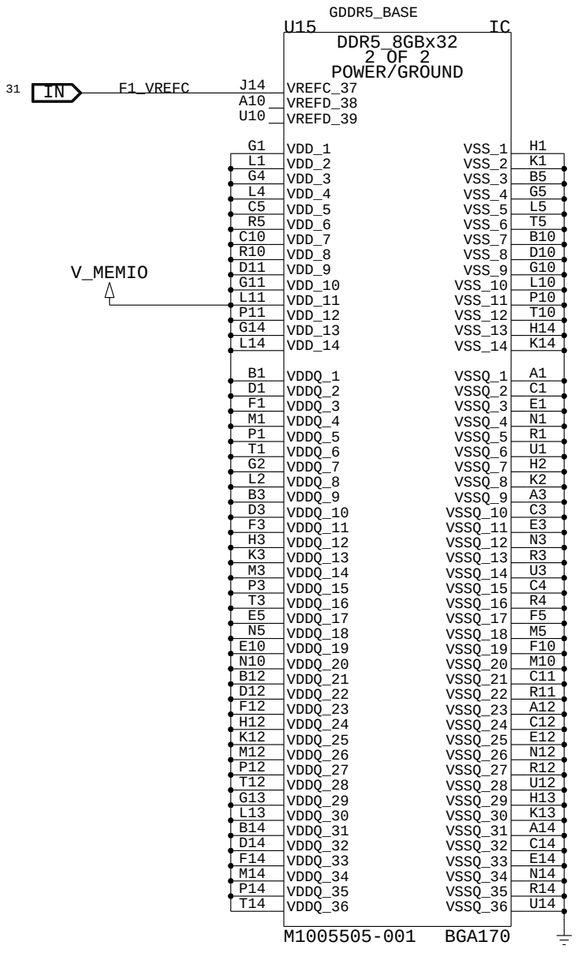


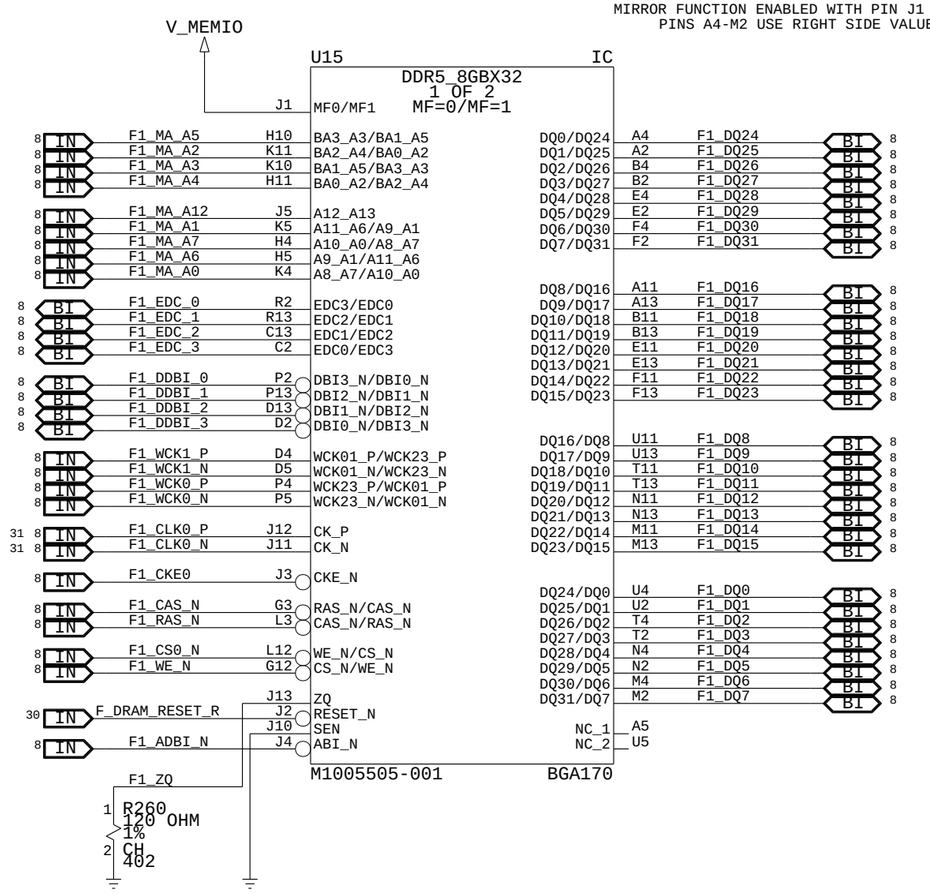



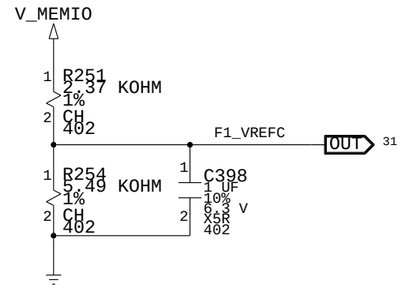



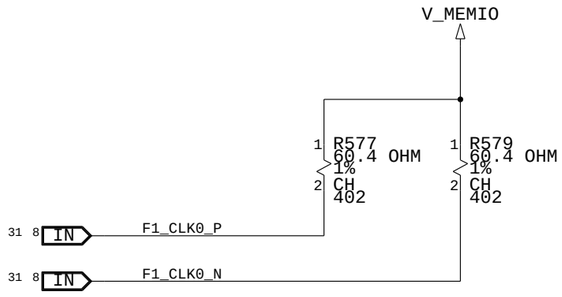

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 30/82 | 30/82 | G-R | 0.991 |

# MEMORY: CHANNEL F1



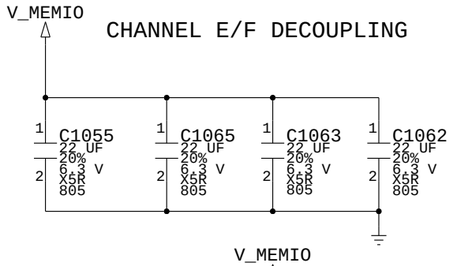



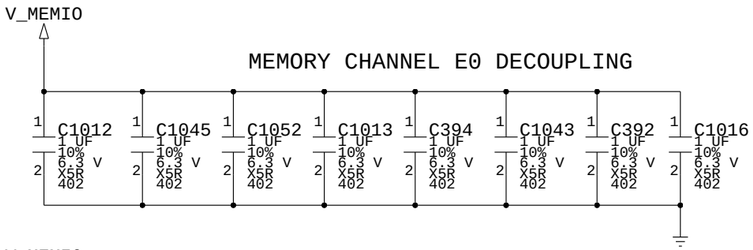



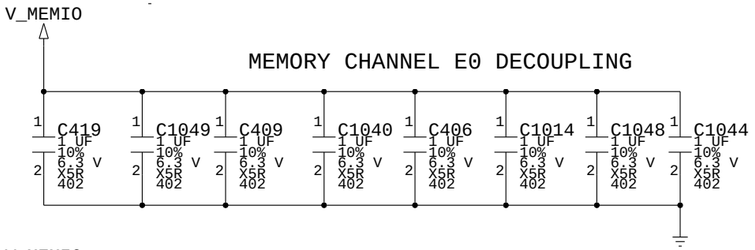



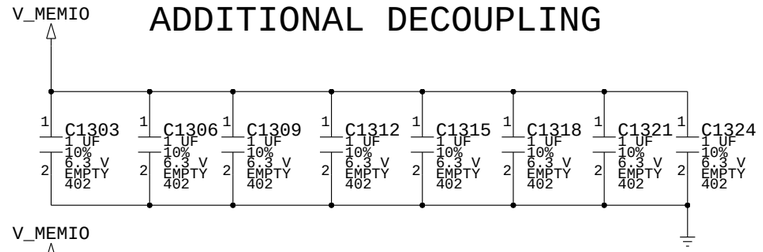

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 31/82 | 31/82 | G-R | 0.991 |

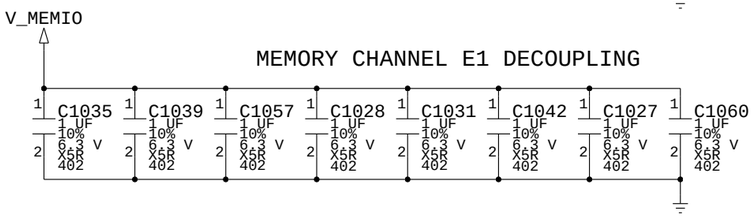
# MEMORY: CHANNEL E/F DECOUPLING

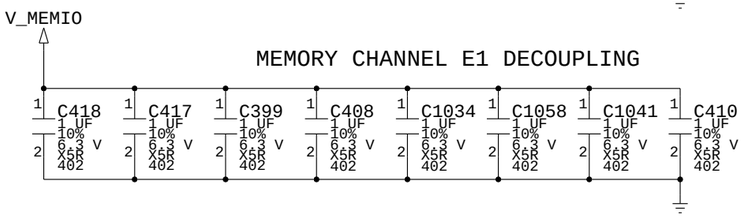


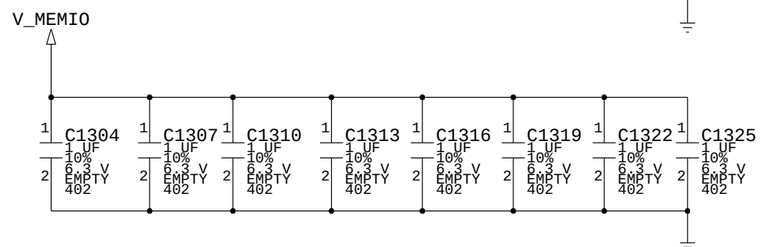



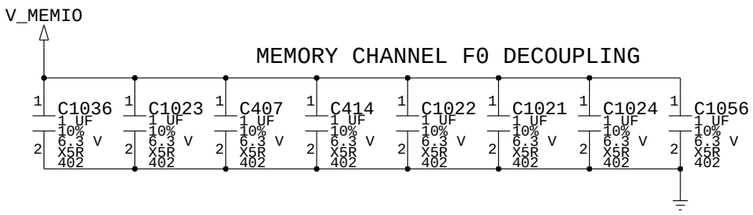



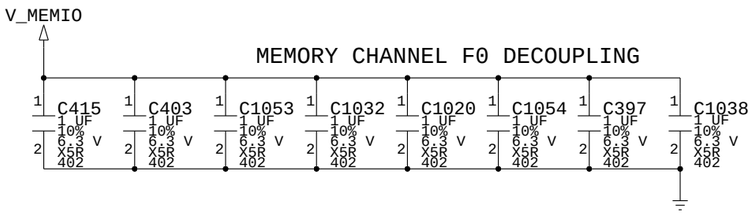



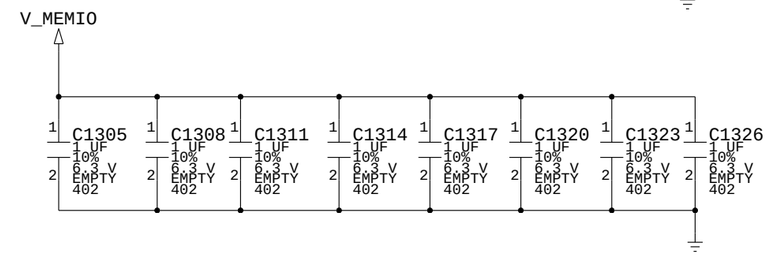



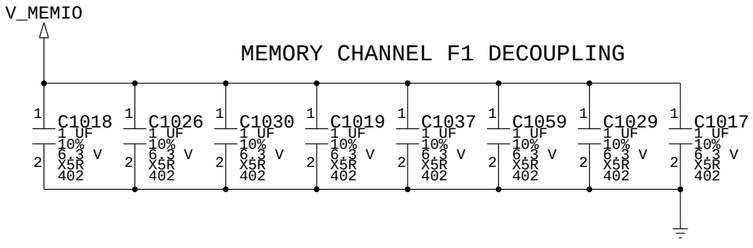



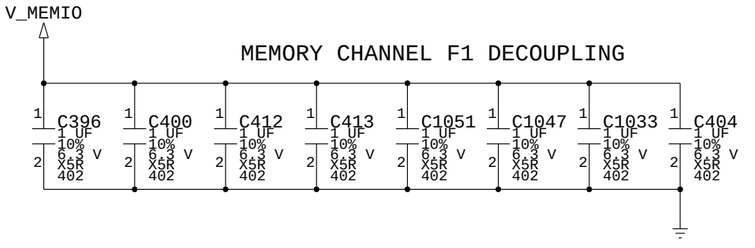



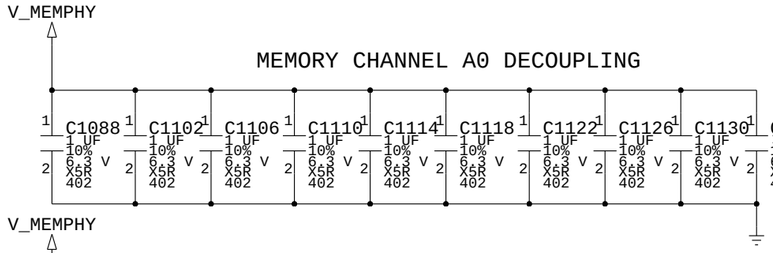




NOTE: ADDITIONAL MEMORY DECOUPLING ON PAGE 33

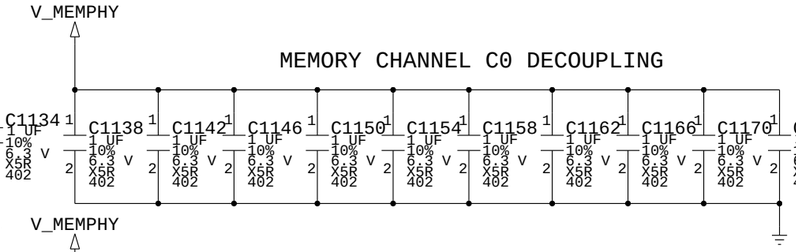
| | | | | | |
|---|---|---|---|---|---|
| MICROSOFT CONFIDENTIAL | PROJECT NAME: Cactus | PAGE: 32/82 | CSA PAGE: 32/82 | FAB: G-R | VER: 0.991 |

# MEMORY: ADDITIONAL DECOUPLING

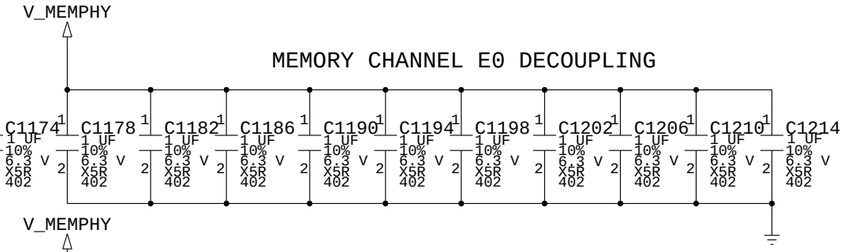
## MEMORY CHANNEL A0 DECOUPLING



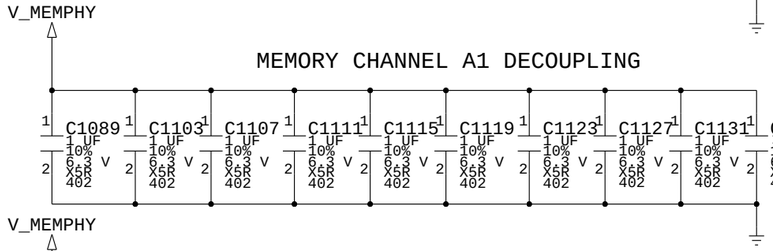

## MEMORY CHANNEL C0 DECOUPLING



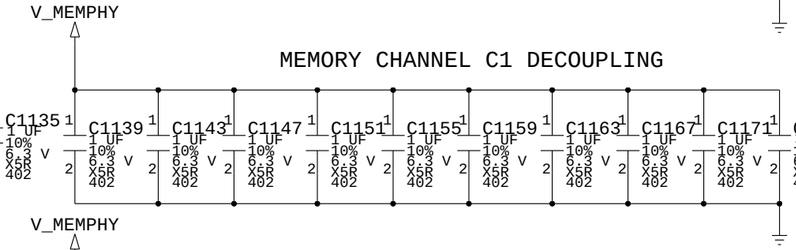

## MEMORY CHANNEL E0 DECOUPLING



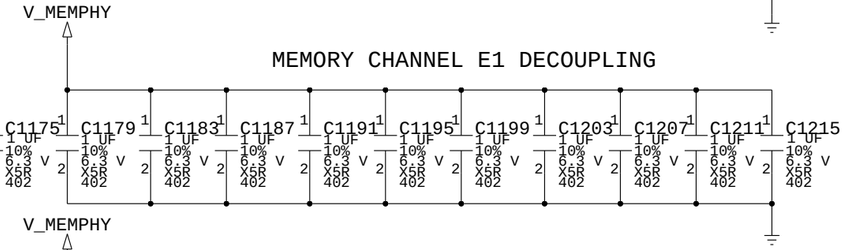

## MEMORY CHANNEL A1 DECOUPLING



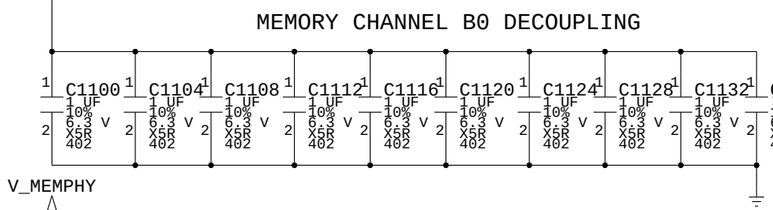

## MEMORY CHANNEL C1 DECOUPLING



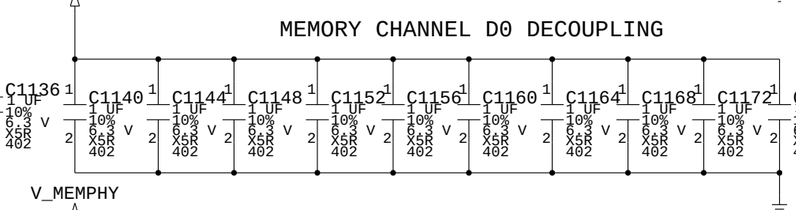

## MEMORY CHANNEL E1 DECOUPLING



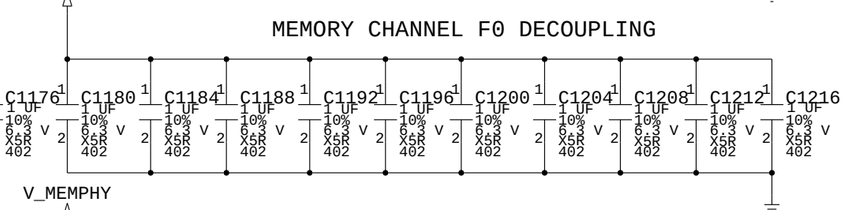

## MEMORY CHANNEL B0 DECOUPLING



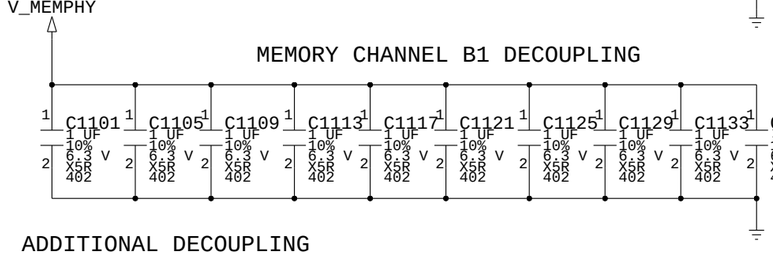

## MEMORY CHANNEL D0 DECOUPLING



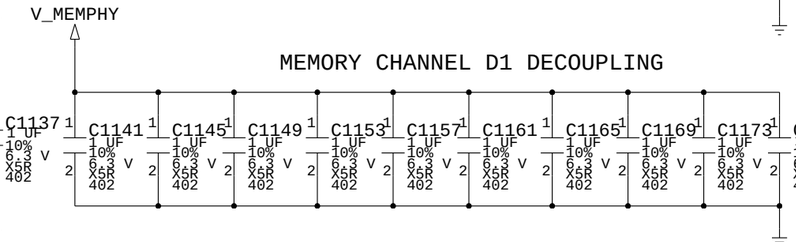

## MEMORY CHANNEL F0 DECOUPLING



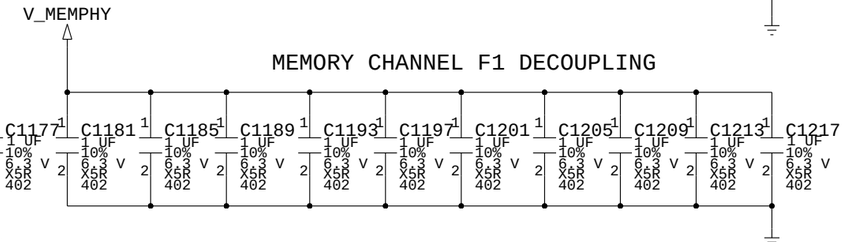

## MEMORY CHANNEL B1 DECOUPLING



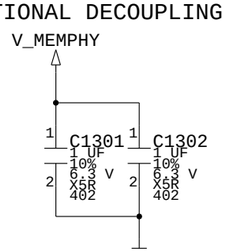

## MEMORY CHANNEL D1 DECOUPLING




## MEMORY CHANNEL F1 DECOUPLING




## ADDITIONAL DECOUPLING




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|---|---|---|---|---|---|
| | Cactus | 33/82 | 33/82 | G-R | 0.991 |

# KIC: USB

U2 KIC IC

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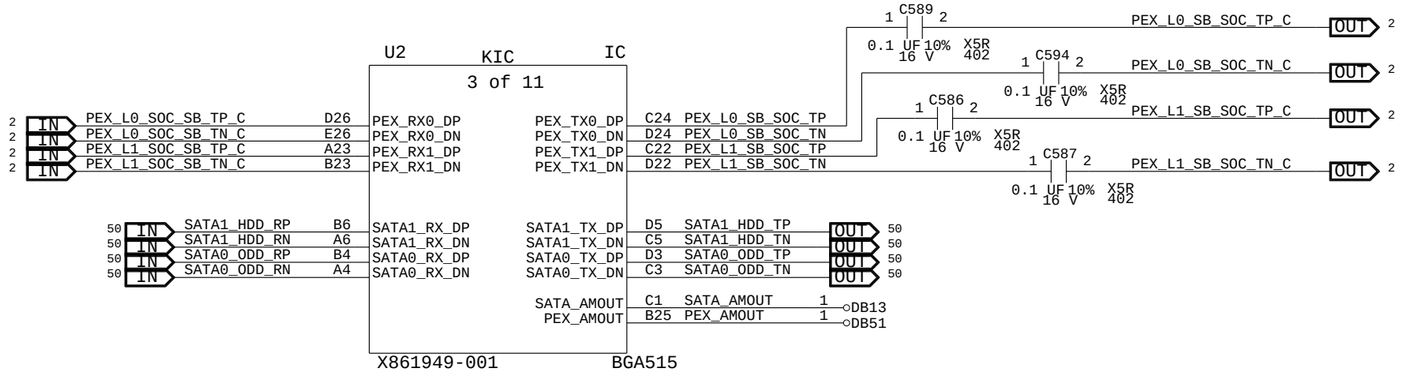
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| USBD_D0_DP | AF8 | | | USB2_WIFI_DP | BI | 46 | WIFI |
| USBD_D0_DN | AE8 | | | USB2_WIFI_DN | BI | 46 | |
| USBC_D0_DP | AC4 | | | USB2_AUX_DP | BI | 45 | REAR 1 |
| USBC_D0_DN | AC3 | | | USB2_AUX_DN | BI | 45 | |
| REAR 1 45 IN USB3_AUX_RP | Y1 | USBC_D1_RXP | USBC_D1_TXP | W3 USB3_AUX_TP_C | C501 0.1 UF 10% X5R 16 V 402 | USB3_AUX_TP OUT | 45 |
| REAR 1 45 IN USB3_AUX_RN | Y2 | USBC_D1_RXN | USBC_D1_TXN | W4 USB3_AUX_TN_C | C500 0.1 UF 10% X5R 16 V 402 | USB3_AUX_TN OUT | 45 |
| REAR 0 45 IN USB3_BP_P1_RP | V1 | USBB_D2_RXP | USBB_D2_TXP | U3 USB3_BP_P1_TP_C | C499 0.1 UF 10% X5R 16 V 402 | USB3_BP_P1_TP OUT | 45 |
| REAR 0 45 IN USB3_BP_P1_RN | V2 | USBB_D2_RXN | USBB_D2_TXN | U4 USB3_BP_P1_TN_C | C498 0.1 UF 10% X5R 16 V 402 | USB3_BP_P1_TN OUT | 45 |
| | | | USBB_D0_DP | AE4 | | USB2_BP_P1_DP BI | 45 REAR 0 |
| | | | USBB_D0_DN | AE3 | | USB2_BP_P1_DN BI | 45 |
| | | | USBB_D1_DP | AD2 | | USB2_BP_P0_DP 1 DB85 | NOT USED |
| | | | USBB_D1_DN | AD1 | | USB2_BP_P0_DN 1 DB84 | |
| NOT USED DB90 1 USB3_BP_P0_RP | T1 | USBB_D3_RXP | USBB_D3_TXP | R3 | | USB3_BP_P0_TP 1 DB14 | |
| DB91 1 USB3_BP_P0_RN | T2 | USBB_D3_RXN | USBB_D3_TXN | R4 | | USB3_BP_P0_TN 1 DB15 | |
| FRONT 45 IN USB3_FP_P0_RP_BUFF | AA8 | USBA_D2_RXP | USBA_D2_TXP | AB10 USB3_FP_P0_TP_C | C1097 0.1 UF 10% X5R 16 V 402 | USB3_FP_P0_TP OUT | 45 |
| FRONT 45 IN USB3_FP_P0_RN_BUFF | AA7 | USBA_D2_RXN | USBA_D2_TXN | AB9 USB3_FP_P0_TN_C | C1096 0.1 UF 10% 16 V 402 X5R | USB3_FP_P0_TN OUT | 45 FRONT |
| | | | USBA_D0_DP | AC7 | | USB2_FP_P0_DP BI | 45 |
| | | | USBA_D0_DN | AC8 | | USB2_FP_P0_DN BI | 45 |
| | | | USBA_D1_DP | AF6 | | USB2_ACC_DP BI | 51 FRONT PANEL ACC |
| | | | USBA_D1_DN | AF5 | | USB2_ACC_DN BI | 51 |
| NOT USED DB18 1 USB3_ACC_RP | AB1 | USBA_D3_RXP | USBA_D3_TXP | AA3 | | USB3_ACC_TP 1 DB20 | NOT USED |
| DB17 1 USB3_ACC_RN | AB2 | USBA_D3_RXN | USBA_D3_TXN | AA4 | | USB3_ACC_TN 1 DB22 | |

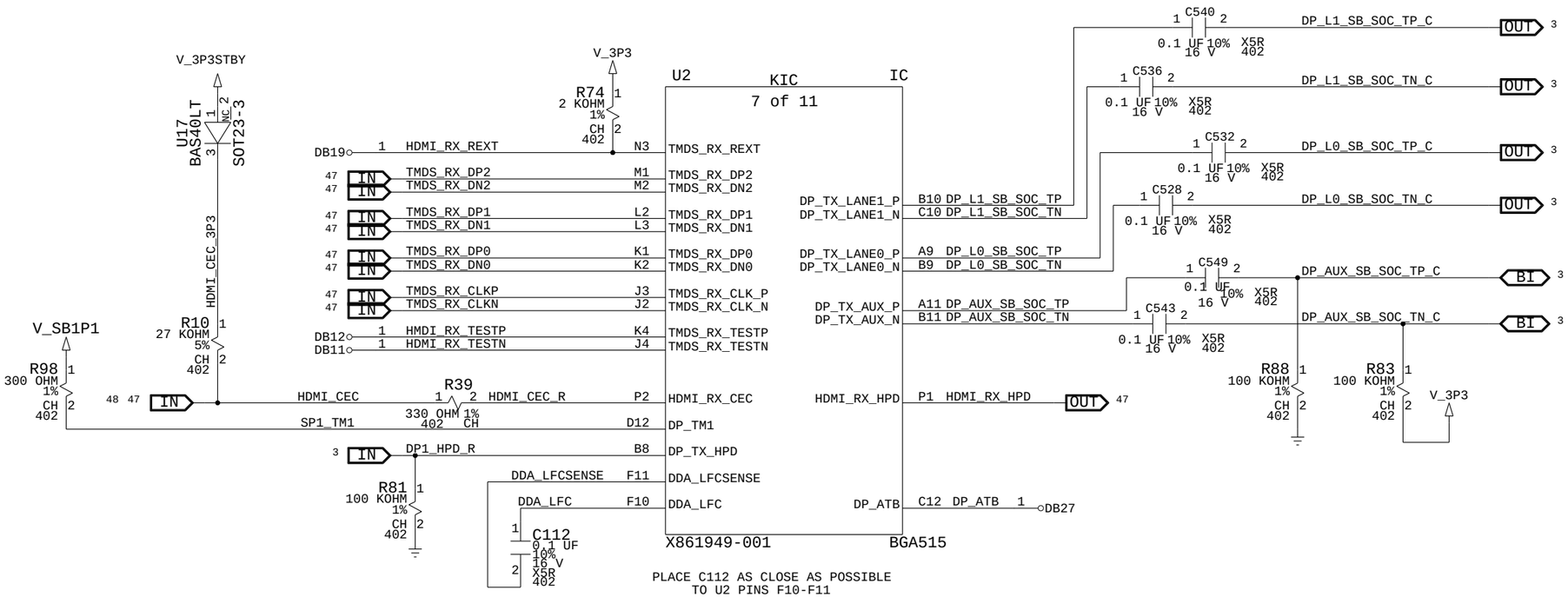
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | USBA_D2_AMOUT | AA5 | USBA_D2_AMOUNT 1 | DB97 | |
| | | | USBA_D1_AMOUT | AE6 | USBA_D1_AMOUNT 1 | DB23 | |
| | | | USBA_D0_AMOUT | AC11 | USBA_D0_AMOUNT 1 | DB100 | |
| | | | USBB_D2_AMOUT | R5 | USBB_D2_AMOUNT 1 | DB10 | |
| | | | USBB_D1_AMOUT | AD5 | USBB_D1_AMOUNT 1 | DB95 | |
| | | | USBB_D0_AMOUT | AF4 | USBB_D0_AMOUNT 1 | DB94 | |
| USBA_D1_TXRTUNE | AB11 | USBA_D1_TXRTUNE | USBC_D1_AMOUT | W5 | USBC_D1_AMOUNT 1 | DB24 | |
| R86 174 OHM 1% CH 402 | | | USBC_D0_AMOUT | AC5 | USBC_D0_AMOUNT 1 | DB21 | |
| | | | USBD_D0_AMOUT | AE9 | USBD_D0_AMOUNT 1 | DB98 | |

X861949-001 BGA515

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 34/82 | 34/82 | G-R | 0.99 |

# KIC: PCIEX,SATA,VIDEO




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 35/82 | 35/82 | G-R | 0.991 |

# KIC: SMC

## U2 KIC IC — 4 of 11

| | | | |
|---|---|---|---|
| SMC_RST_N_IN | AF23 | SMC_RST_N | 73 39 IN |
| 3P3V_5V_OK | | | |
| SMC_P4_GPIO7 | V23 | PWRSW_N | 74 51 IN |
| SMC_P4_GPIO6 | V22 | EJECTSW_N | 74 51 IN |
| SMC_P4_GPIO5 | T23 | BINDSW_N_R | 45 IN |
| SMC_P4_GPIO4 | U23 | AUDIO_PWM_OUT | 51 OUT |
| SMC_P4_GPIO3 | U22 | VREG_PWRGPC_EN | 64 OUT |
| SMC_P4_GPIO2 | U21 | ACC_RESET_N | 51 OUT |
| SMC_P4_GPIO1 | T21 | PCIE_RST_N | 76 42 OUT |
| SMC_P4_GPIO0 | R23 | VREG_3_3_EN | 63 40 OUT |
| I21 1P8V_NO_5V | | | |
| SMC_P3_GPIO7 | L21 | TCK | 75 14 OUT |
| SMC_P3_GPIO6 | M22 | TMS | 75 14 OUT |
| SMC_P3_GPIO5 | N22 | VREG_V5P0_PWRGD | 62 IN |
| SMC_P3_GPIO4 | N21 | VREG_V5P0_EN | 62 OUT |
| SMC_P3_GPIO3 | P21 | PSU_FAN_EN | |
| SMC_P3_GPIO2 | R21 | VREG_PWRGPB_EN | 64 54 OUT |
| SMC_P3_GPIO1 | R22 | FAN_SPEED | 51 IN |
| SMC_P3_GPIO0 | P23 | VREG_PWRGPA_EN | 63 58 65 OUT |
| 3P3V_5V_OK | | | |
| SMC_UART1_RXD | V25 | SMC_RXD | FT52 FTP |
| SMC_UART1_CTS | V26 | ODD_STATUS | 50 IN |
| SMBUS3_SDA | AE21 | SB_SOC_DATA | 14 BI |
| SMBUS3_SCK | AD21 | SB_SOC_CLK | 14 OUT |
| SMBUS2_SDA | AE20 | SMBUS_DATA | 78 73 70 58 54 48 BI |
| SMBUS2_SCK | AF20 | SMBUS_CLK | 78 73 70 58 54 48 OUT |
| SMBUS1_SDA | H6 | HDMI_TX_DDC_DATA_SB | |
| SMBUS1_SCK | G6 | HDMI_TX_DDC_CLK_SB | |
| SMBUS0_SDA | G5 | HDMI_RX_DDC_DATA_SB | |
| SMBUS0_SCK | F5 | HDMI_RX_DDC_CLK_SB | |
| SECURITY_BYPASS | AC14 | SECURITY_BYPASS | |
| SMC_IR_IN | N26 | IR_DATA | 51 IN |

X861949-001 BGA515

| | | | |
|---|---|---|---|
| SMC_UART1_TXD | T24 | SMC_TXD | FT48 FTP |
| SMC_UART1_RTS | V24 | ODD_3P3V_STBY_EN_N | 50 OUT |
| 3P3V_5V_OK | | | |
| SMC_P2_GPIO7 | G2 | VREG_PWRGPAB_CPUGFX_PWRGD | 54 58 61 64 74 IN |
| SMC_P2_GPIO6 | F2 | COLD_RESET_N | 74 75 IN |
| SMC_P2_GPIO5 | G1 | HDD_PWR_EN | 50 OUT |
| SMC_P2_GPIO4 | F1 | SOC_THERMTRIP_R | R125 36 |
| SMC_P2_GPIO3 | G3 | ODD_PWR_EN | 50 OUT |
| SMC_P2_GPIO2 | H5 | ODD_OPEN | 50 OUT |
| SMC_P2_GPIO1 | J5 | SP_SMC_INT_R | R105 36 |
| SMC_P2_GPIO0 | J6 | ODD_WAKE_N | 50 IN |
| 1P8V_NO_5V | | | |
| SMC_P1_GPIO7 | J21 | TDI_R | R133 14 75 |
| SMC_P1_GPIO6 | H21 | SOC_PWR_OK_R | R127 |
| SMC_P1_GPIO5 | J22 | SOC_RST_N_R | R136 |
| SMC_P1_GPIO4 | K22 | TDO_R | |
| SMC_P1_GPIO3 | J23 | FAN_PWM | 51 OUT |
| SMC_P1_GPIO2 | H25 | IR_MOD_TIMER_IN | 1 DB53 |
| SMC_P1_GPIO1 | J25 | IR_BLAST_SMC_OUT | 69 OUT |
| SMC_P1_GPIO0 | H26 | LED_NEXUS | 51 OUT |
| 3P3V_NO_5V | | | |
| 3P3V_5V_OK | | | |
| SMC_P0_GPIO7 | M23 | HDMI_V_5P0_EN | 48 OUT |
| SMC_P0_GPIO6 | M24 | IR_MOD_TIMER_OUT | 1 DB52 |
| SMC_P0_GPIO5 | L23 | DP0_HPD_SB | 48 IN |
| SMC_P0_GPIO4 | K24 | WIFI_RESET_N | 46 OUT |
| SMC_P0_GPIO3 | L25 | M2_PWR_EN | 42 76 OUT |
| SMC_P0_GPIO2 | M25 | HDMI_RX_DDC_5V_E_R | R491 47 |
| SMC_P0_GPIO1 | L26 | SMC_DBG_LED0_SWO | 73 OUT |
| SMC_P0_GPIO0 | K26 | VREG_1P1_ACTIVE_EN | 65 68 OUT |

## 3.3V LEVEL SHIFTER

V_SOC1P8 — R514 10 KOHM 1% CH 402 — 36 14 IN

V_3P3STBY — R205 100 KOHM 1% CH 402

R122 0 OHM 5% CH 402

SP_SMC_INT_N

SP_SMC_INT_N_R

U37 XSTR

R110 10 KOHM 1% CH 402 — SP_SMC_INT

R126 10 KOHM 1% CH 402 — SOC_THERMTRIP

SOC_THERMTRIP_N

R123 0 OHM 5% CH 402

SOC_THERMTRIP_N_R

R517 4.7 KOHM 5% CH 402 — FT60 FTP — 36 14 IN

R518 4.7 KOHM 5% CH 402

R128 4.7 KOHM 5% CH 402

FT39 FTP — 14 54 75 OUT

FT61 FTP — 14 75 OUT

R129 0 OHM 5% 402 EMPTY — SOC_THERMTRIP_N — 14 36 IN

R139 0 OHM 5% 402 EMPTY — SP_SMC_INT_N — 14 36 IN

R137 0 OHM 5% 402 — TDO — 14 75 IN

R492 10 KOHM 1% CH 402

## Passives

R486 100 KOHM 1% CH 402

C466 0.01 UF 10% 16 V EMPTY 402

R472 4.7 KOHM 5% EMPTY 402

R483 10 KOHM 1% CH 402

R380 10 KOHM 1% CH 402

R488 100 OHM 1% 402 CH — FAN_TACH_IN

DB8 — DB122

R467 1.27 KOHM 1% CH 402 — R469 1.27 KOHM 1% CH 402

R515 2 KOHM 1% CH 402 — R516 2 KOHM 1% CH 402

R471 243 OHM 1% 402 CH — R473 243 OHM 1% 402 CH

R470 0 OHM 5% 402 CH

R468 0 OHM 5% 402 CH

R20 49.9 OHM 1% 402 CH

R27 49.9 OHM 1% 402 CH

R390 49.9 OHM 1% 402 CH

R389 10 OHM 1% 402 CH

FT53 FTP — FT43 FTP — FT413 FTP — FT51 FTP — FT41 FTP — FT42 FTP — FT38 FTP — FT36 FTP

DDC = HDMI OUT

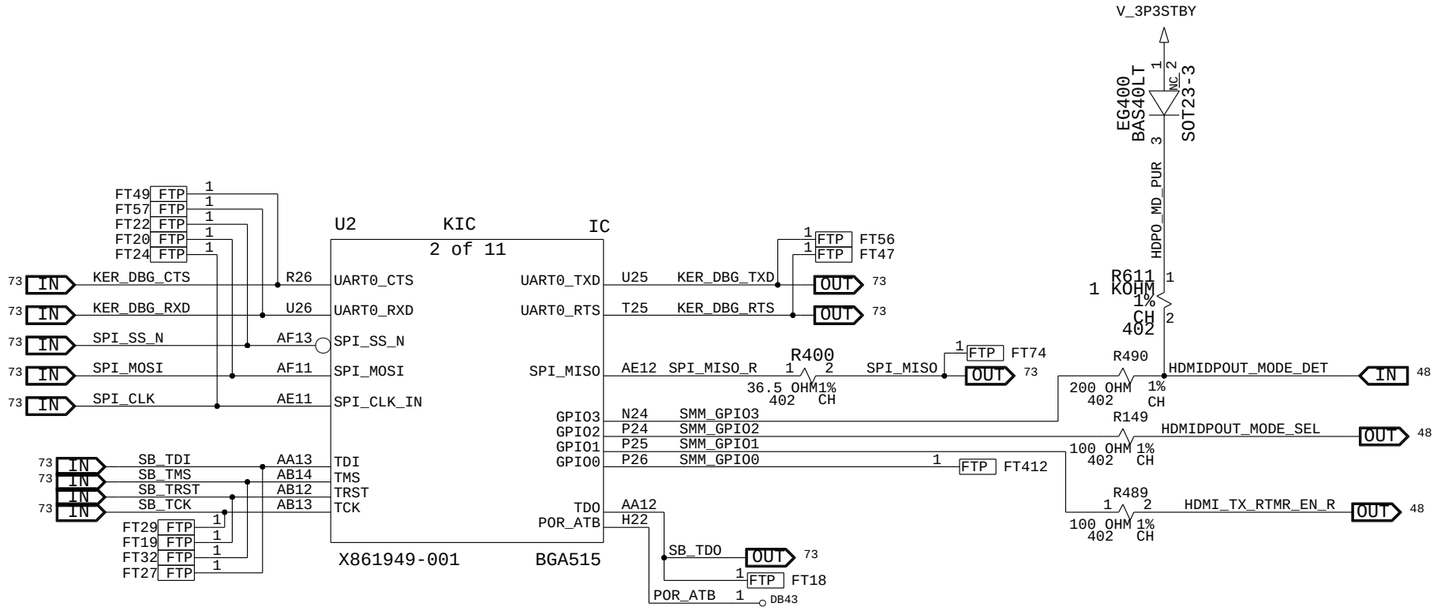
RX_DDC = HDMI IN

USED BY SMC IR BLASTER SOLUTION
SMC_IR_BLASTER_GPIO'S

SMC_P1_GPIO1 > IR_SMC_BLAST_OUT (TIMER 0)
SMC_P0_GPIO6 > IR_MOD_TIMER_OUT (TIMER 1)
SMC_P4_GPIO2 > IR_MOD_TIMER_IN (TIMER 2)

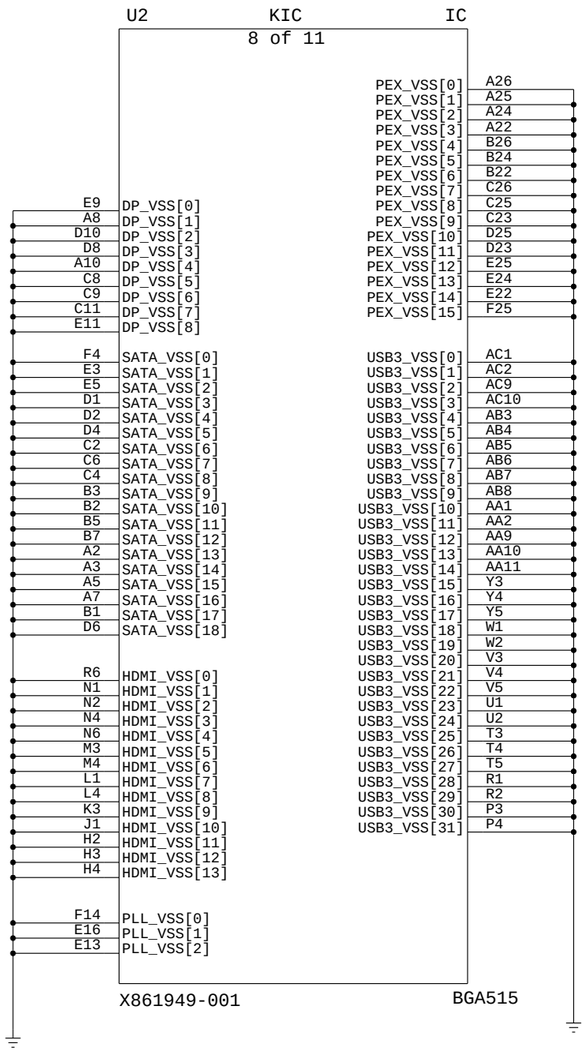
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 36/82 | 36/82 | G-R | 0.991 |

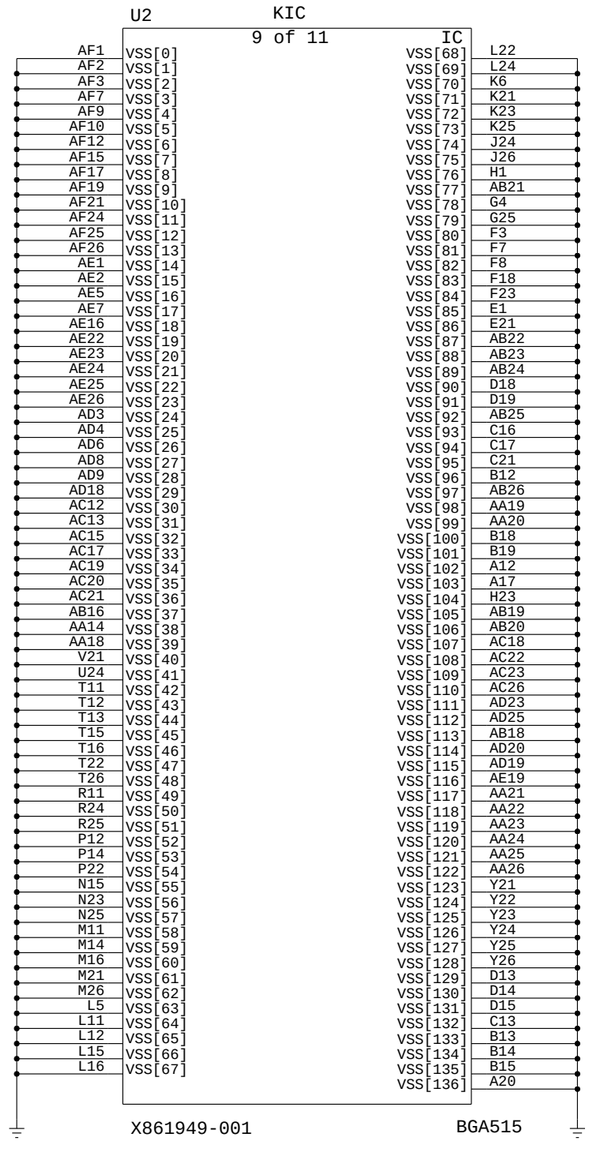
# KIC: FACET




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 37/82 | 37/82 | G-R | 0.991 |

# KIC: POWER




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 38/82 | 38/82 | G-R | 0.991 |

# KIC: CLOCKS,STRAPPING,POR

PEX REF CLOCKS CONTROLLED BY SMC

V_3P3
R94 10 KOHM 1% CH 402
R90 10 KOHM 1% EMPTY 402
R108 10 KOHM 1% CH 402
R107 10 KOHM 1% EMPTY 402
V_1P8STBY

SMC_XTAL_OUT
R461 249 KOHM 1% 402 CH
R457 1 KOHM 1% 402 CH
SMC_XTAL_OUT_R
SMC_XTAL_IN
R465 49.9 OHM 1% EMPTY 402
Y2 25 MHZ XTAL M1020244-001 SM
C582 30 PF 5% 50 V NPO 402
C585 30 PF 5% 50 V NPO 402
R121 10 KOHM 1% 402 CH
SMC_BYPASS_CLK_IN

KIC_BASE
U2 KIC 1 of 11 IC

| | | | |
|---|---|---|---|
| | G22 | V_VDD18_DET | |
| PEX_REF_CLK1_REQN | C18 | PEX_REF_CLK1_REQN | |
| PEX_REF_CLK2_REQN | C19 | PEX_REF_CLK2_REQN | |
| | AD22 | VPGM | |
| DB36 1 BKUP_SB_SYNC_CLKP | E18 | 100M_SYNC_CLK_BYPASS_DP | |
| DB42 1 BKUP_SB_SYNC_CLKN | E19 | 100M_SYNC_CLK_BYPASS_DN | |
| SMC_XTAL_OUT_R | A18 | XTAL_CLK_OUT | |
| SMC_XTAL_IN | A19 | XTAL_CLK_IN | |
| | AD24 | TEST | |
| SMC_BYPASS_CLK_IN | D21 | SMC_BYPASS_CLK_IN | |
| DB25 1 BKUP_SB_PEX_REF_CLKP | E20 | PCIE_BYPASS_CLK_DP | |
| DB28 1 BKUP_SB_PEX_REF_CLKN | D20 | PCIE_BYPASS_CLK_DN | |
| DB46 1 250M_CLK_BYPASS_DP | F20 | 250M_CLK_BYPASS_DP | |
| DB47 1 250M_CLK_BYPASS_DN | F21 | 250M_CLK_BYPASS_DN | |
| DB31 1 BKUP_SB_AV_CLKP | D16 | AV_CLK_BYPASS_DP | |
| DB32 1 BKUP_SB_AV_CLKN | D17 | AV_CLK_BYPASS_DN | |

X861949-001

| | | | |
|---|---|---|---|
| V_EXT_DET | G21 | V_EXT_DET | |
| V_EXT_OK | F22 | V12P0_PWRGD | OUT 64 → DB49 |
| SMC_RST_N_OUT | AF22 | SMC_RST_N_OUT_R | → DB29, DB30 |
| CLK_MONITOR | E17 | SB_CLK_MONITOR | |
| RTC_CLK_OUT | A21 | RTC_CLK_OUT | R120 0 OHM 5% 402 CH → X32K_X2_R OUT 14 |
| EMMC_CLK | AC16 | MMC_CLK | OUT 43 |
| ENET_REF_CLK | C20 | | ENET_REF_CLK OUT 42 |
| SOC_NS_SYNC_CLK_DP | B16 | SOC_SYNC_NS_100M_CLKP | OUT 2 |
| SOC_NS_SYNC_CLK_DN | B17 | SOC_SYNC_NS_100M_CLKN | OUT 2 |
| SOC_NS_AV_CLK_DP | A13 | SOC_AV_NS_100M_CLKP | OUT 2 |
| SOC_NS_AV_CLK_DN | A14 | SOC_AV_NS_100M_CLKN | OUT 2 |
| PEX_REF_CLK2_DP | B20 | SOC_NB_CLKP | NO_M2 R456 0 OHM 5% 402 CH SOC_NB_CLKP_R; NO_M2 R460 0 OHM 5% 402 CH SOC_NB_PEX_SS_100M_CLKP OUT 2,77 |
| PEX_REF_CLK2_DN | B21 | SOC_NB_CLKN | OUT 77; R464 NO_M2 0 OHM 5% 402 CH SOC_NB_CLKN_R; R459 NO_M2 0 OHM 5% 402 CH SOC_NB_PEX_SS_100M_CLKN OUT 2,77 |
| PEX_REF_CLK1_DP | C14 | ENET_CLKP | NO_M2 R411 0 OHM 5% 402 CH ENET_CLKP_R; NO_M2 R423 0 OHM 5% 402 CH ENET_PEX_SS_100M_CLKP OUT 42,77 |
| PEX_REF_CLK1_DN | C15 | ENET_CLKN | OUT 77; R419 NO_M2 0 OHM 5% 402 CH ENET_CLKN_R; R428 NO_M2 0 OHM 5% 402 CH ENET_PEX_SS_100M_CLKN OUT 42,77 |
| PEX_REF_CLK0_DP | A15 | SOC_CLKP | NO_M2 R427 0 OHM 5% 402 CH SOC_CLKP_R; NO_M2 R443 0 OHM 5% 402 CH SOC_PEX_SS_100M_CLKP OUT 2,77 |
| PEX_REF_CLK0_DN | A16 | SOC_CLKN | R435 NO_M2 0 OHM 5% 402 CH SOC_CLKN_R; R449 NO_M2 0 OHM 5% 402 CH SOC_PEX_SS_100M_CLKN OUT 2,77 |

BGA515

V_12P0_GATED
C212 1 UF 10% 16 V X5R 603
R494 10 KOHM 1% CH 402
R493 1.13 KOHM 1% CH 402
R494/V_EXT_DET IS NOW CONNECTED TO V_12P0_GATED
NOMINAL TRIP POINT = 9.55V

R476 1 KOHM 1% 402 CH SMC_RST_N OUT 36,73
C593 270 PF 10% 50 V X7R 402
KEEP DB29 AND DB30 CLOSE TO EACH OTHER IN LAYOUT
R112 69.8 KOHM 1% EMPTY 402

M2_ONLY R432 10 KOHM 1% CH 402
M2_ONLY R462 10 KOHM 1% CH 402
TODO; IF SMCFW CHANGES BASED ON M2_ONLY STUFFING OPTION, THEN TURN OFF CLOCKS INSTEAD OF TERMINATING

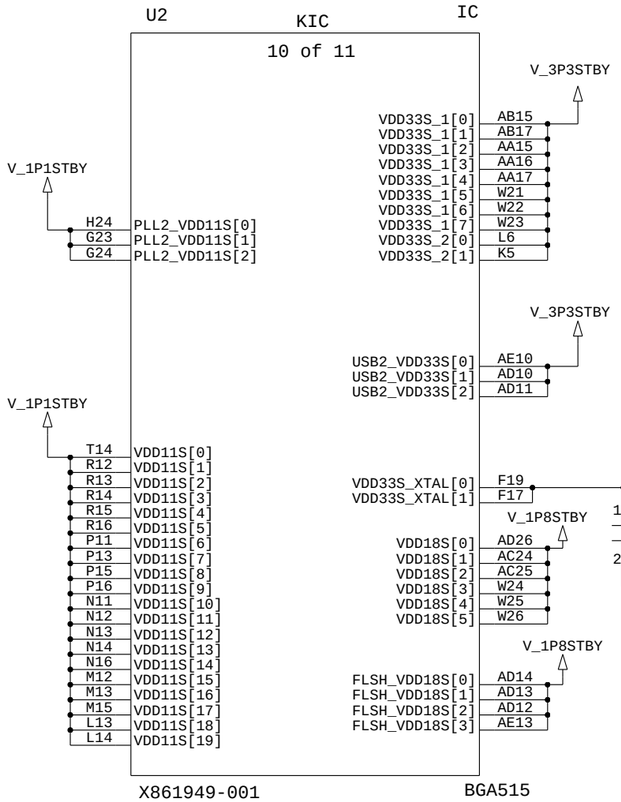
DO NOT STUFF RESISTORS IF BOTH HDD AND M2 ARE STUFFED

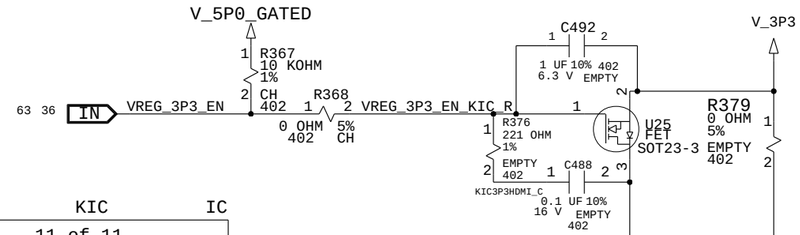
| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| X861949-007 | IC | U2 | IC,KRAKEN SB,BGA515 | KIC_RETAIL |
| X861949-008 | IC | U2 | IC,KRAKEN SB,BGA515 | KIC_DEV |
| X861949-008 | EMPTY | U2 | IC,KRAKEN SB,BGA515 | KIC_EMPTY |

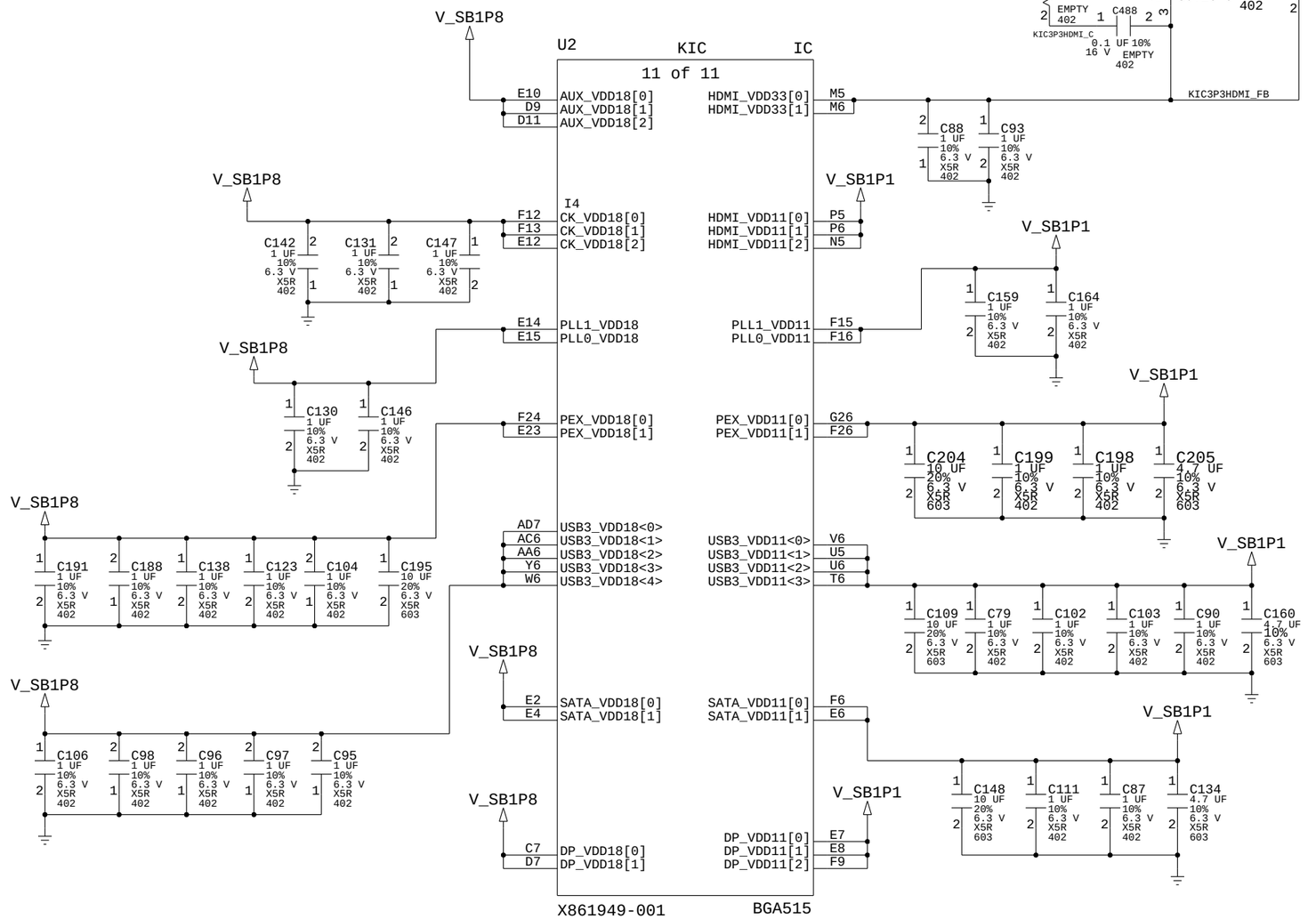
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 39/82 | 39/82 | G-R | 0.991 |

# KIC: POWER

FET PREVENTS LEAKAGE THROUGH KRAKEN
FROM STANDBY RAILS TO 3P3 RAIL

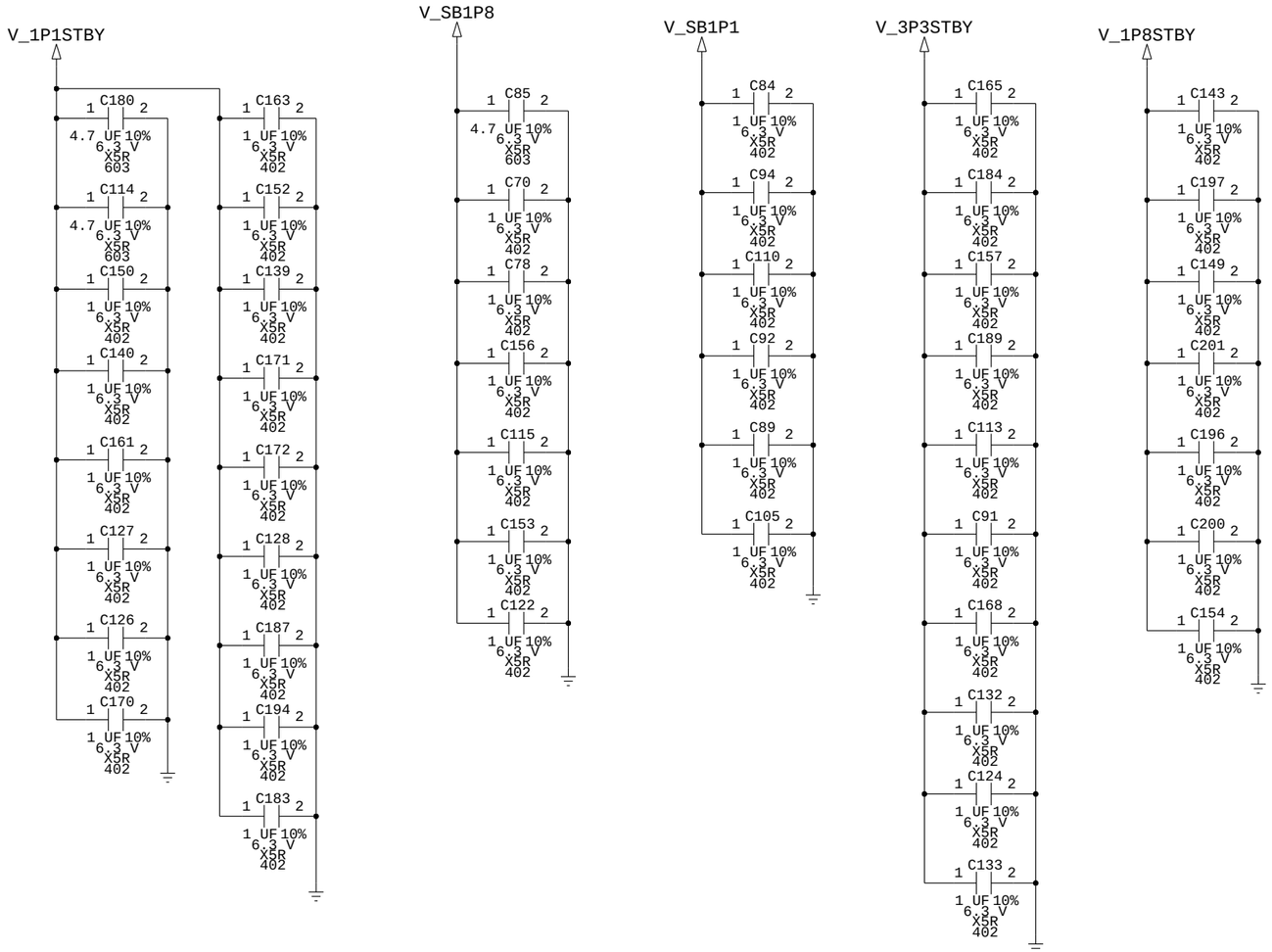


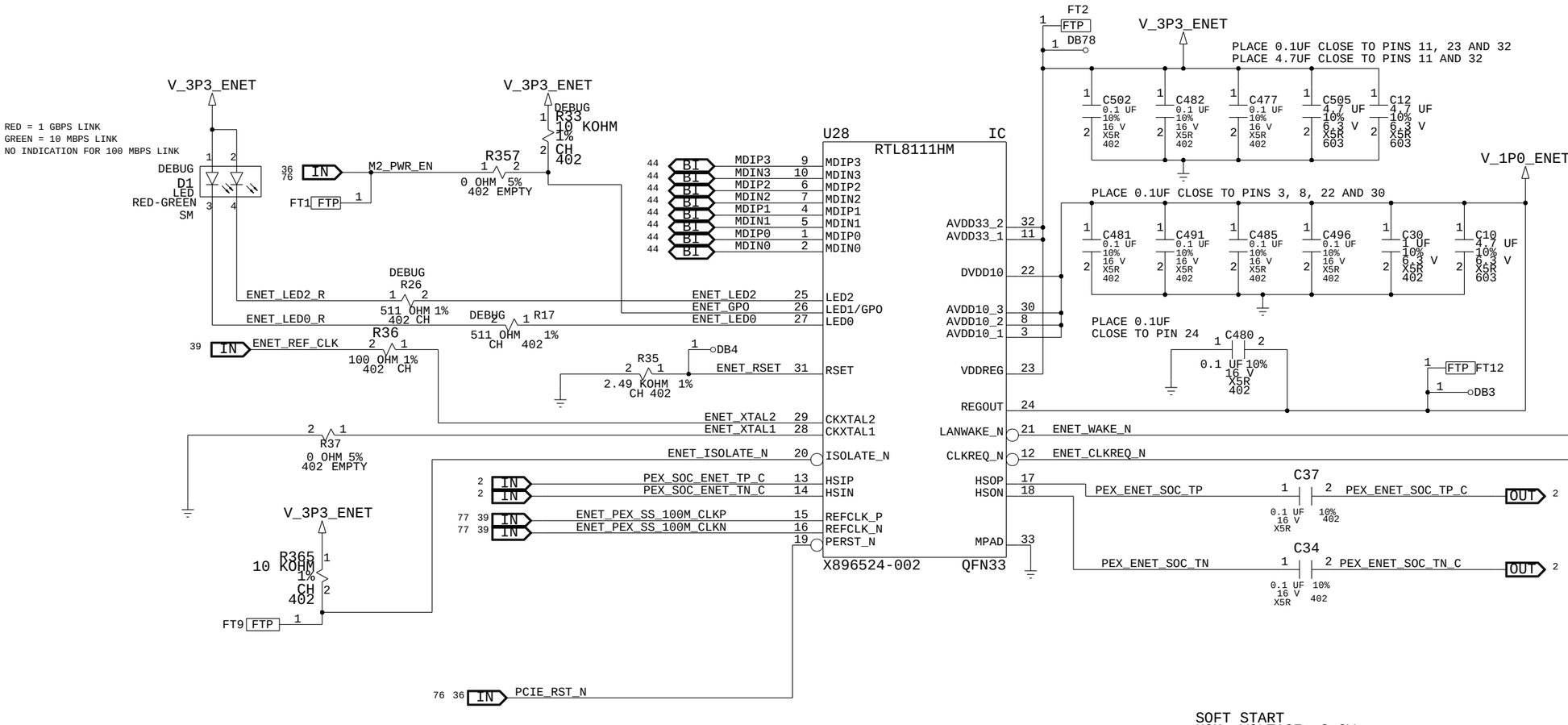

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 40/82 | 40/82 | G-R | 0.991 |

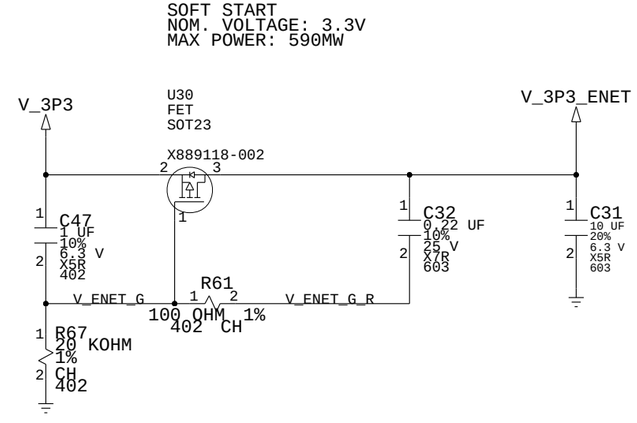
# KIC: DECOUPLING




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 41/82 | 41/82 | G-R | 0.991 |

# ETHERNET CONTROLLER

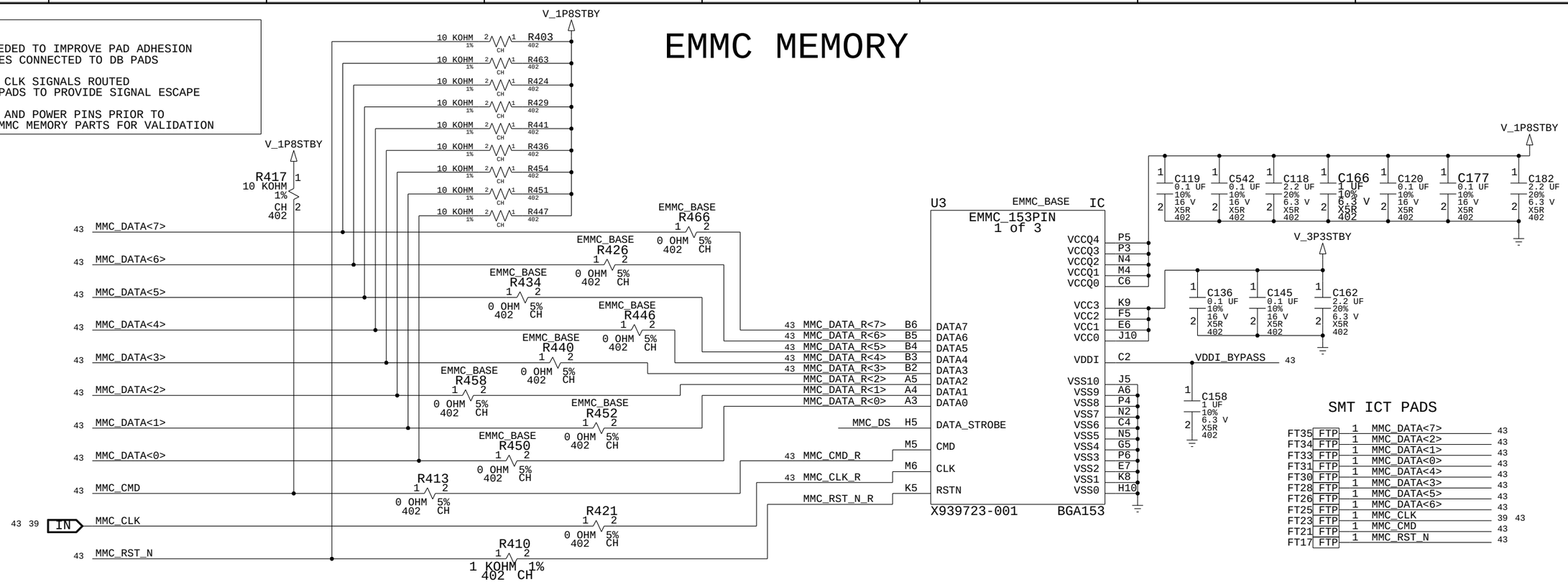


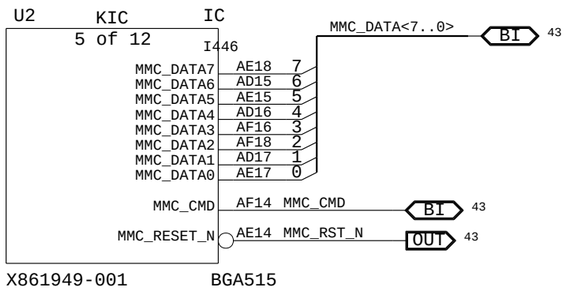



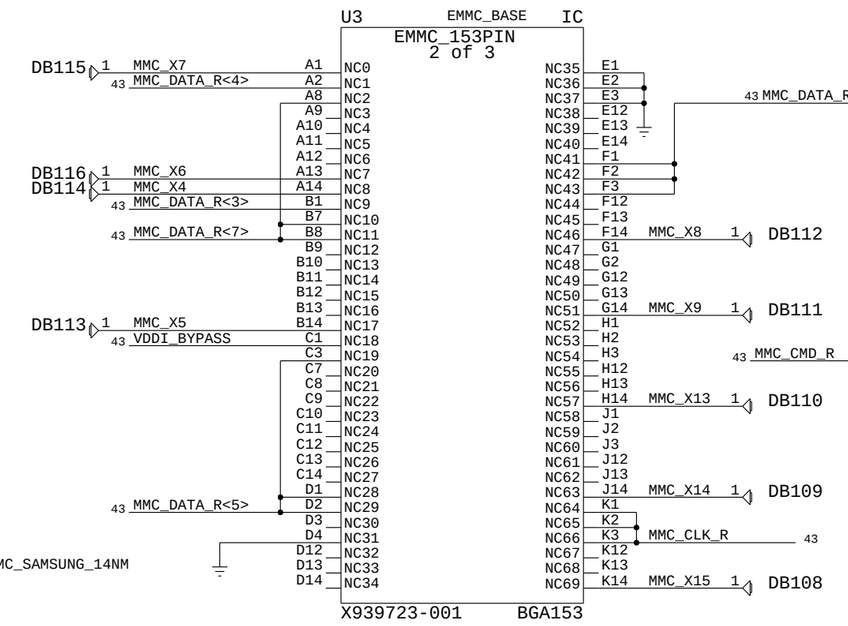

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 42/82 | 42/82 | G-R | 0.991 |

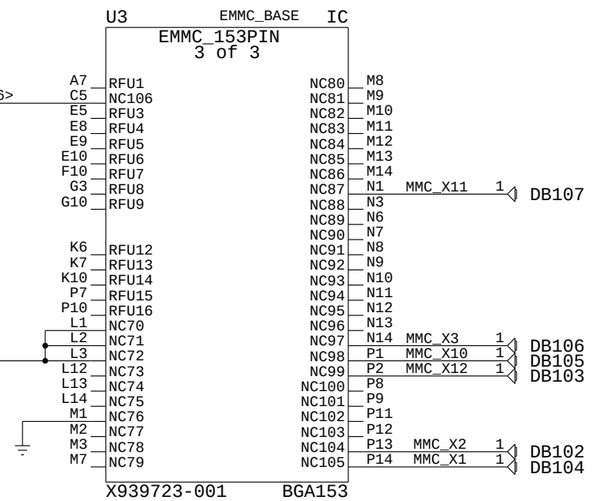
# EMMC MEMORY

NOTES:
1. MMC_X NETS ARE NEEDED TO IMPROVE PAD ADHESION USING 13 MIL TRACES CONNECTED TO DB PADS
2. EMMC DATA,CMD AND CLK SIGNALS ROUTED THROUGH MULTIPLE PADS TO PROVIDE SIGNAL ESCAPE
3. VERIFY ALL SIGNAL AND POWER PINS PRIOR TO CONSIDERING NEW EMMC MEMORY PARTS FOR VALIDATION



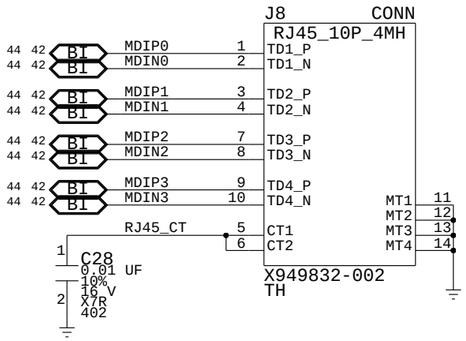



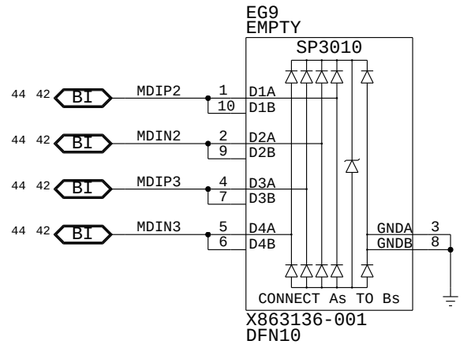



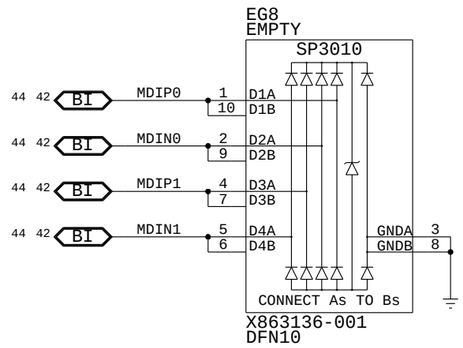

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| X934261-001 | IC | U3 | MEM,8GB,EMMC,FLASH,15NM,BGA153,TOSHIBA | EMMC_TOSHIBA_15NM |
| X945540-002 | IC | U3 | MEM,8GB,EMMC,FLASH,14NM,BGA153,SAMSUNG | EMMC_SAMSUNG_14NM |
| X939723-001 | IC | U3 | MEM,8GB,EMMC,FLASH,16NM,BGA153,HYNIX | EMMC_HYNIX_16NM |
| X800617-001 | RES | R426,R434,R440,R446,R450,R452,R458,R466 | 49.9 OHM 1%,1/16W,0402 | EMMC_TOSHIBA_15NM,EMMC_SAMSUNG_14NM |
| X809936-001 | RES | R426,R434,R440,R446,R450,R452,R458,R466 | 100 OHM 1%,1/16W,0402 | EMMC_HYNIX_16NM |

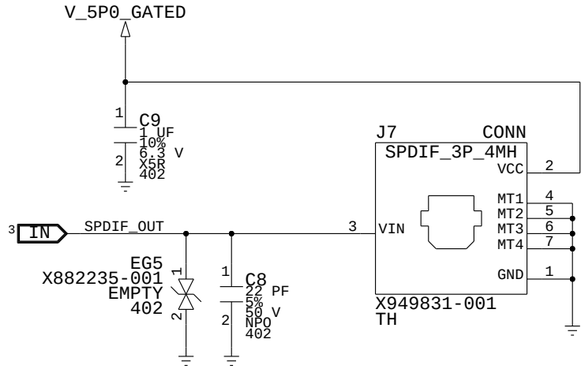
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 43/82 | 43/82 | G-R | 0.991 |

# CONN: RJ45,TOSLINK




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 44/82 | 44/82 | G-R | 0.991 |

# CONN: USB (FRONT & REAR)

EG26
ESD ARRAY
SP3010
X863136-001
DFN10

EG25
ESD ARRAY
SP3010
X863136-001
DFN10

EG27
EMPTY
SP3010
X863136-001
DFN10

REAR 1 (KINECT)
J4 CONN
USB3_9P_4MH
X949829-004
TH
TX/RX FROM
HOST PERSPECTIVE

REAR 0
J5 CONN
USB3_9P_4MH
X949829-004
TH
TX/RX FROM
HOST PERSPECTIVE

FRONT USB
MTG6
STANDOFF
M1015028-001
SM
MTG7
STANDOFF
M1015028-001
SM

FRONT USB CONNECTION
J27
2X10RCPT
SM
R348
2.49 KOHM 1%
402 CH
EG14
X882235-001
EMPTY
402

U79 IC
PTN36001
X949696-001
QFN12

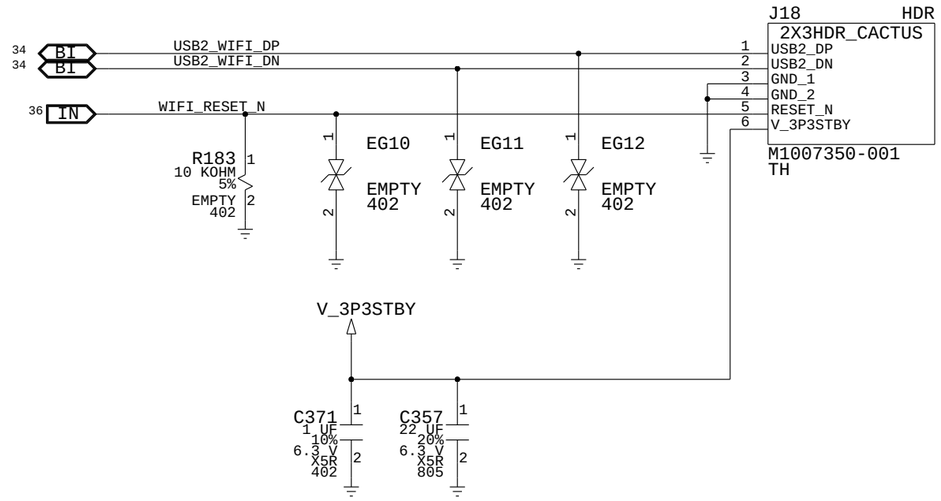
FILT U76
ECMF02_2HSMX6
X900532-002
DFN6

FILT U75
ECMF02_2HSMX6
X900532-002
DFN6

FILT U80
ECMF02_2HSMX6
X900532-002
DFN6

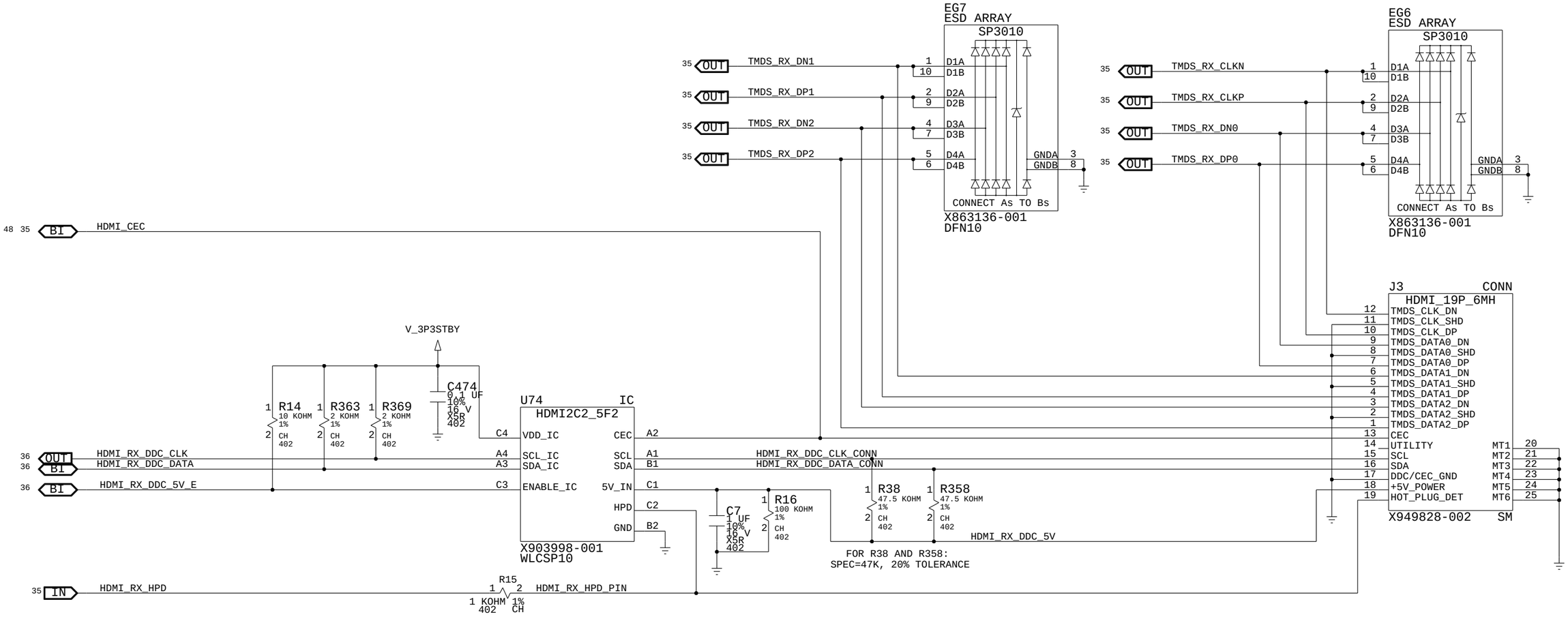
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 45/82 | 45/82 | G-R | 0.99 |

# CONN: WIFI




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 46/82 | 46/82 | G-R | 0.991 |

# CONN: HDMI IN




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 47/82 | 47/82 | G-R | 0.991 |

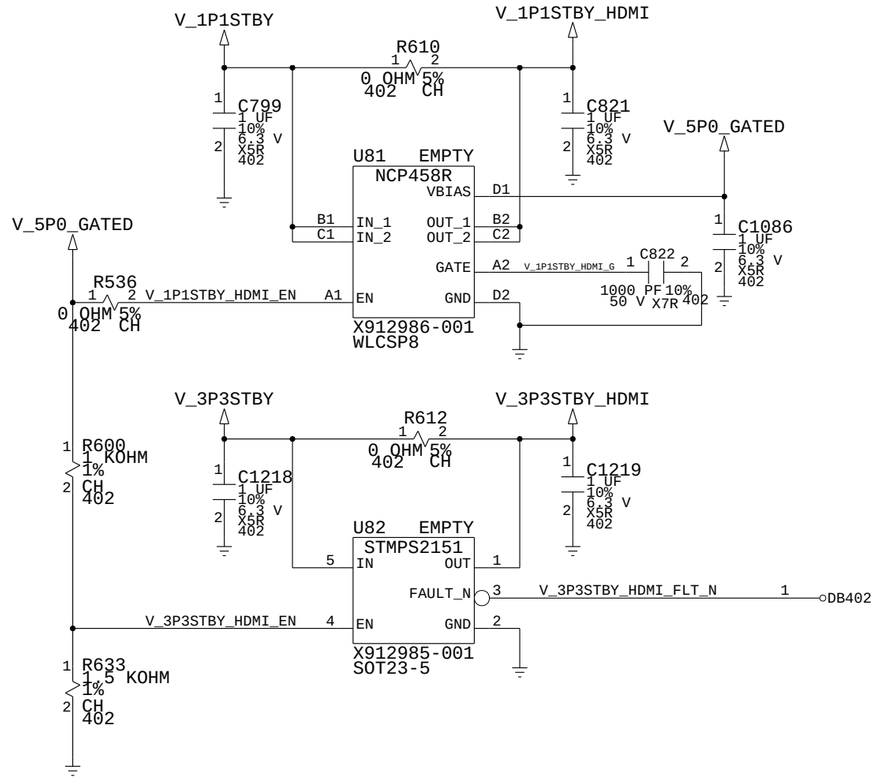
# CONN: HDMI OUT

| TDP158 CONTROL SLAVE ADDRESS | | | |
|---|---|---|---|
| SELECTED | A1/A0 | 7-BIT ADDRESS | 8-BIT ADDRESS |
| --> | 00 | 0X5E | BC/BD |
| | 01 | 0X5D | BA/BB |
| | 10 | 0X5C | B8/B9 |
| | 11 | 0X5B | B6/B7 |

LANE_SWAP MUST BE ENABLED DUE TO INVERTED TMDS PIN-ORDER IMPLEMENTATION (0X09[7]=1)

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 48/82 | 48/82 | G-R | 0.991 |

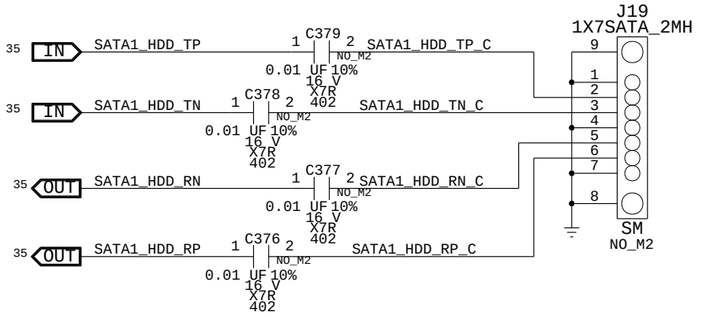
# HDMI LOAD SWITCHES



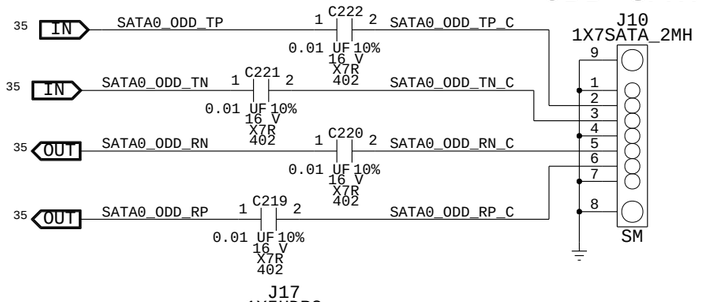

| RE-DRIVER | U29 | R610 | R612 | U81 | U82 |
|---|---|---|---|---|---|
| PG1.0 | M1003966-001 | EMPTY | EMPTY | STUFF | STUFF |
| PG1.1 | M1003966-002 | STUFF | STUFF | EMPTY | EMPTY |

# CONN: ODD & HDD

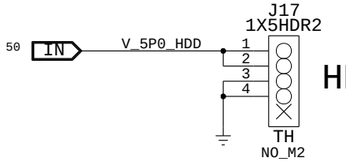
## HDD SATA



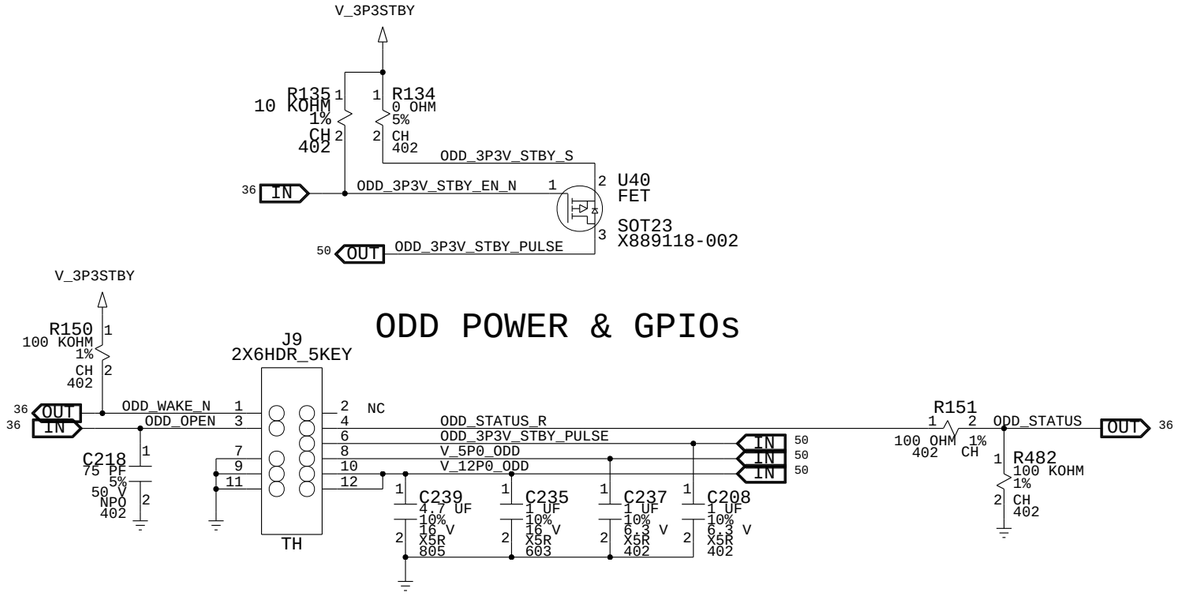

## ODD SATA




## HDD POWER

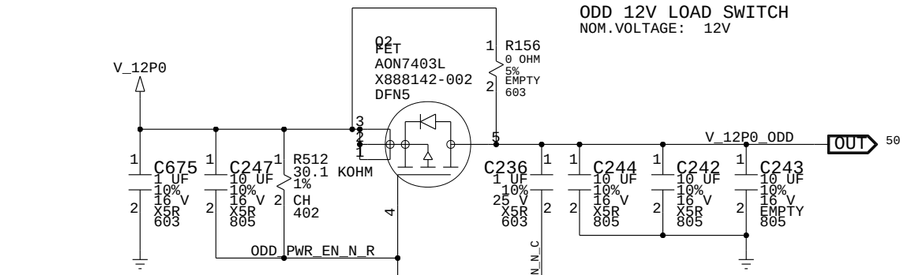



## ODD POWER & GPIOs



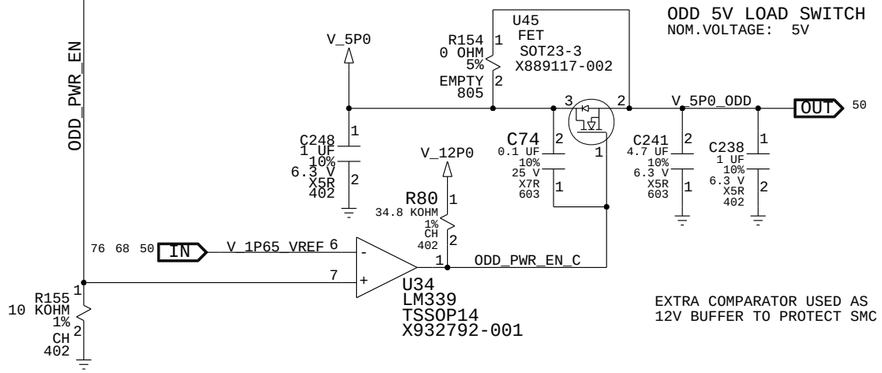

## ODD 12V LOAD SWITCH

NOM.VOLTAGE: 12V



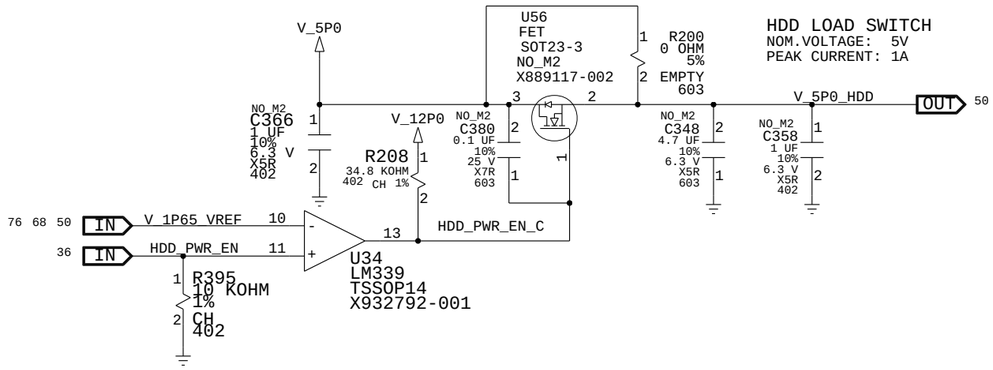

## ODD 5V LOAD SWITCH

NOM.VOLTAGE: 5V



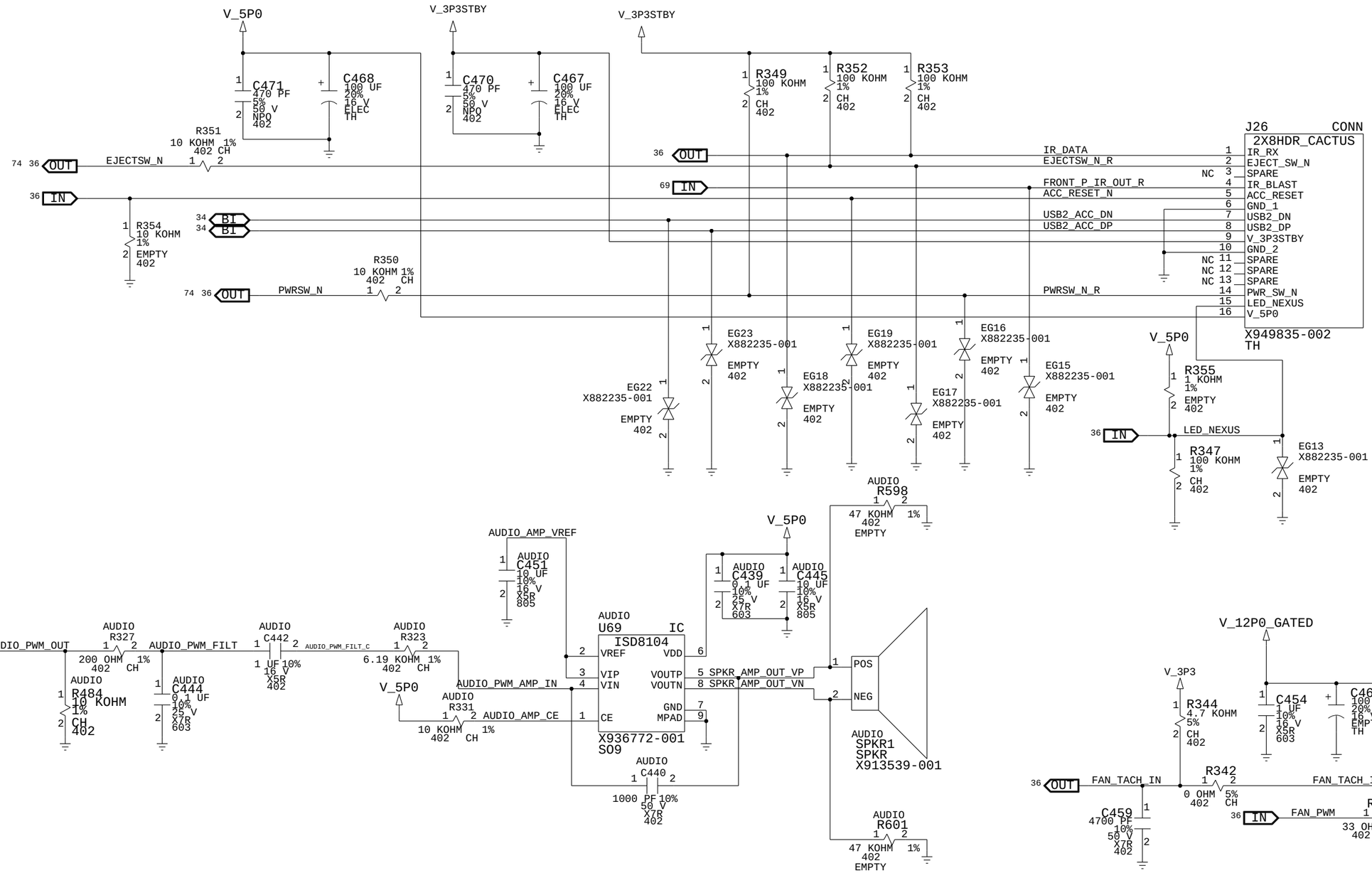

## HDD LOAD SWITCH

NOM.VOLTAGE: 5V
PEAK CURRENT: 1A



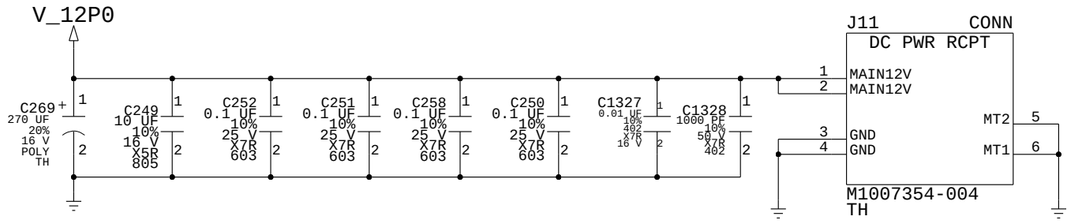

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 50/82 | 50/82 | G-R | 0.991 |

# CONN: FRONT PANEL,FAN, AUDIO




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 51/82 | 51/82 | G-R | 0.99 |

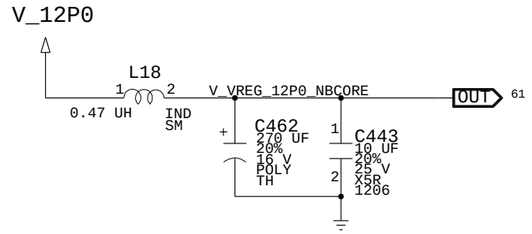
# CONN: POWER



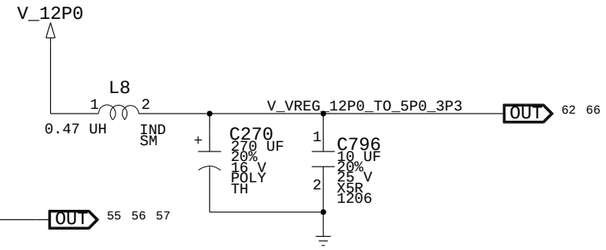

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 52/82 | 52/82 | G-R | 0.991 |

# VREGS: INPUT AND OUTPUT FILTERS

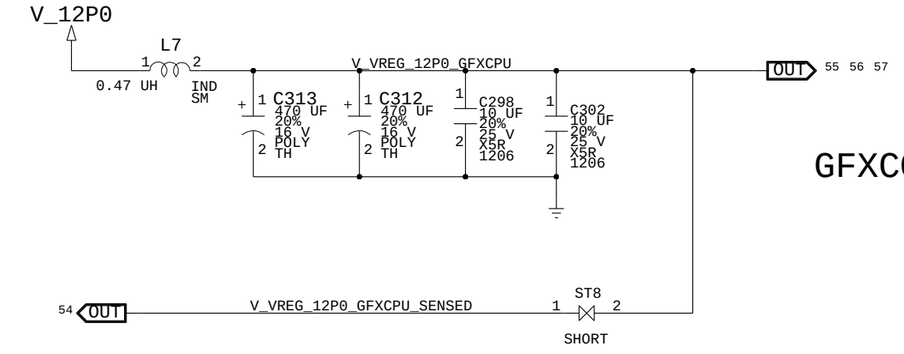
## NBCORE INPUT FILTER



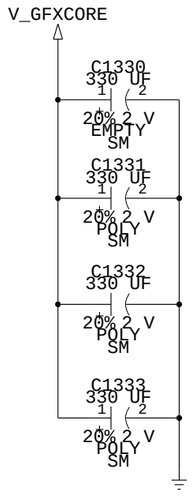

## V_5P0 AND V_3P3 INPUT FILTER



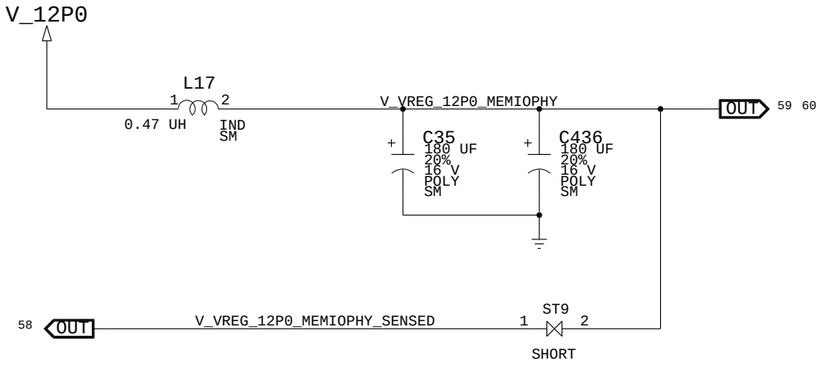

## GFX/CPU INPUT FILTER




## GFXCORE OUTPUT FILTER

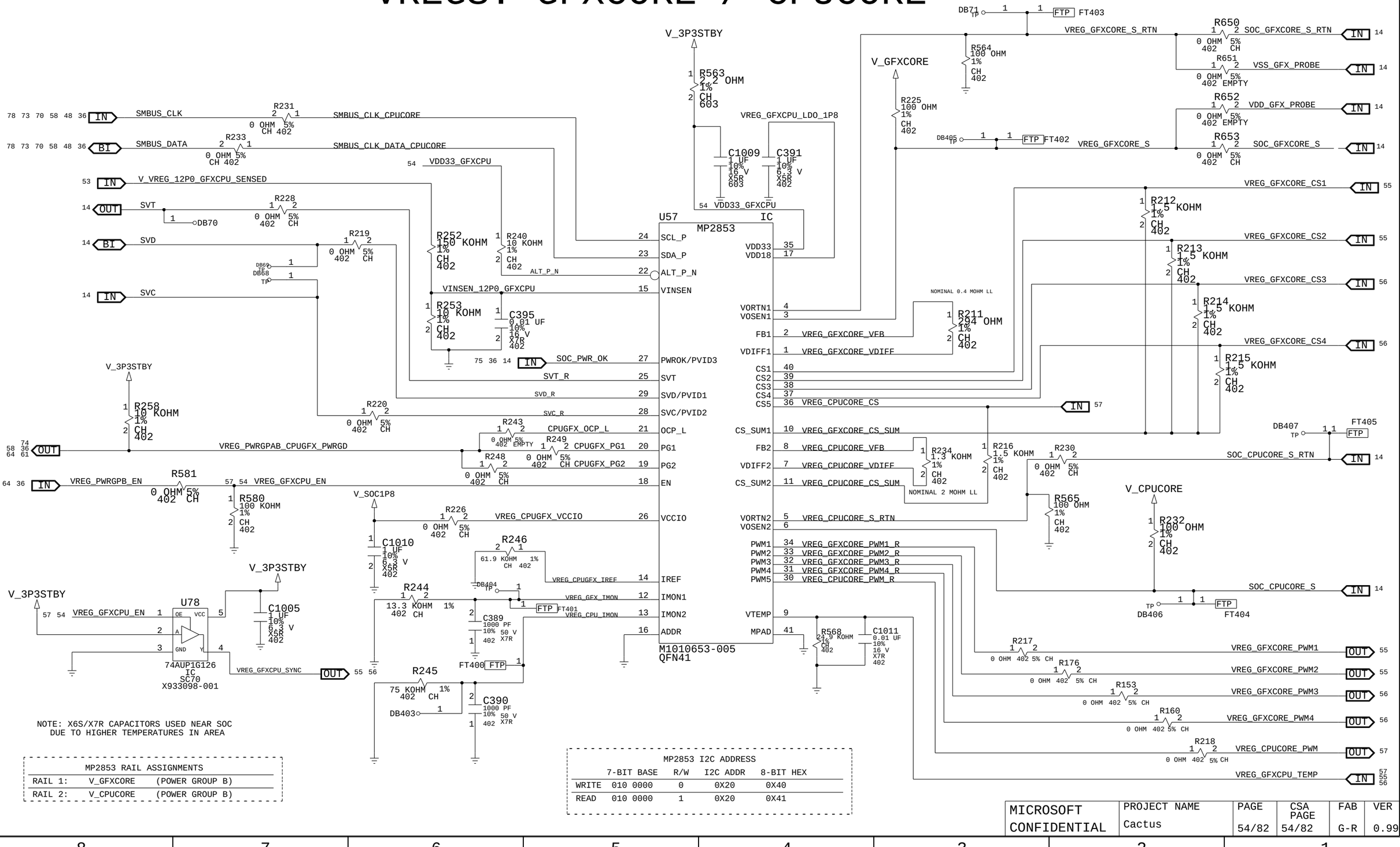



## MEMIO/MEMPHY INPUT FILTER




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 53/82 | 53/82 | G-R | 0.991 |

# VREGS: GFXCORE / CPUCORE




NOTE: X6S/X7R CAPACITORS USED NEAR SOC DUE TO HIGHER TEMPERATURES IN AREA

MP2853 RAIL ASSIGNMENTS

| | | |
|---|---|---|
| RAIL 1: | V_GFXCORE | (POWER GROUP B) |
| RAIL 2: | V_CPUCORE | (POWER GROUP B) |

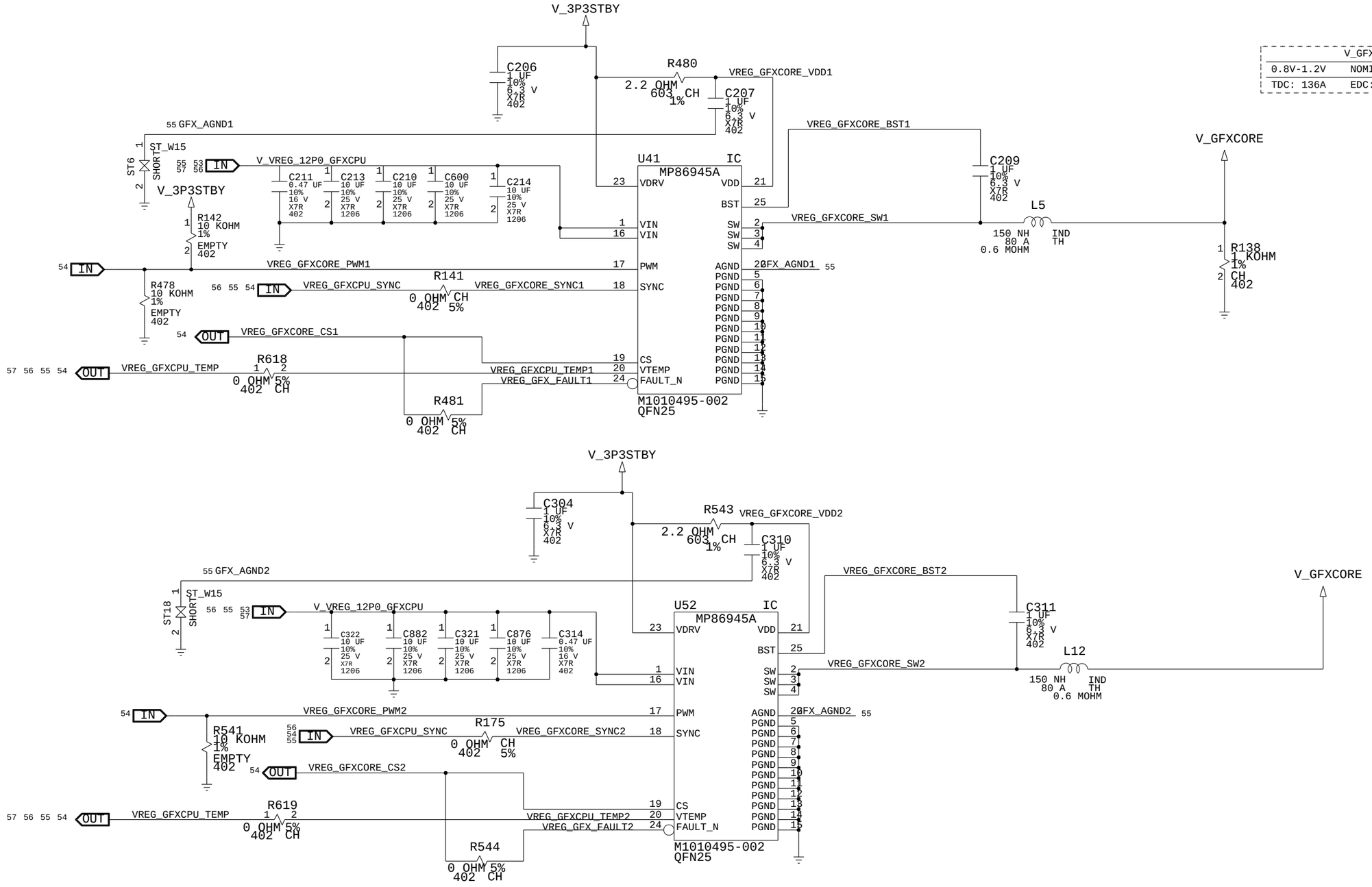
MP2853 I2C ADDRESS

| | 7-BIT BASE | R/W | I2C ADDR | 8-BIT HEX |
|---|---|---|---|---|
| WRITE | 010 0000 | 0 | 0X20 | 0X40 |
| READ | 010 0000 | 1 | 0X20 | 0X41 |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 54/82 | 54/82 | G-R | 0.991 |

# VREGS: GFXCORE OUTPUT PHASE 1 & 2

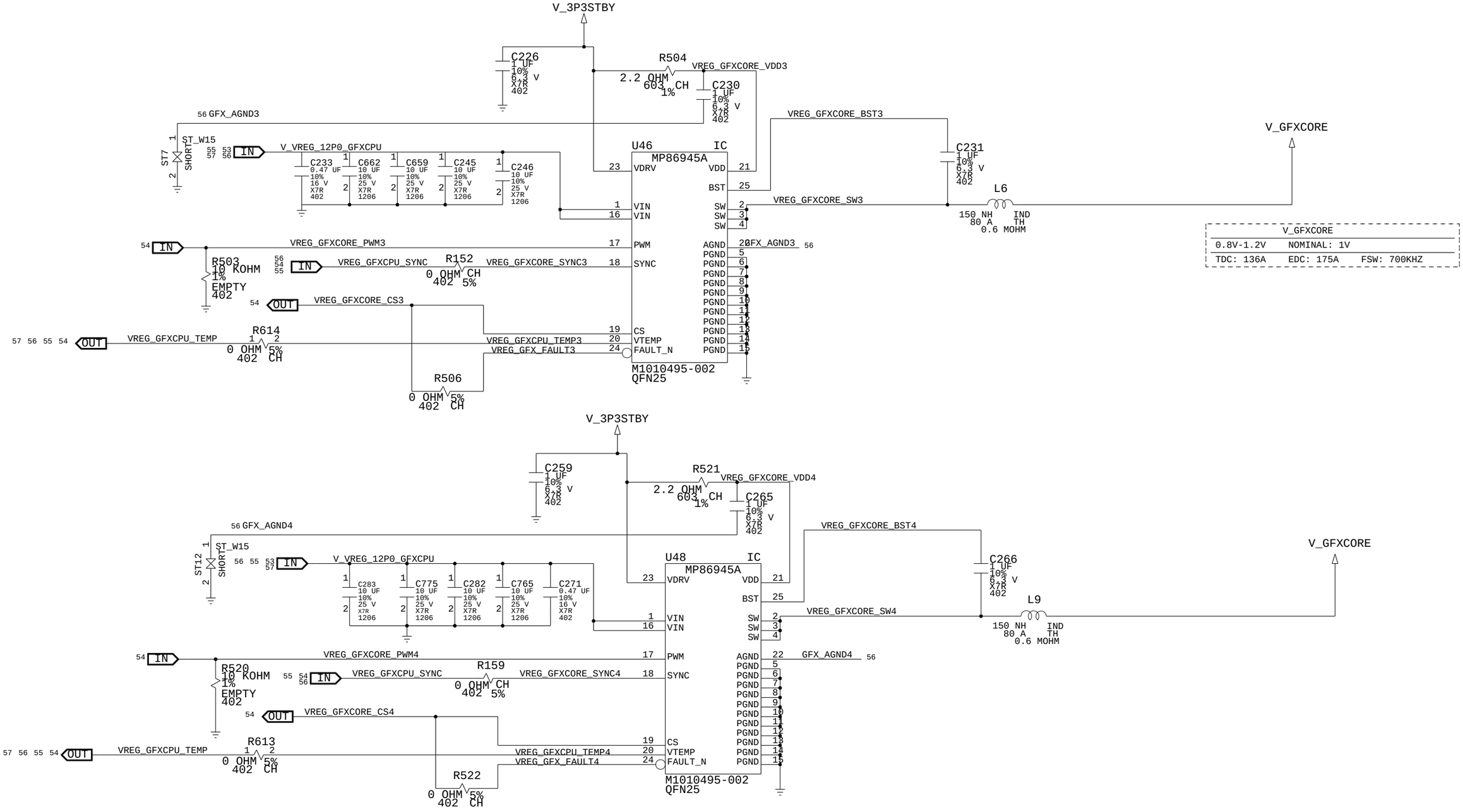
| V_GFXCORE | | |
|---|---|---|
| 0.8V-1.2V | NOMINAL: 1V | |
| TDC: 136A | EDC: 175A | FSW: 700KHZ |




NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 55/82 | 55/82 | G-R | 0.991 |

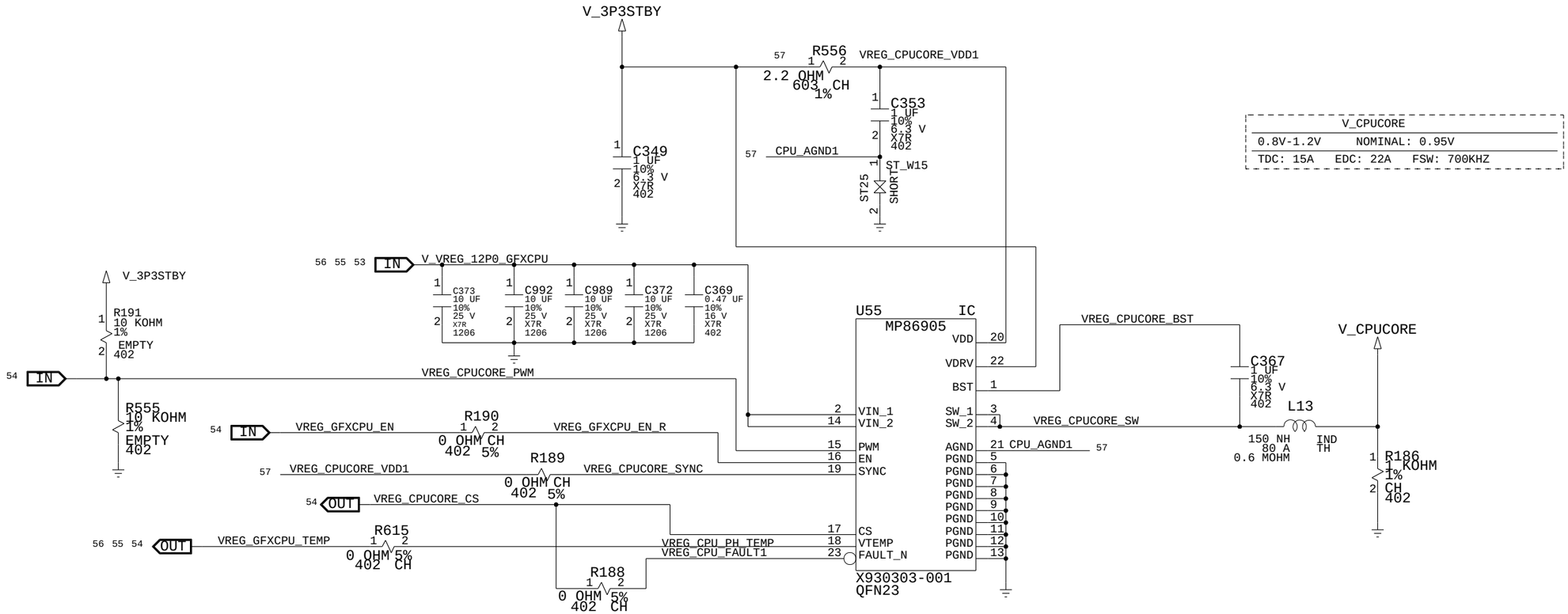
# VREGS: GFXCORE OUTPUT PHASE 3 & 4




NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 56/82 | 56/82 | G-R | 0.991 |

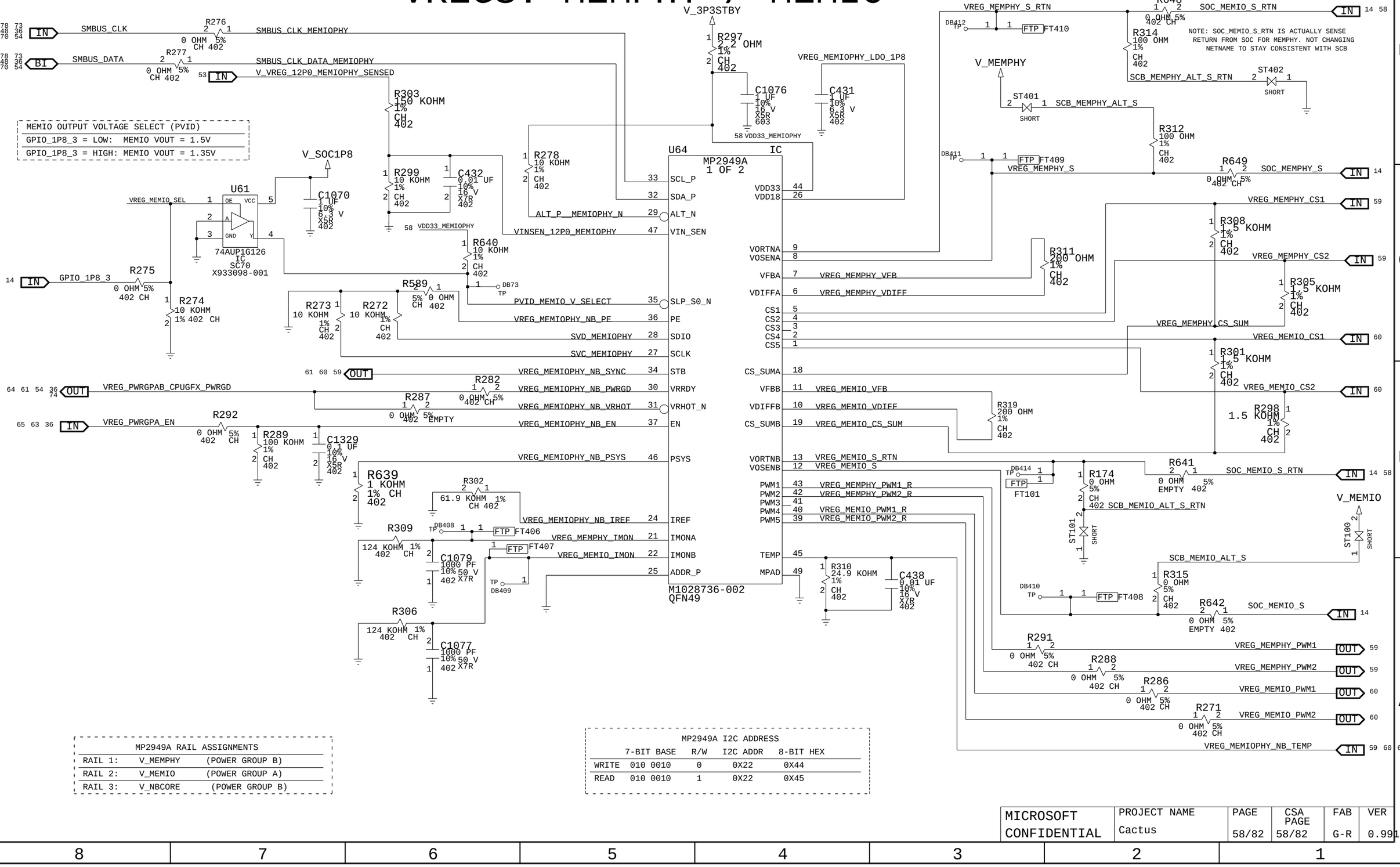
# VREGS: CPUCORE OUTPUT PHASE




NOTE: X6S/X7R CAPACITORS USED NEAR SOC
DUE TO HIGHER TEMPERATURES IN AREA

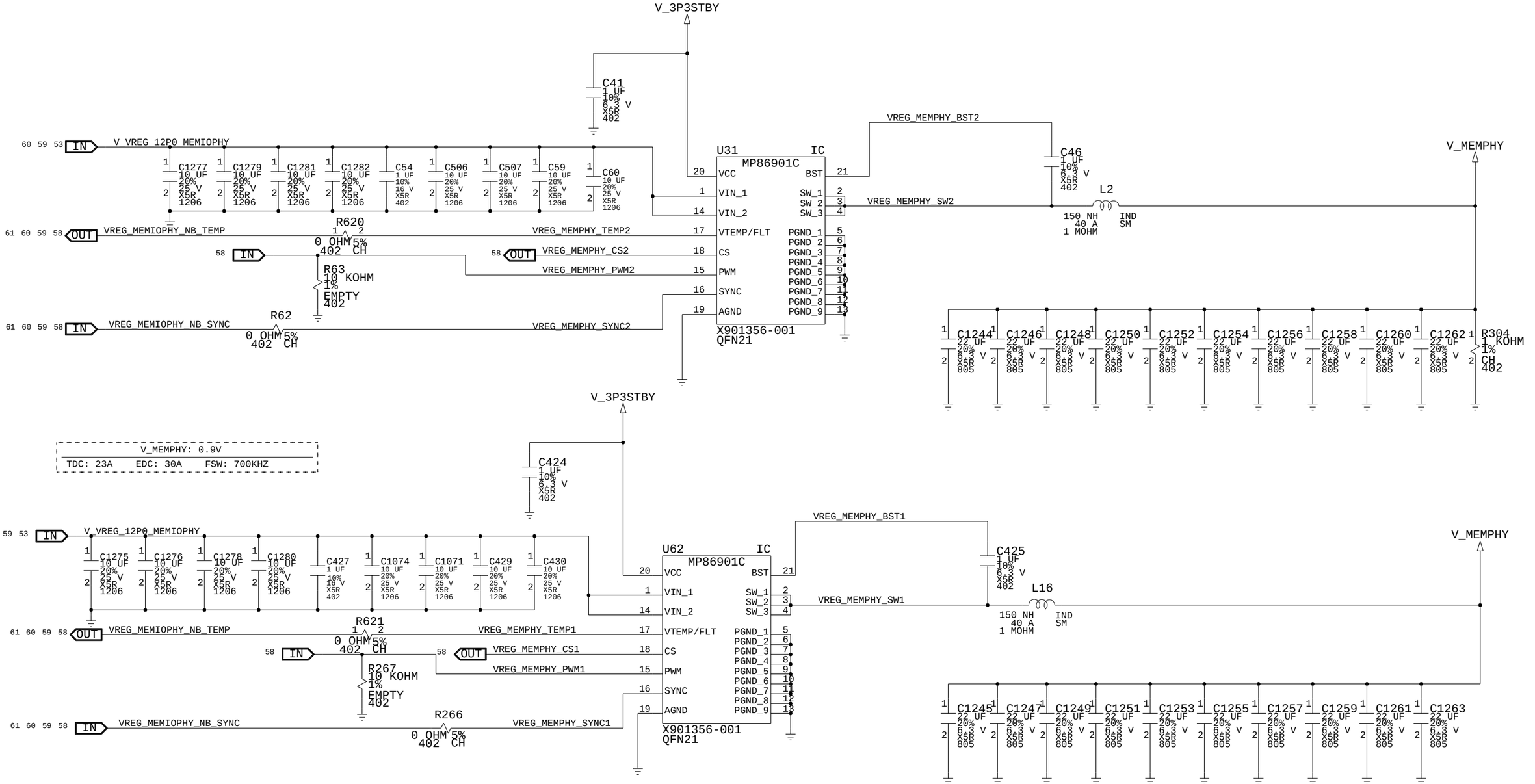
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 57/82 | 57/82 | G-R | 0.991 |

# VREGS: MEMPHY / MEMIO




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 58/82 | 58/82 | G-R | 0.991 |

# VREGS: MEMPHY OUTPUT



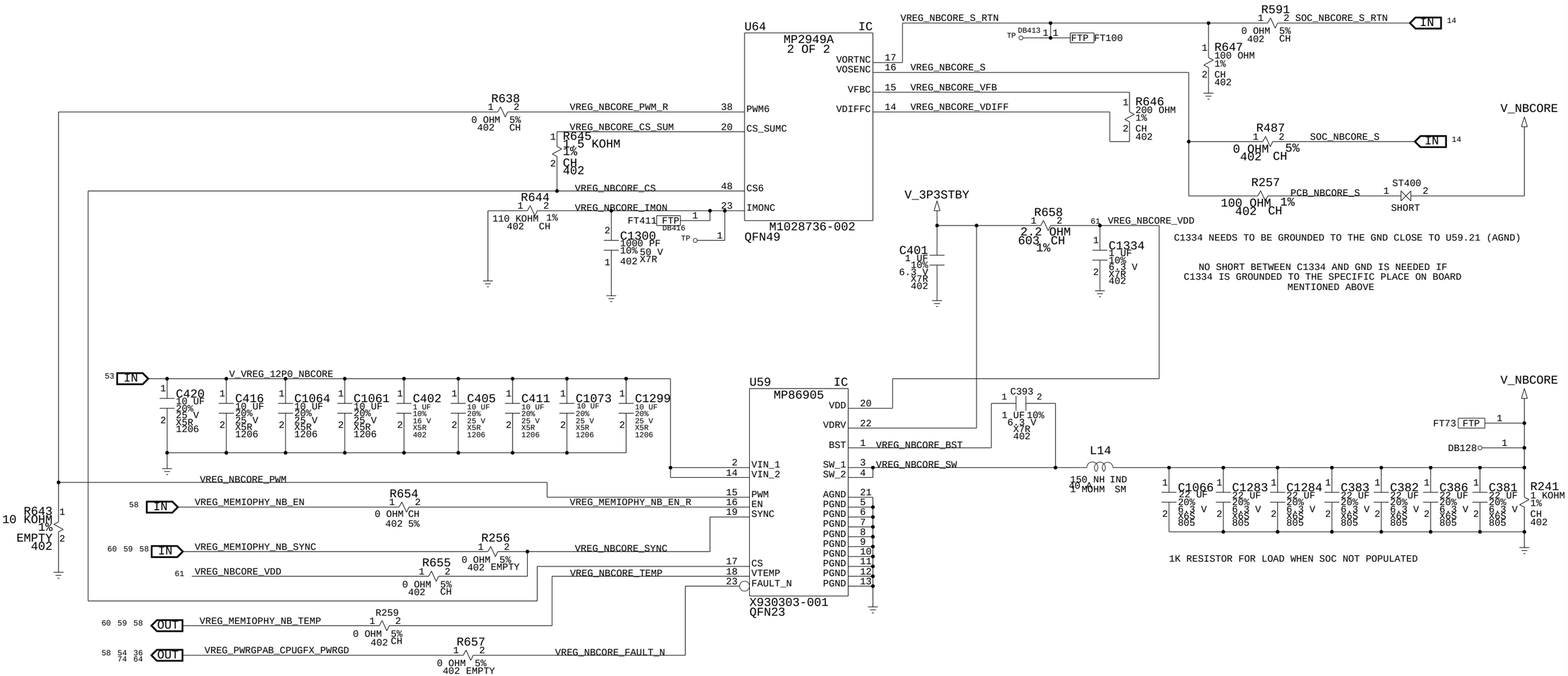

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 59/82 | 59/82 | G-R | 0.991 |

# VREGS: MEMIO OUTPUT

| V_MEMIO | | | |
|---|---|---|---|
| 1.35-1.5V | NOMINAL: 1.5V | | |
| TDC(RETAIL): | 20A | EDC(RETAIL): 23A | FSW: 700KHZ |
| TDC(XDK): | 25A | EDC(XDK): 30A | FSW: 700KHZ |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 60/82 | 60/82 | G-R | 0.991 |

# VREGS: NBCORE

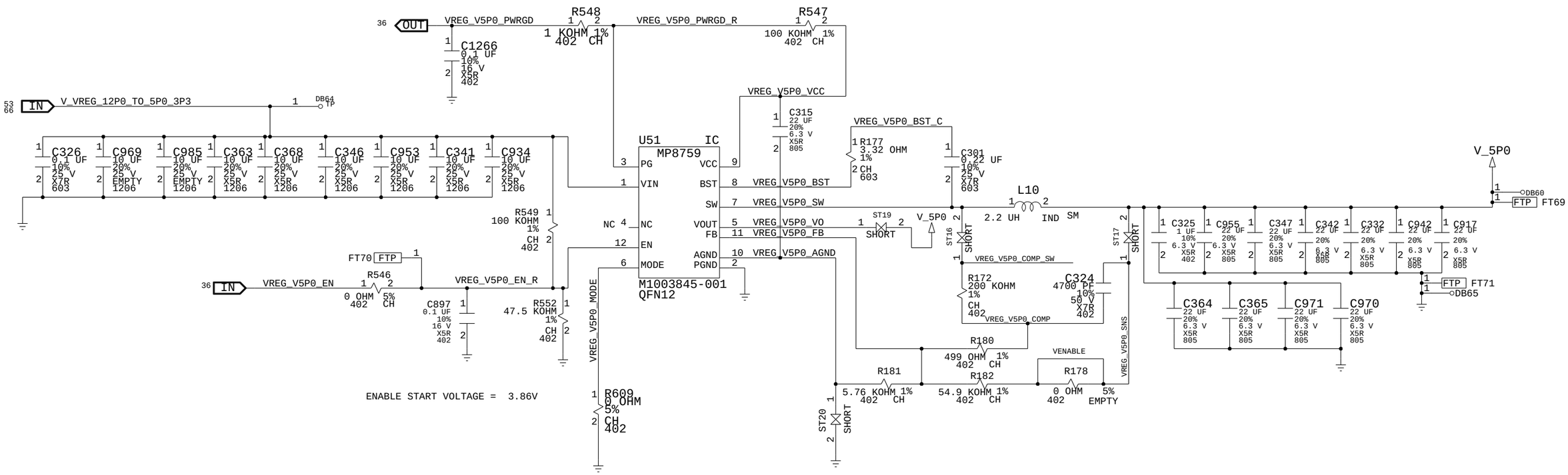



| V_NBCORE | | |
|---|---|---|
| 0.8V-1.2V | NOMINAL: 0.95V | |
| TDC: 16A | EDC: 25A | FSW: 700KHZ |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 61/82 | 61/82 | G-R | 0.991 |

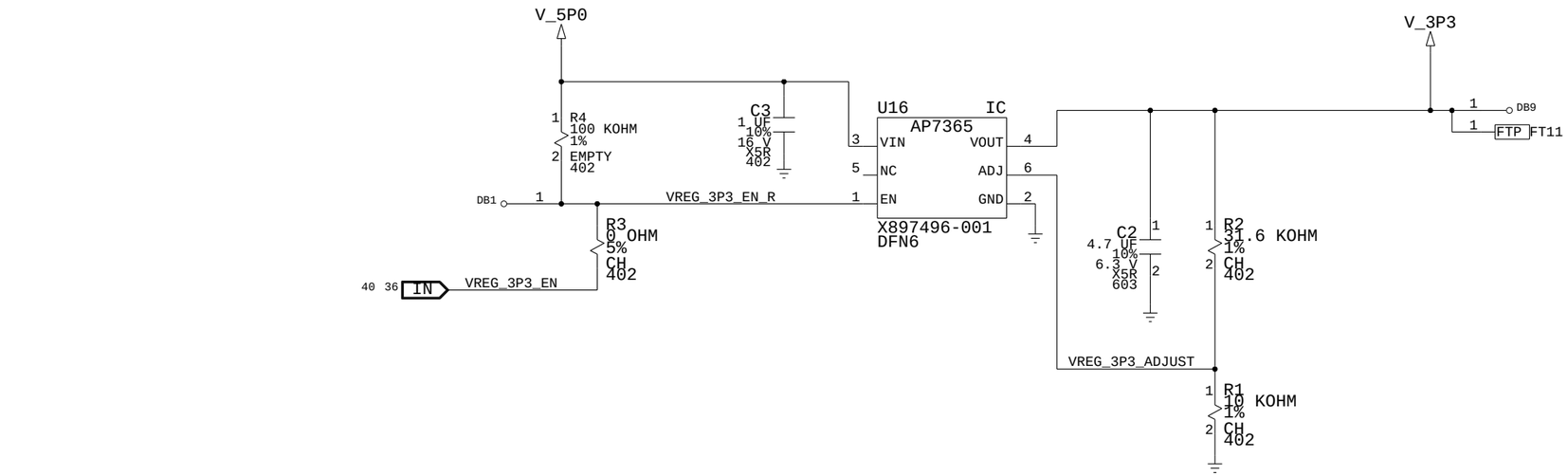
# VREGS: V5P0

V_5P0
NOM. VOLTAGE: 5.1



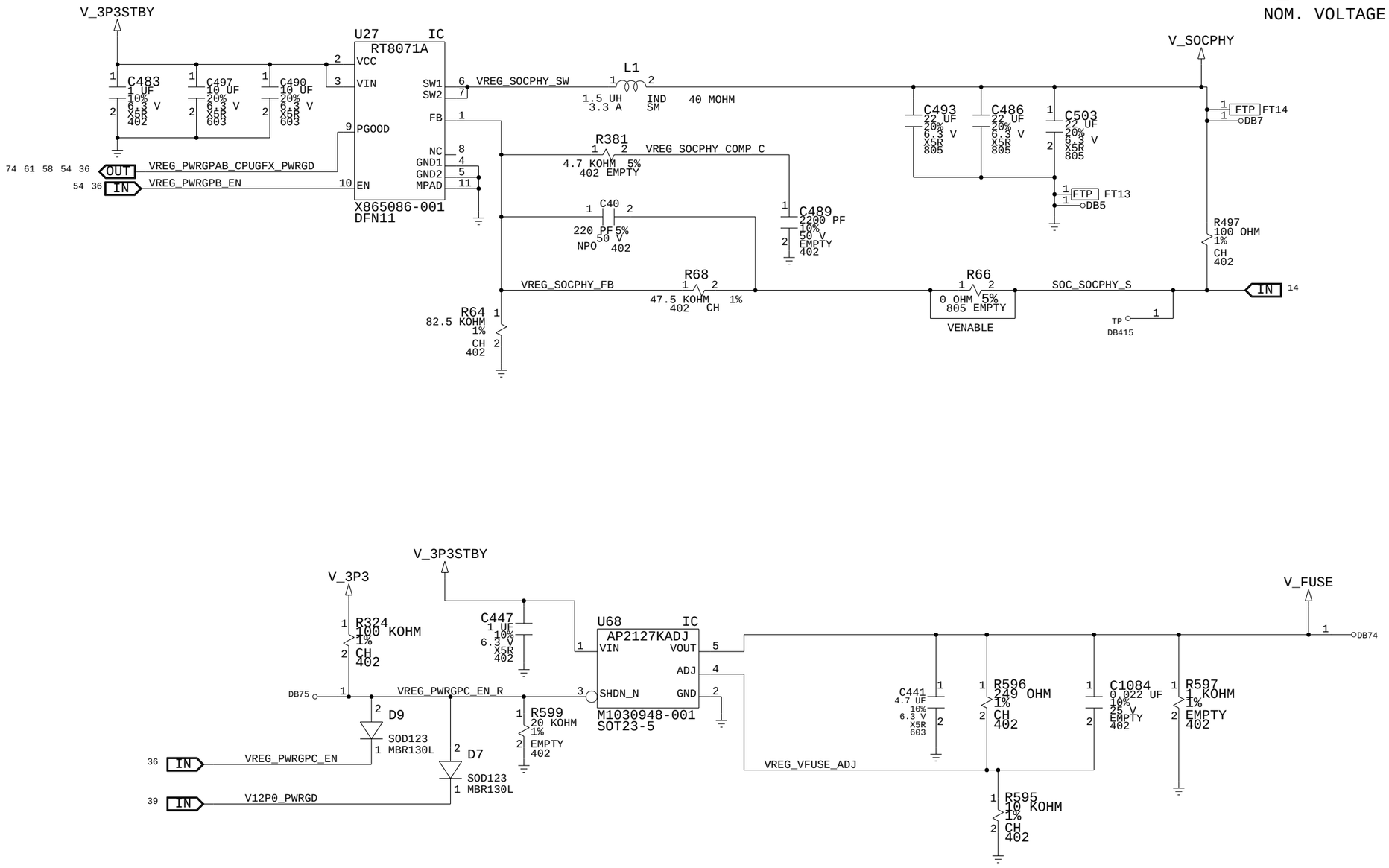

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 62/82 | 62/82 | G-R | 0.991 |

# VREGS: V3P3, VSOC1P8



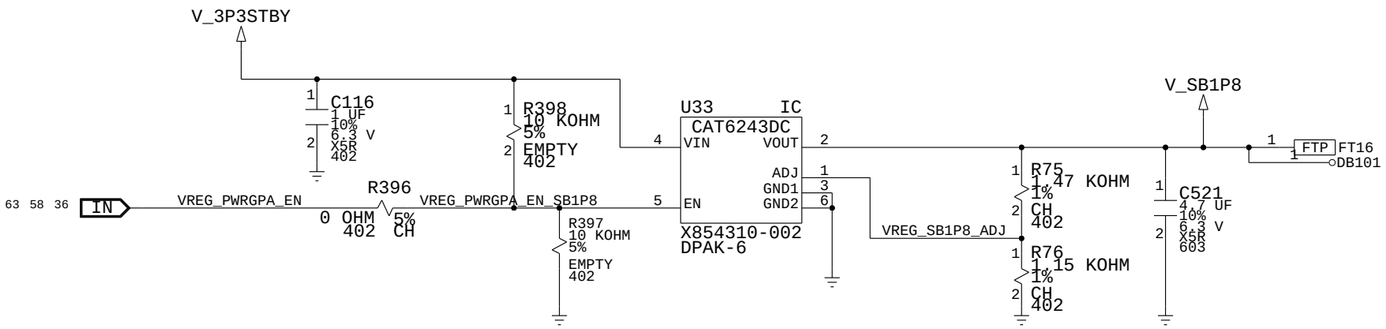

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 63/82 | 63/82 | G-R | 0.991 |

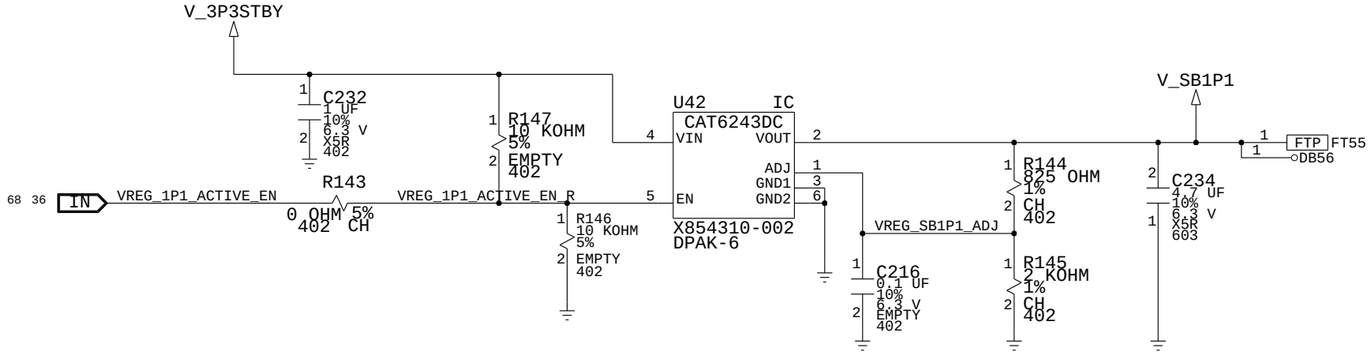
# VREGS: VSOCPHY/VFUSE




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 64/82 | 64/82 | G-R | 0.991 |

# VREGS: V_SB1P8, V_SB1P1

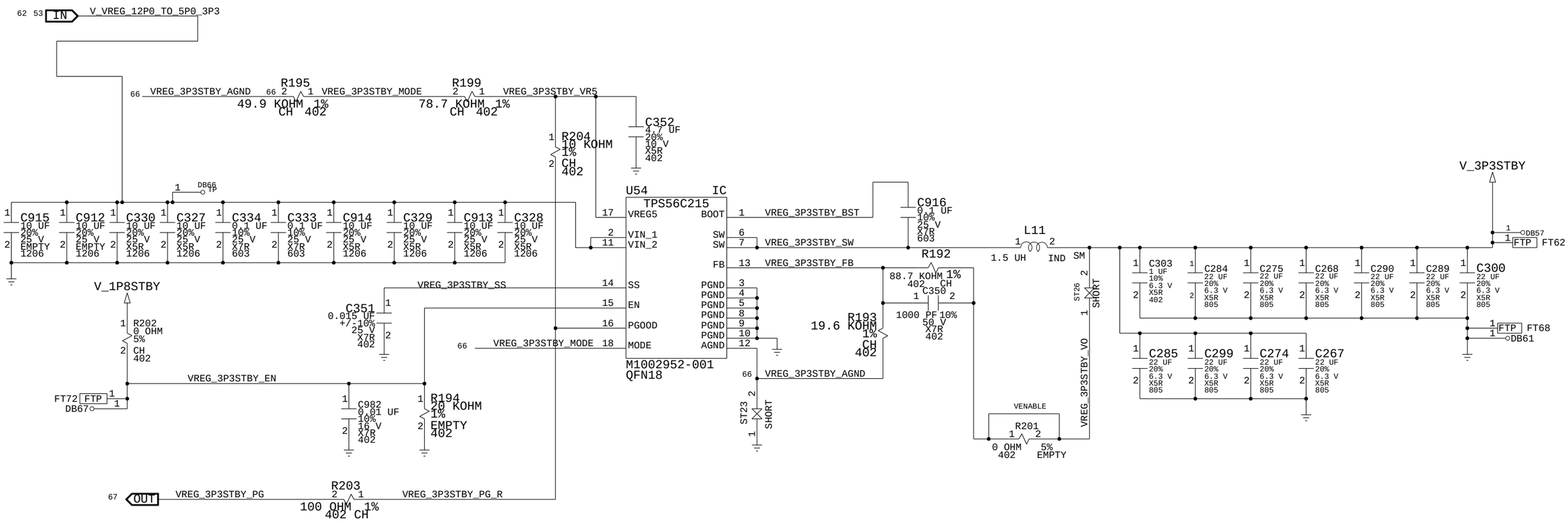



| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 65/82 | 65/82 | G-R | 0.991 |

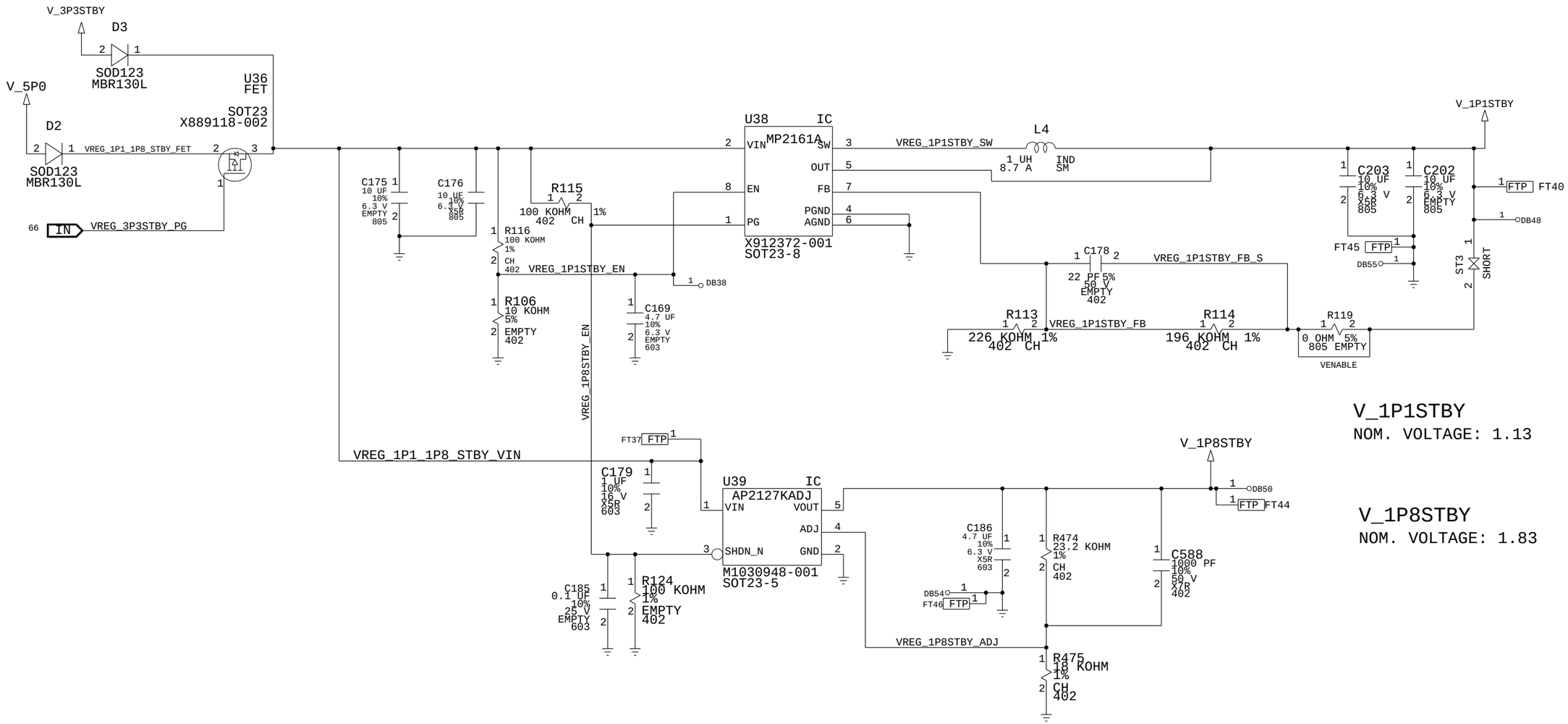
# VREGS: V3P3 STANDBY

V_3P3STBY
NOM. VOLTAGE: 3.32




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 66/82 | 66/82 | G-R | 0.991 |

# VREGS: V1P1 STANDBY, V1P8 STANDBY

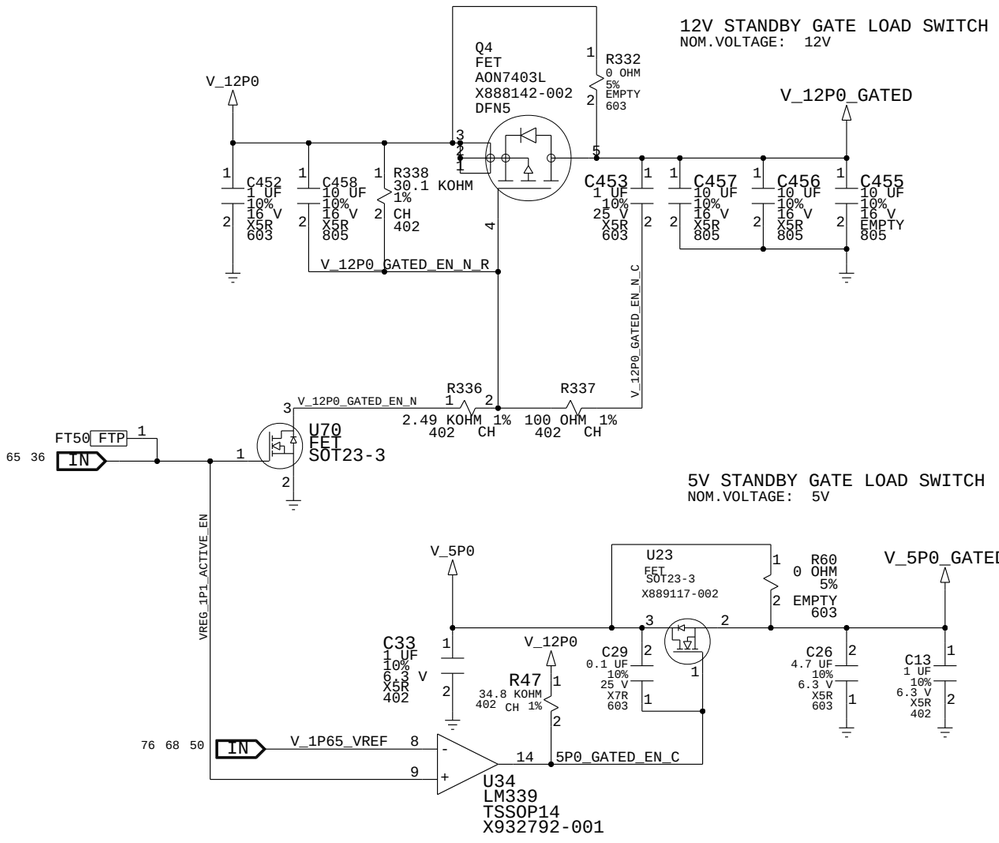


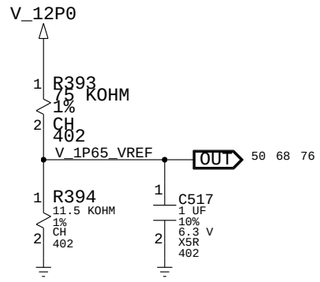

**V_1P1STBY**
NOM. VOLTAGE: 1.13

**V_1P8STBY**
NOM. VOLTAGE: 1.83

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 67/82 | 67/82 | G-R | 0.991 |

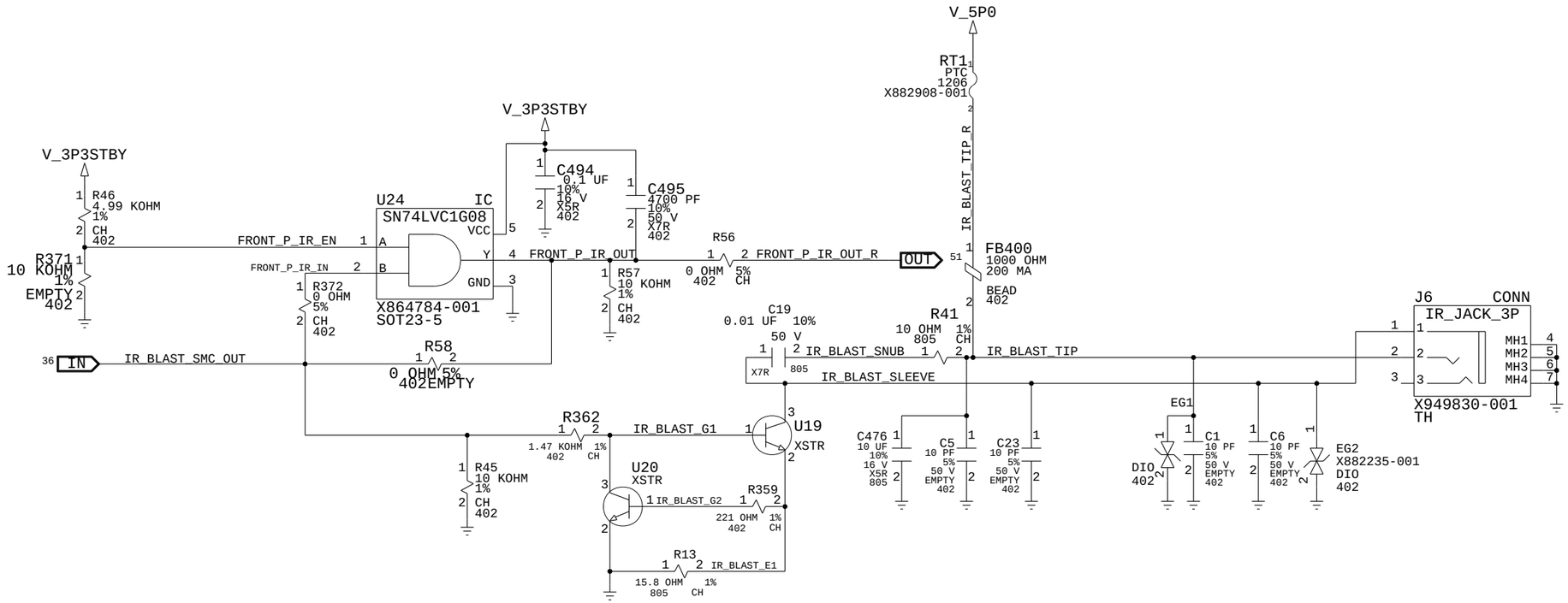
# STANDBY GATES



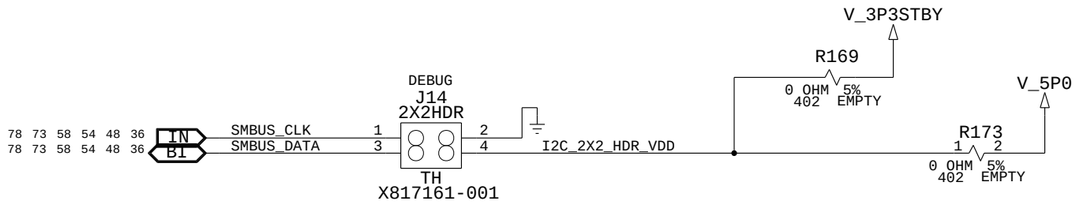

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 68/82 | 68/82 | G-R | 0.991 |

# IR BLASTER




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 69/82 | 69/82 | G-R | 0.99 |

# I2C




FOR AARDVARK, CAN CHOOSE 3.3V OR 5V VDD

MARGIN: SOCPHY,SOC1P8

# BLANK

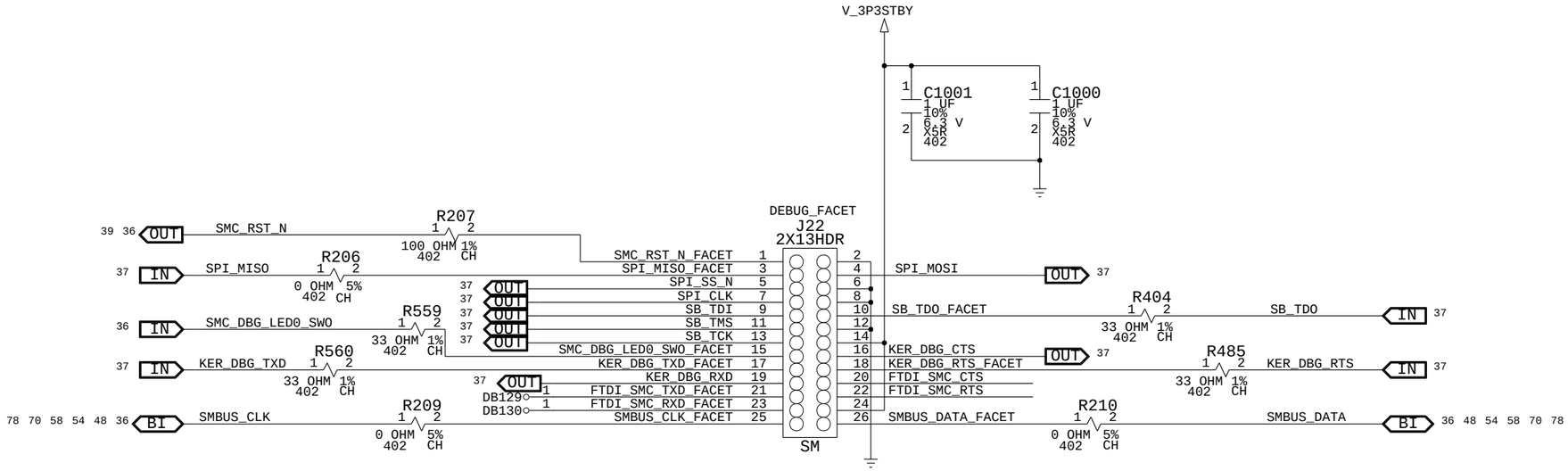
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 71/82 | 71/82 | G-R | 0.991 |

MONITOR: VSOC1P8,VSOCPHY,V12P0

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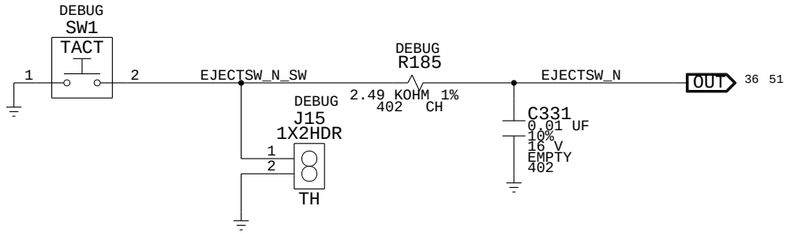
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 72/82 | 72/82 | G-R | 0.991 |

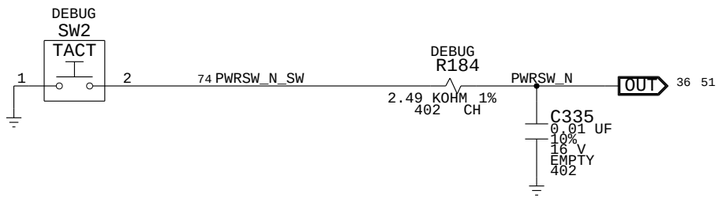
# CONN: FACET BOARD

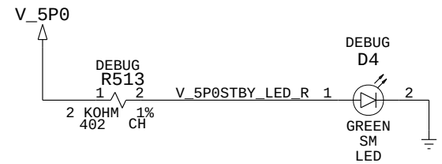


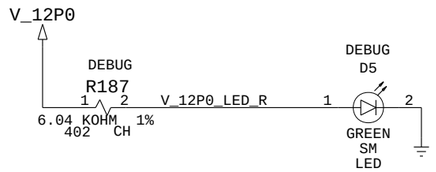

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 73/82 | 73/82 | G-R | 0.991 |

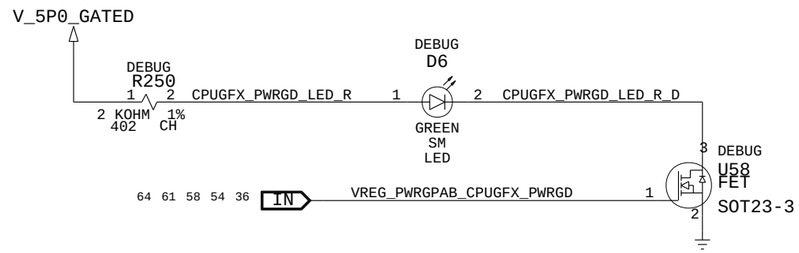
# CONN: SWITCHES

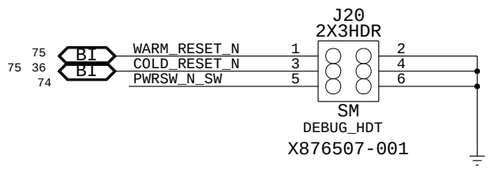


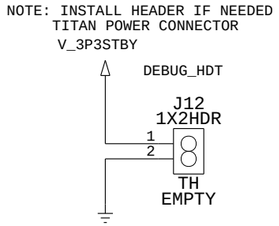



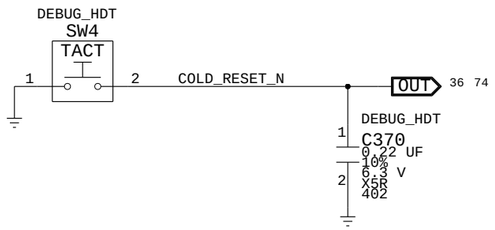



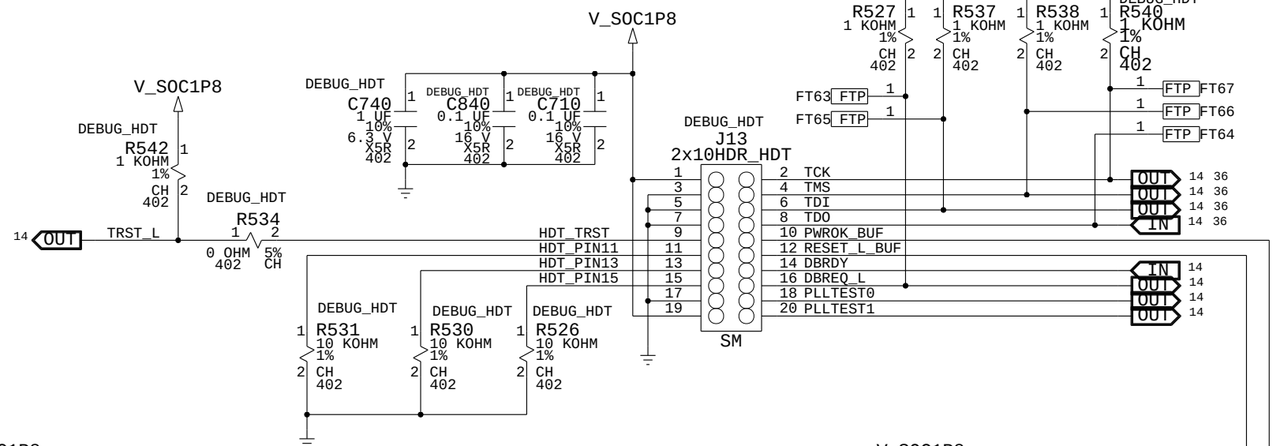

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 74/82 | 74/82 | G-R | 0.991 |

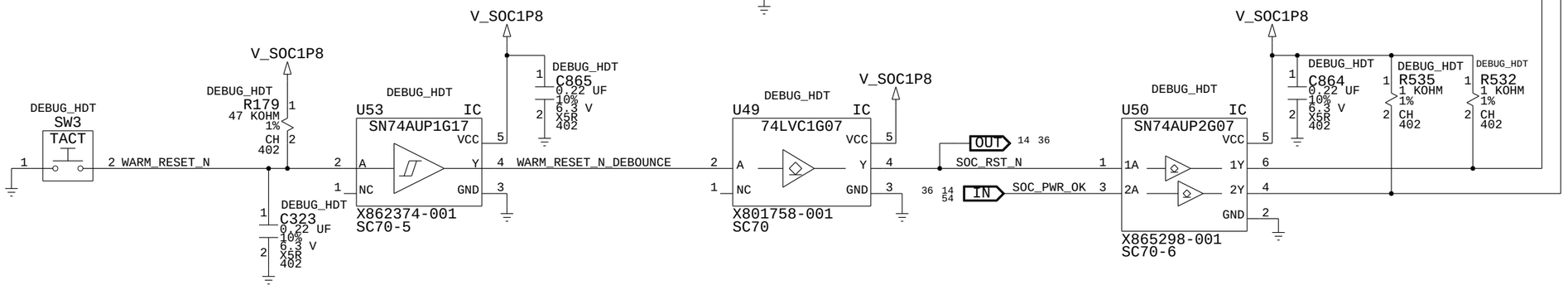
# CONN: HDT

NOTE: INSTALL HEADER IF NEEDED
TITAN POWER CONNECTOR




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 75/82 | 75/82 | G-R | 0.991 |

# CONN: M.2

MTG5
SPACER_1P
M1008483-001
TH
M2_ONLY
SSD MOUNTING HARDWARE

SSD LOAD SWITCH
NOM.VOLTAGE: 3.3V
PEAK CURRENT: 2.5A

NOTE: R328 AND D8 ARE DEBUG COMPONENTS AND NEED TO BE REMOVED FROM RETAIL CONFIGS

R325 0 OHM 5% 805 EMPTY
Q3 NTTFS4C50N X889324-001 DFN5 FET
V_3P3STBY
V_12P0
V_3P3_M2
C434 1 UF 10% 6.3 V X5R 402
C437 0.1 UF 10% 25 V X7R 603
C450 4.7 UF 10% 6.3 V X5R 603
C448 1 UF 10% 6.3 V X5R 402
C1075 1 UF 10% 6.3 V X5R 402
C1078 1 UF 10% 6.3 V X5R 402
C449 1 UF 10% 6.3 V X5R 402
C446 22 UF 20% 6.3 V X5R 805
C519 0.1 UF 10% 25 V X7R 603
R637 10 KOHM 1% CH 402
R328 511 OHM 1% 402 CH
D8 GREEN SM LED
M2_LED
M2_LED_R

M2_ONLY
J32
PCIE_M2_M_RCPT_67P_2MH
M1007401-001
SM

R78 34.8 KOHM 402 CH 1%
M2_ENABLE_C
68 50 IN V_1P65_VREF
V_3P3
U34 LM339 TSSOP14 X932792-001
R634 10 KOHM 1% EMPTY 402
M2_PWR_EN_R
36 42 IN M2_PWR_EN
R635 0 OHM 5% 402 CH M2_ONLY
R636 10 KOHM 1% CH 402

R283 10 KOHM 1% CH 402 M2_ONLY
R293 10 KOHM 1% CH 402 M2_ONLY
M2_PCIE_RESET_N
77 OUT M2_CLKREQ_N
REF CLK CONTROLLED BY SSD TO SUPPORT POWER MODE EXPERIMENTS
U63 FET SOT23-3 M2_ONLY
RST_G_C
42 36 IN PCIE_RST_N
R318 100 OHM 1% 402 CH M2_ONLY
U66 FET SOT23-3 M2_ONLY
C433 0.022 UF 10% 50 V X7R 402 M2_ONLY
R307 100 OHM 1% 402 CH M2_ONLY
RST_G
RST_G_R

PEX_L3_M2_SOC_TN
PEX_L3_M2_SOC_TP
PEX_L3_SOC_M2_TN_C
PEX_L3_SOC_M2_TP_C
PEX_L2_M2_SOC_TN
PEX_L2_M2_SOC_TP
PEX_L2_SOC_M2_TN_C
PEX_L2_SOC_M2_TP_C
PEX_L1_M2_SOC_TN
PEX_L1_M2_SOC_TP
PEX_L1_SOC_M2_TN_C
PEX_L1_SOC_M2_TP_C
PEX_L0_M2_SOC_TN
PEX_L0_M2_SOC_TP
PEX_L0_SOC_M2_TN_C
PEX_L0_SOC_M2_TP_C
M2_PEX_SS_100M_CLKN
M2_PEX_SS_100M_CLKP

THE SSD HAS AN INTERNAL PULL-UP ON ITS PCIE RESET INPUT.
ISOLATION OF SIGNAL PCIE_RST_N IS NEEDED TO ISOLATE
SOUTHBRIDGE OUTPUT FROM SSD PULLUP

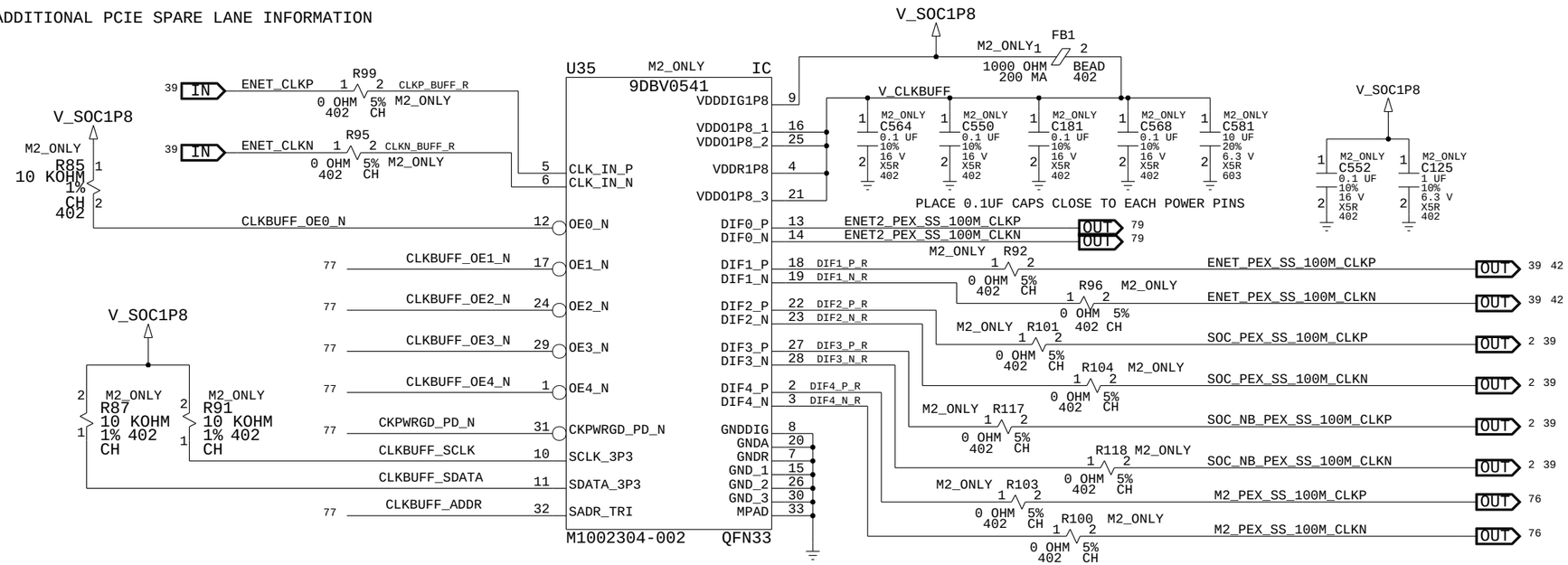
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 76/82 | 76/82 | G-R | 0.991 |

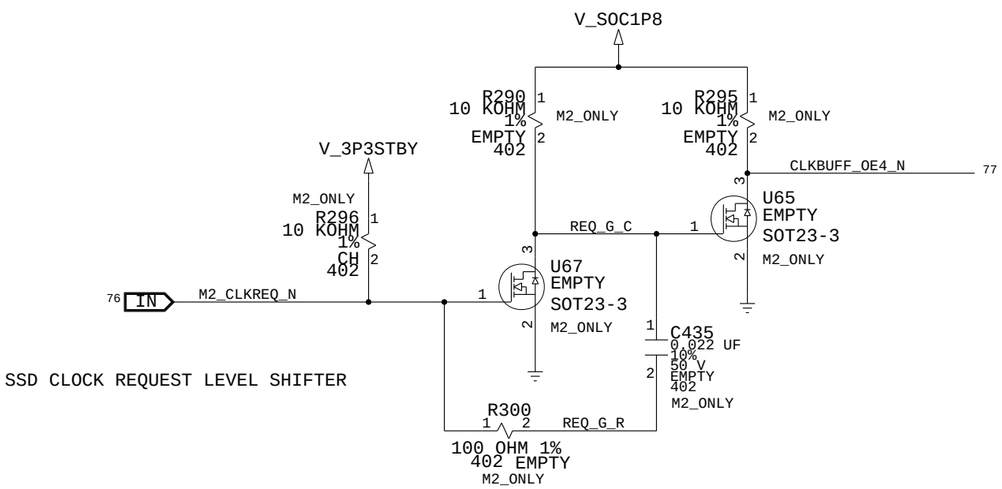
# CLOCK BUFFER

NOTES:

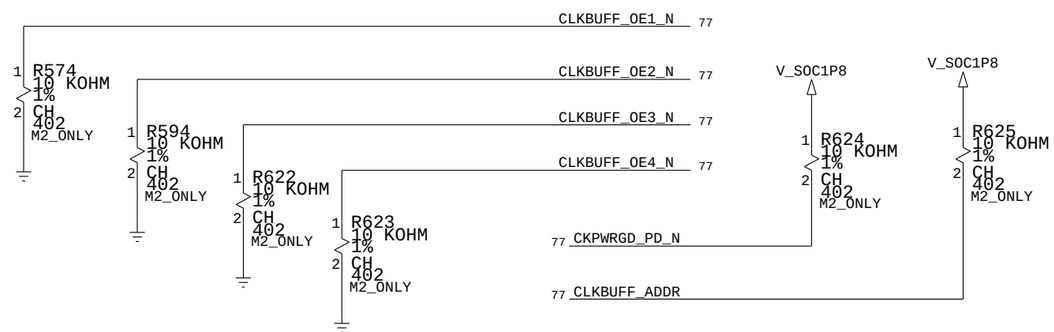
1.TO ENABLE ENET2 CLOCK ON J28 HEADER, UNSTUFF R85. U35 HAS INTERNAL PULL DOWN.

2.SEE PAGES 2 AND 79 FOR ADDITIONAL PCIE SPARE LANE INFORMATION



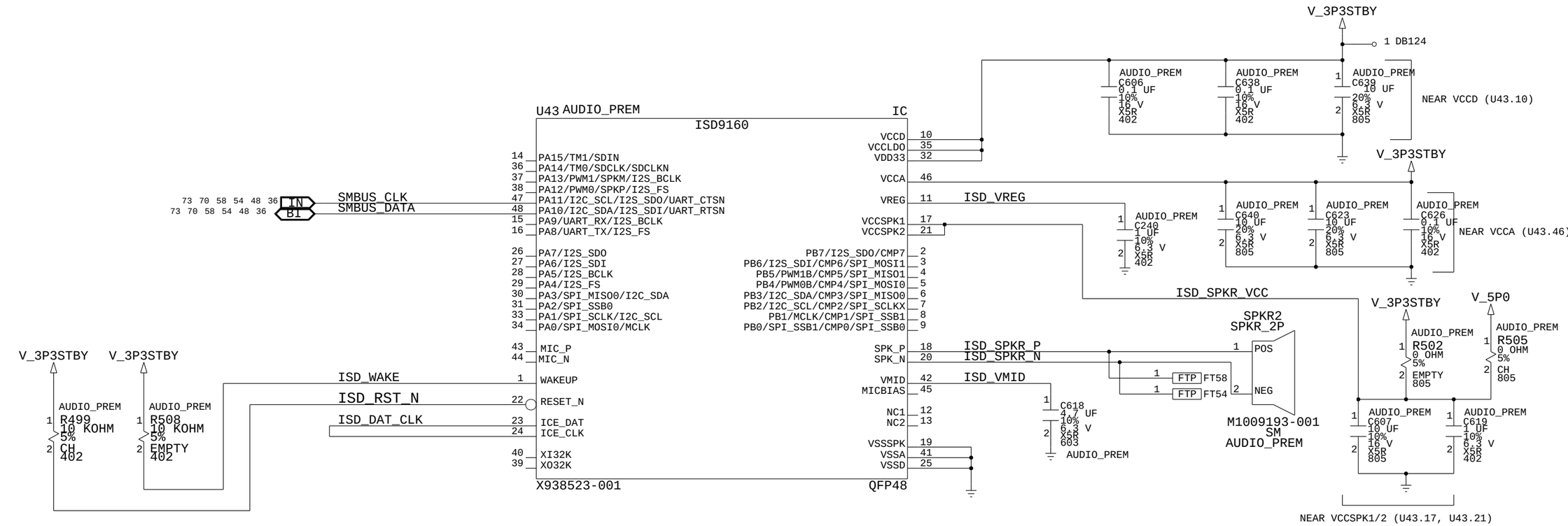

TO ADDRESS LONG TERM RELIABILITY CONCERNS
PERMANENT PULL UP/DOWN RESISTORS HAVE BEEN ADDED
EVEN THOUGH THIS PART HAS ITS OWN INTERNAL 100K EQUIVALENTS




| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 77/82 | 77/82 | G-R | 0.99 |

# PREMIUM SPEAKER (SE/LE)

## PREMIUM/SE/LE SKU ONLY




ISD9160FIMS05 - REMOVED CAP TOUCH FUNCTIONALITY

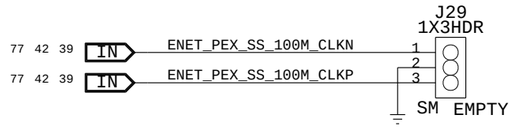
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | VER |
|---|---|---|---|---|
| | Cactus | 78/82 | 78/82 | 0.991 |

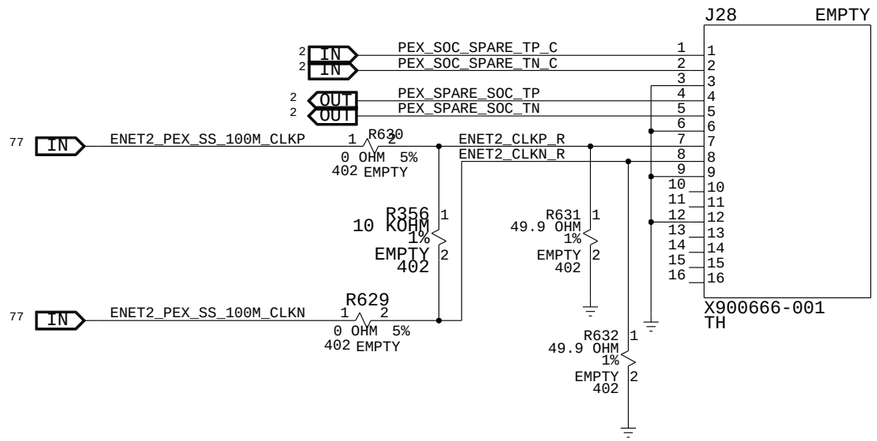
# DEBUG: VR HEADERS, TEST POINTS, CONNECTORS

V_12P0 — DEBUG TP17 — 1 — SM
V_5P0 — DEBUG TP11 — 1 — SM
V_3P3 — DEBUG TP1 — 1 — SM
V_1P8STBY — DEBUG TP7 — 1 — SM
V_SOC1P8 — DEBUG TP16 — 1 — SM
V_1P1STBY — DEBUG TP6 — 1 — SM
V_SOCPHY — DEBUG TP2 — 1 — SM
V_BAT — DEBUG TP3 — 1 — SM
V_SB1P1 — DEBUG TP8 — 1 — SM
V_FUSE — DEBUG TP15 — 1 — SM
V_SB1P8 — DEBUG TP5 — 1 — SM
DEBUG TP12 — 1 — SM
V_3P3STBY — DEBUG TP10 — 1 — SM
DEBUG TP18 — 1 — SM
V_GFXCORE — DEBUG TP9 — 1 — SM
V_CPUCORE — DEBUG TP19 — 1 — SM
V_NBCORE — DEBUG TP13 — 1 — SM
V_MEMIO — DEBUG TP4 — 1 — SM
V_MEMPHY — DEBUG TP14 — 1 — SM

NOTE: THESE TEST POINTS ARE NOT TO BE USED FOR VOLTAGE REGULATOR QUALIFICATION TEST POINTS

## PCIE CONNECTORS



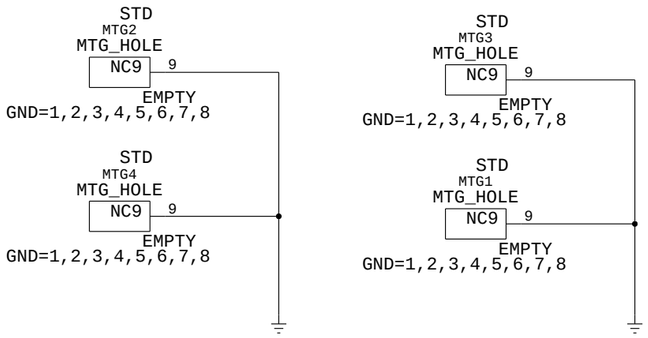

## NOTES: TO ENABLE 100 MHZ REF CLOCK ON HEADER J28:

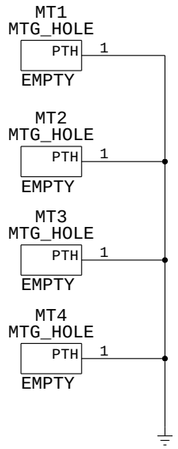
1. WHEN CLOCK BUFFER U35 IS INSTALLED: POPULATE R629 AND R630.
2. WITHOUT CLOCK BUFFER INSTALLED:
   A. POPULATE R631 AND R632.
   B. WIRE JUMPER ETHERNET 100 MHZ CLOCK FROM HEADER J29.1 AND J29.3 TO J28.8 AND J28.7 RESPECTIVELY.
3. SEE PAGES 2 AND 77 FOR ADDITIONAL PCIE SPARE LANE INFORMATION.

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | FAB | VER |
|---|---|---|---|---|---|
| | Cactus | 79/82 | 79/82 | G-R | 0.991 |

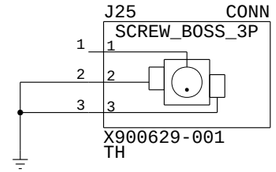
# LABELS AND MOUNTING

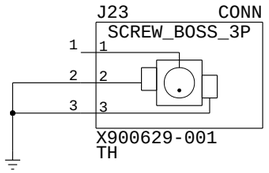
HEAT SINK MOUNTING HOLES



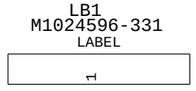




SCREW BOSS

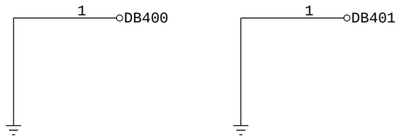



INTELLIGENT SERIAL NUMBER TARGET




GND PADS FOR HEATSINK ALIGNMENT PINS

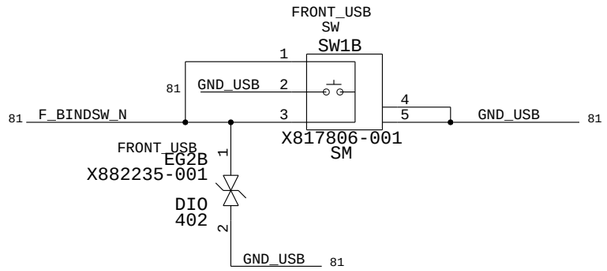


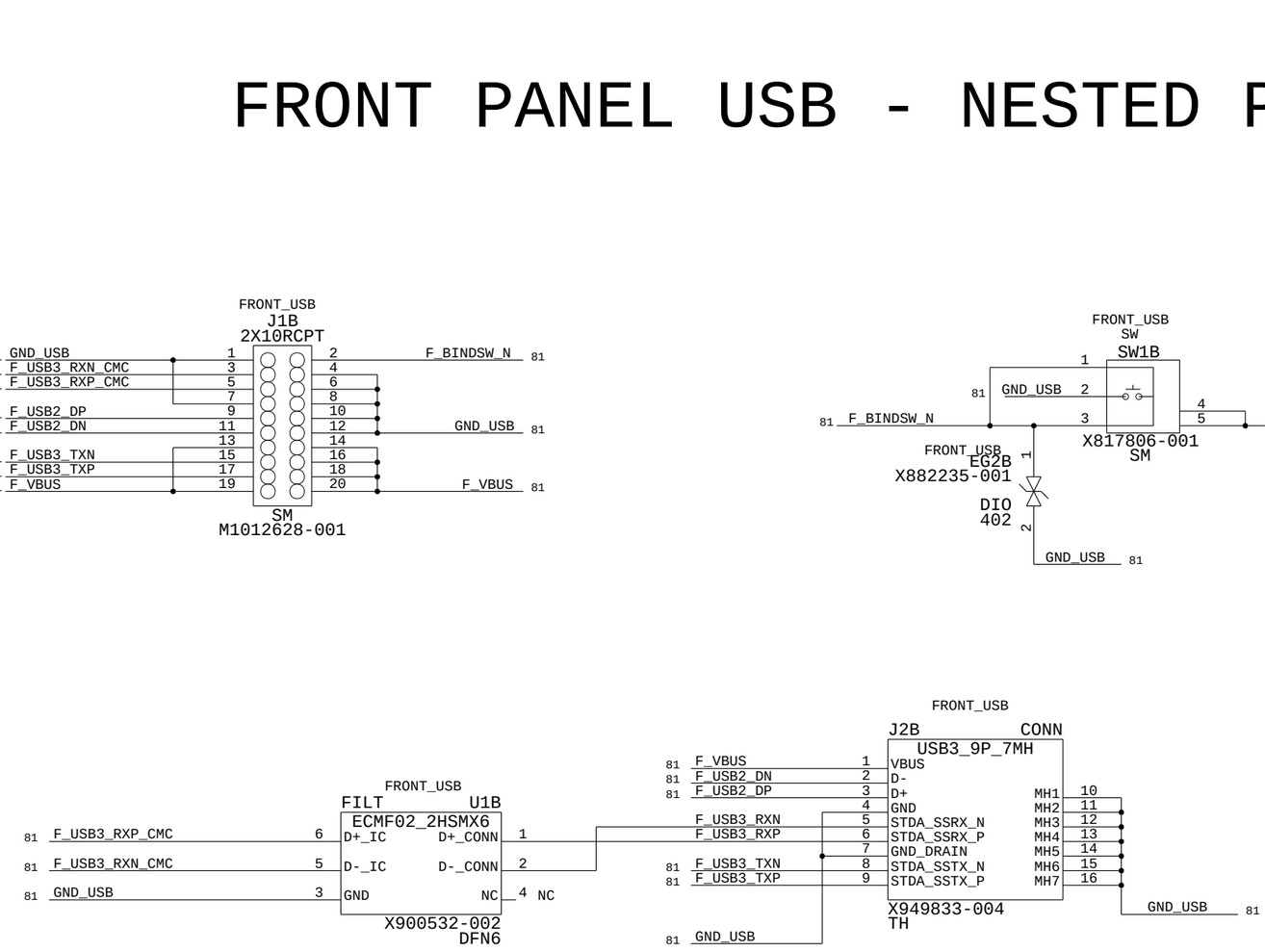

| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| NA | FR4 | PCB1 | PCB,CACTUS,FAB G RETAIL,8 LAYERS,GI | PCB_GI |
| M1037358-002 | FR4 | PCB1 | PCB,CACTUS,FAB G RETAIL,8 LAYERS,OSP | PCB_OSP |

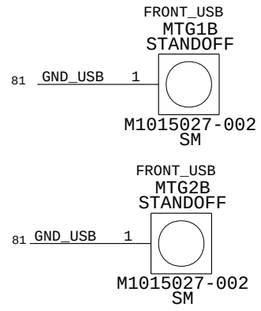
| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | VER |
|---|---|---|---|---|
| | Cactus | 80/82 | 80/82 | 0.991 |

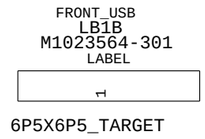
# FRONT PANEL USB - NESTED PCB

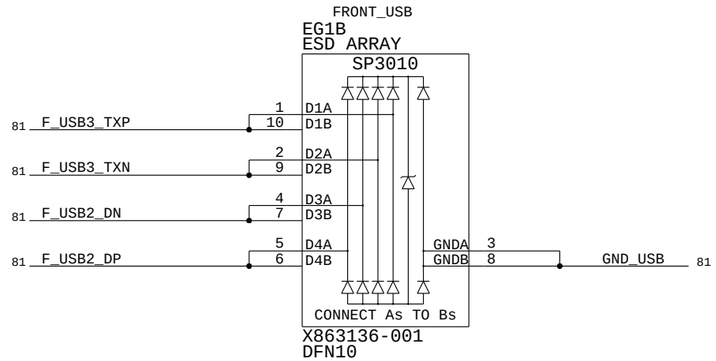



| MS_PART# | MATL | REF_DES | DESCR. | BOM PROPERTY |
|---|---|---|---|---|
| M1033394-001 | FR4 | PCB1B | PCB,CACTUS,FAB E RETAIL,FRONT USB,8 LAYERS,G1 | FRONT_USB |

| MICROSOFT CONFIDENTIAL | PROJECT NAME | PAGE | CSA PAGE | VER |
|---|---|---|---|---|
| | Cactus | 81/82 | 81/82 | 0.991 |

# BOM DEFINITIONS

| BOM | DEFINTION |
|---|---|
| AUDIO | INCLUDES COMPONENTS FOR THE STANDARD AUDIO SOLUTION |
| AUDIO_PREM | INCLUDES COMPONENTS FOR THE PREMIUM SE/LE SPEAKER SOLUTION |
| COMMON | ALL COMPONENTS WITH NO BOM PROPERTY |
| DEBUG | COMPONENTS REQUIRED FOR BRING UP & DEBUG |
| DEBUG_HDT | HDT-RELATED DEBUG COMPONENTS |
| DEBUG_SHUNT | COMPONENTS WHICH ARE ON DEBUG BOARDS, BUT ARE REMOVED/SHORTED ON RETAIL |
| EMMC_BASE | DUMMY PLACE HOLDER FOR EMMC DEVICE & RESISTORS.  NEVER USE THIS IN THE RECIPE FILE.  SELECT ONE OF THESE INSTEAD: EMMC_HYNIX_16NM, EMMC TOSHIBA_15NM, EMMC SAMSUNG_14NM |
| EMMC_HYNIX_16NM | HYNIX EMMC DEVICE |
| EMMC_SAMSUNG_14NM | SAMSUNG EMMC DEVICE |
| EMMC_TOSHIBA_15NM | TOSHIBA EMMC DEVICE |
| GDDR5_BASE | DUMMY PLACE HOLDER FOR GDDR5. NEVER USE THIS IN THE RECIPE FILE.  SELECT ONE OF THESE INSTEAD: EMMC_HYNIX_16NM, EMMC TOSHIBA_15NM, EMMC SAMSUNG_14NM |
| GDDR5_HYNIX | HYNIX GDDR5 MEMORY |
| GDDR5_SAMSUNG | SAMSUNG GDDR5 MEMORY |
| FRONT_USB | COMPONENTS ON THE FRONT PANEL USB |
| KIC_BASE | DUMMY PLACE HOLDER FOR KIC.  NEVER USE THIS IN THE RECIPE FILE.  USE ONE OF THESE INSTEAD: KIC_DEV OR KIC_RETAIL |
| KIC_DEV | DEBUG VERSION OF KRAKEN |
| KIC_RETAIL | RETAIL VERSION OF KRAKEN |
| M2_ONLY | POPULATE TO SUPPORT AN M.2 INTERFACE |
| NO_M2 | POPULATE WHEN THERE IS NO M2. INTERFACE |
| PCB_GI | FAB TYPE: GOLD |
| PCB_OSP | FAB TYPE: ORGANIC SOLDERABILITY PRESERVATIVE GREEN SOLDERMASK |
| PCB_OSP_BLACK | FAB TYPE: ORGANIC SOLDERABILITY PRESERVATIVE BLACK SOLDERMASK |
| RTC_RETAIL | RTC CIRCUIT IMPLEMENTATION FOR RETAIL BOARDS |
| RTC_XDK | RTC CIRCUIT IMPLEMENTATION FOR XDK BOARDS |
| SOC_BASE | DUMMY PLACE HOLDER FOR SOC. NEVER USE THIS IN THE RECIPE FILE.  SELECT ONE OF THESE INSTEAD: EMMC_HYNIX_16NM, EMMC TOSHIBA_15NM, EMMC SAMSUNG_14NM |
| SOC_EMPTY | DOES NOT STUFF ANUBIS |
| SOC_INCLUDE | STUFFS ANUBIS |
| VR_FIXED | SET ALL VRS TO FIXED VOLTAGES (NON-MARGINED).  EXCLUDES V_MEMIO.  MUST BE USED IN CONJUNCTION WITH NOT VR_MM |
| VR_MM | ALLOWS MOST VRS TO BE MARGINED FOR M&M BOARDS.  EXCLUDES V_MEMIO.  MUST BE USED IN CONJUNCTION WITH NOT VR_FIXED |